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(54) **CSI REPORTING FOR ENHANCED TYPE-I CODEBOOK**

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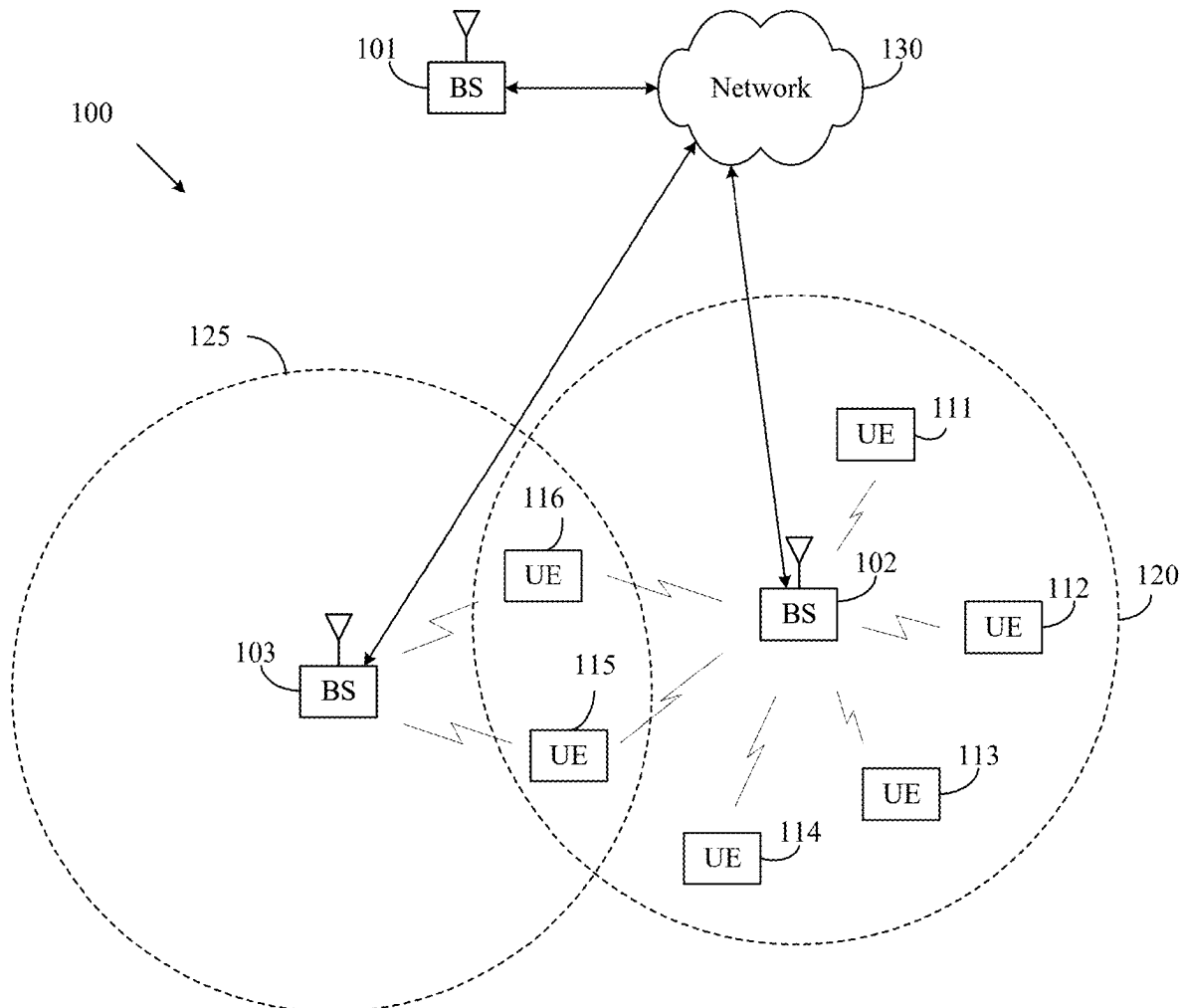
10, 2024, provisional application No. 63/680,425, filed on Aug. 7, 2024, provisional application No. 63/681,540, filed on Aug. 9, 2024, provisional application No. 63/684,746, filed on Aug. 19, 2024.

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(57) **ABSTRACT**

Apparatuses and methods for channel state information (CSI) reporting for an enhanced Type-I codebook a method performed by a user equipment (UE) includes receiving information about a CSI report. The information indicates codebookType set to 'typeI-SinglePanel-r19', codebookMode set to either 'ModeA' or 'ModeB', and P_{CSI-RS} CSI reference signal (CSI-RS) antenna ports, where $P_{CSI-RS} > 32$. The method further includes, based on the information, determining the CSI report associated with the CSI-RS antenna ports, the CSI report including a precoding matrix indicator (PMI), and transmitting the CSI report.



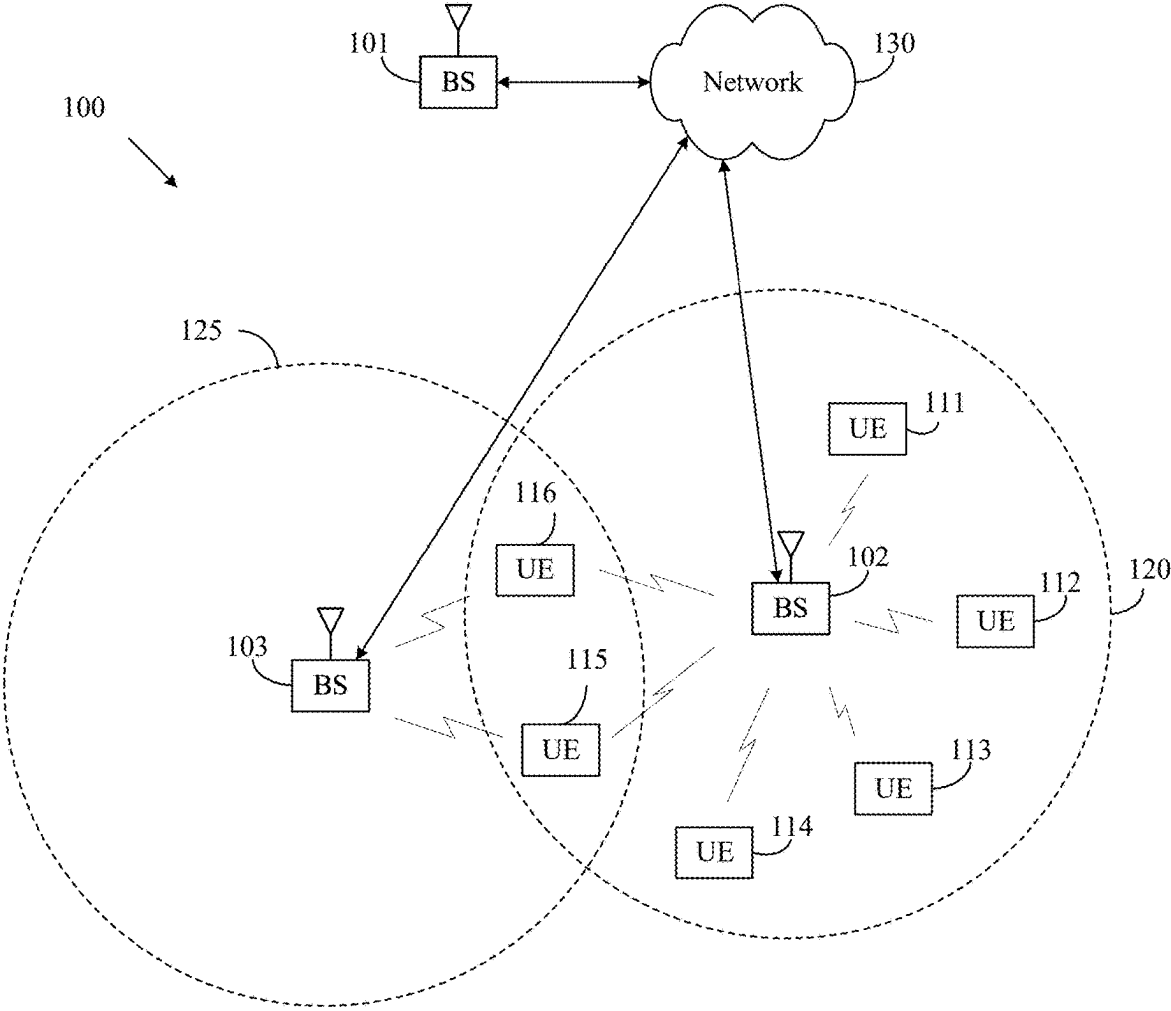


FIG. 1

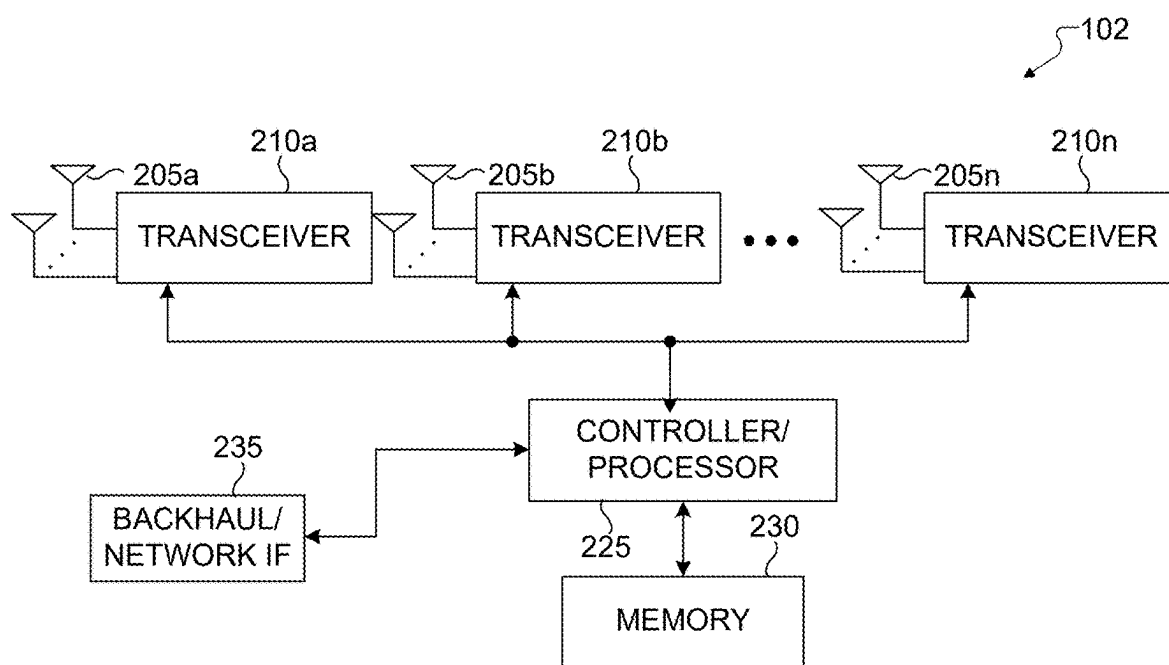


FIG. 2

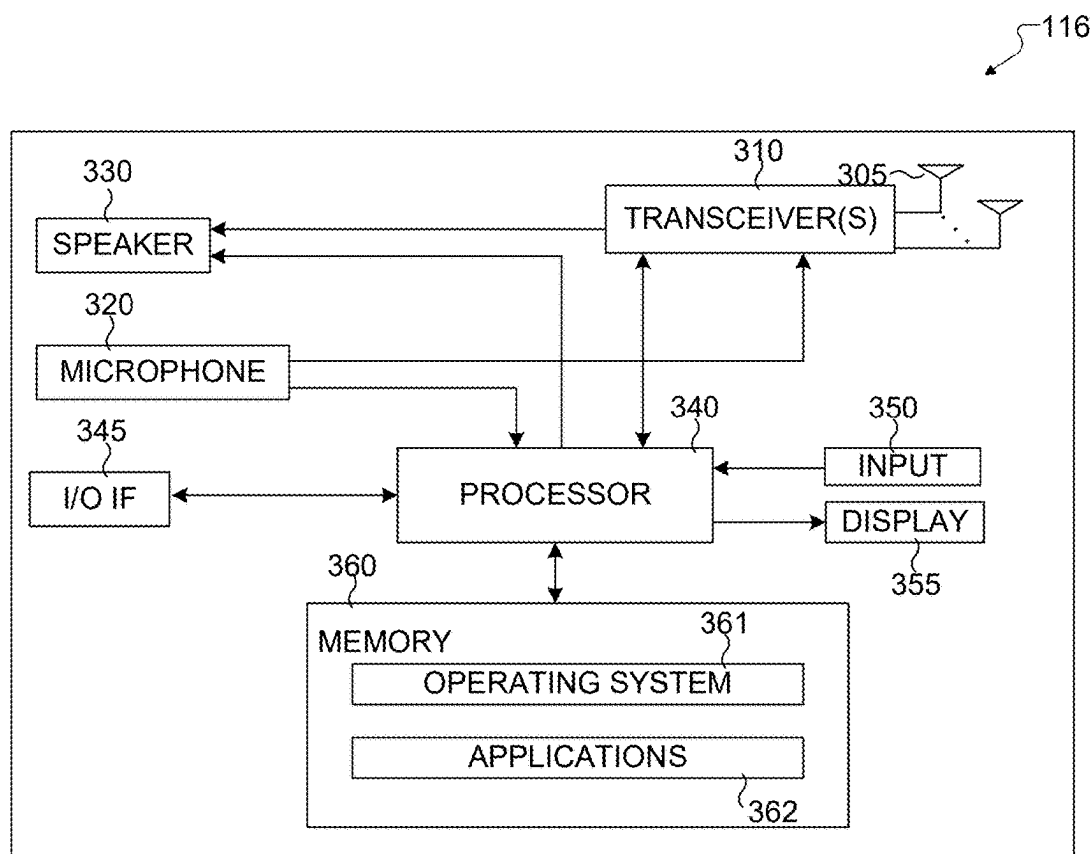


FIG. 3

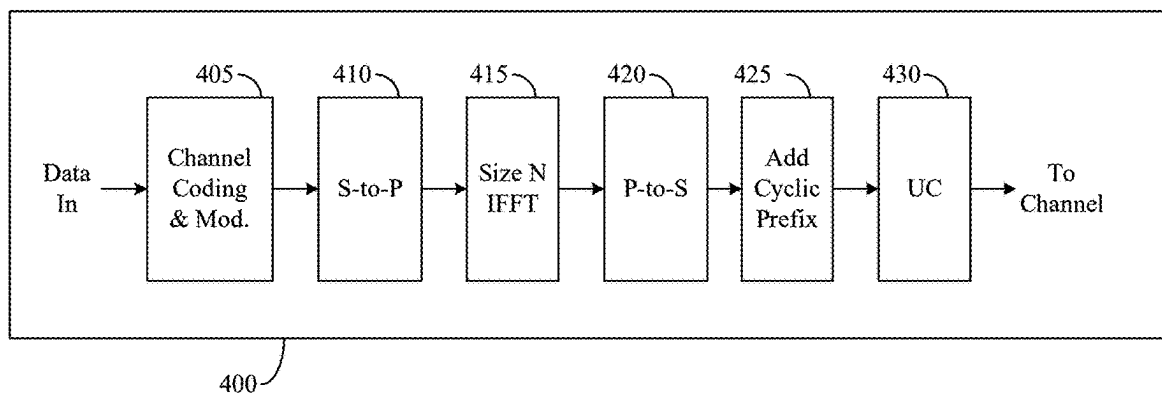


FIG. 4A

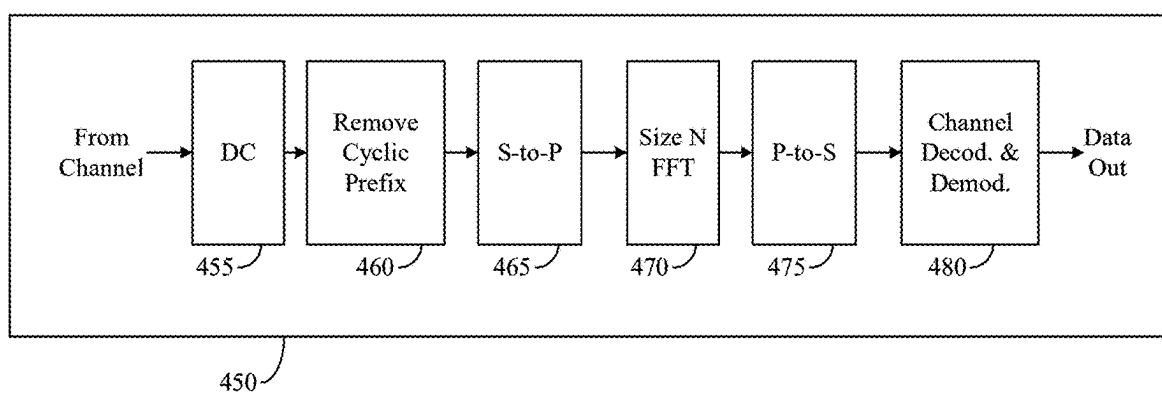


FIG. 4B

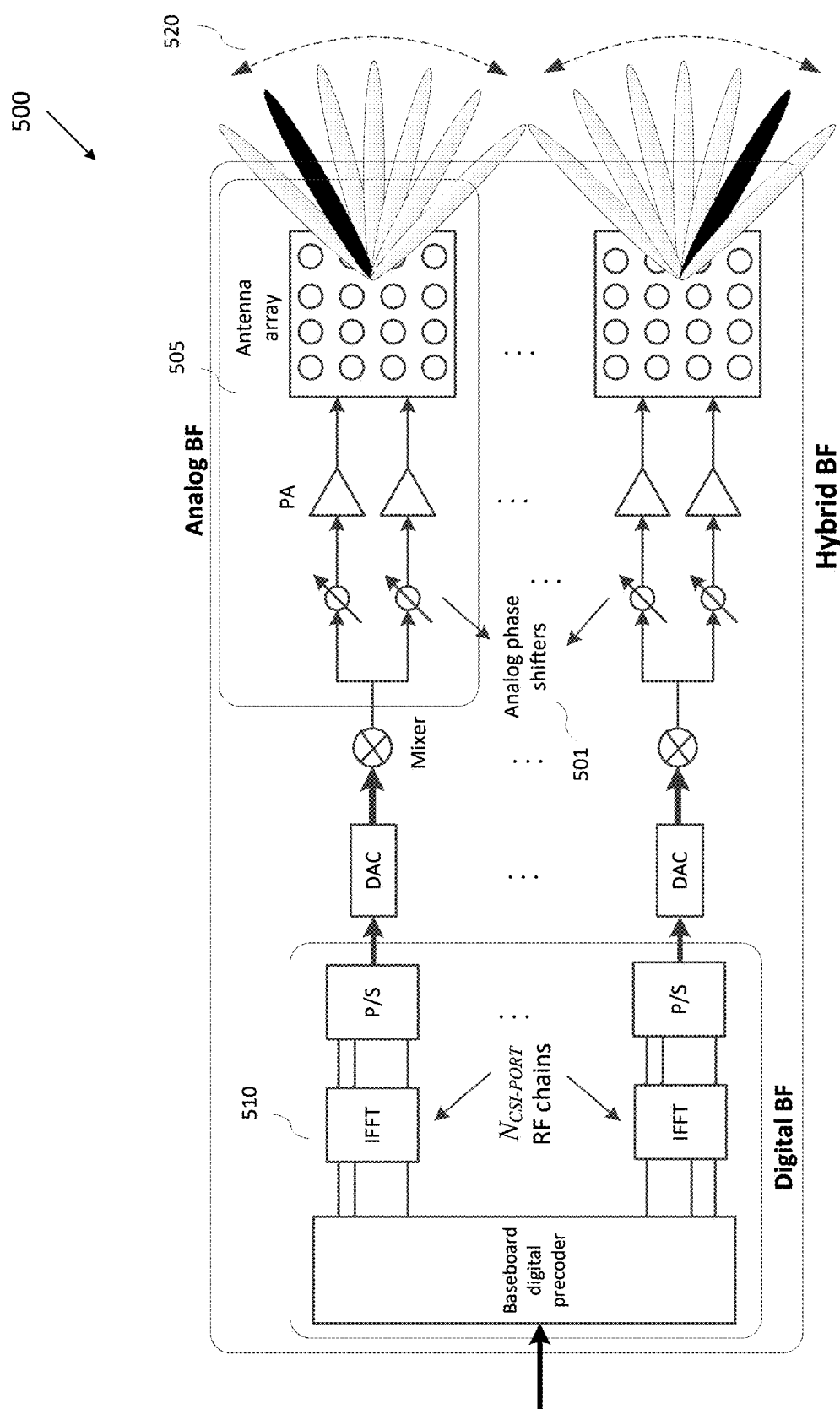


FIG. 5

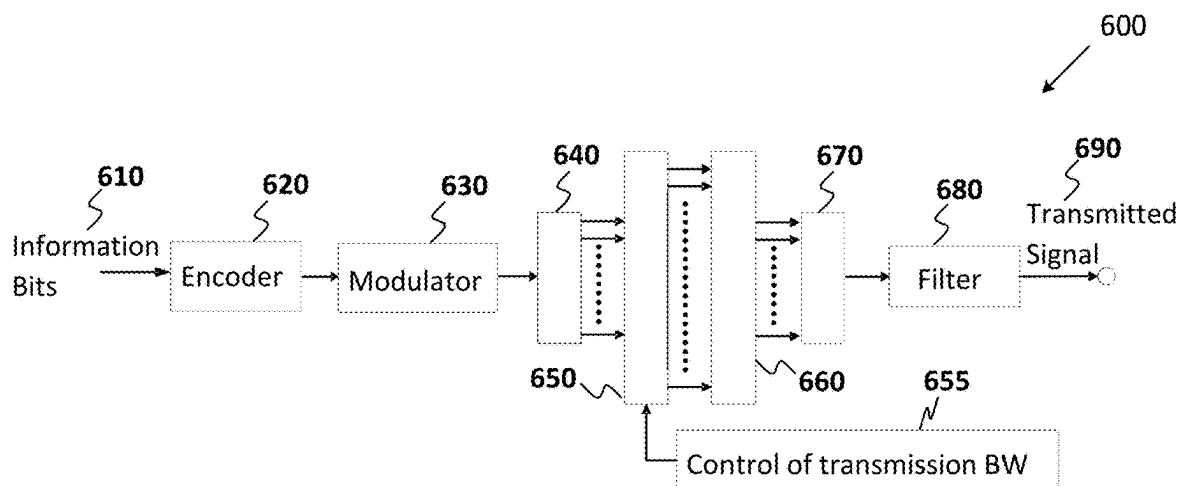


FIG. 6

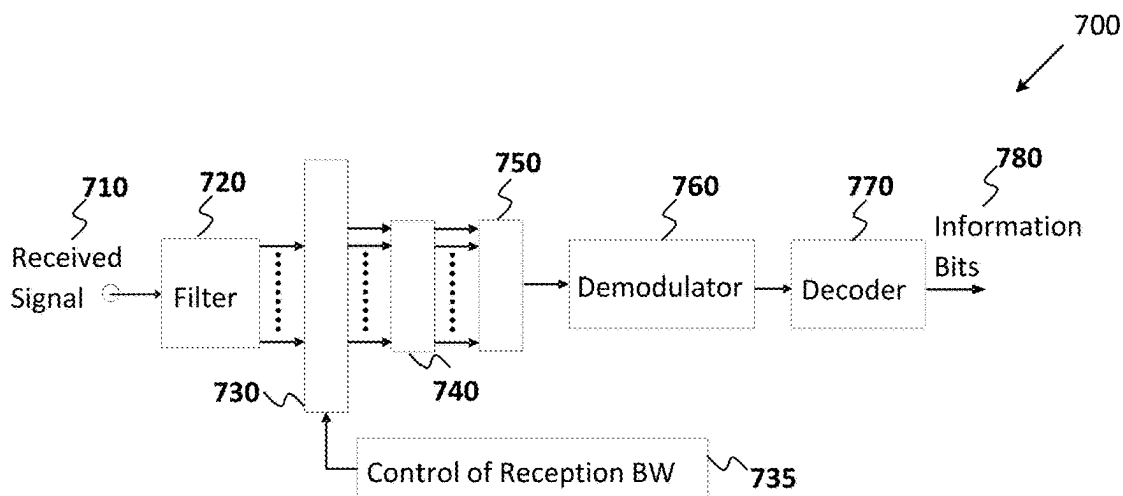


FIG. 7

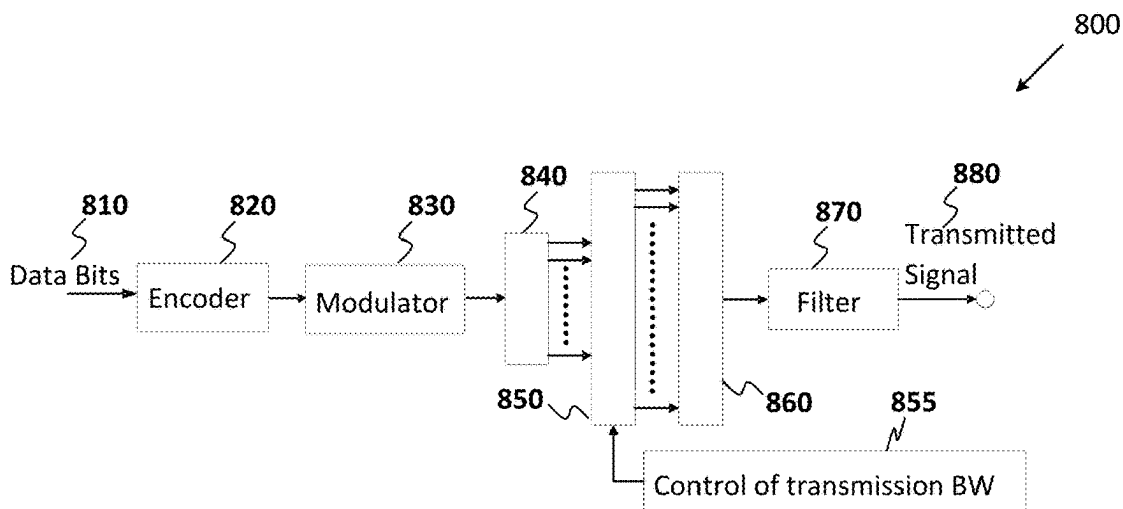


FIG. 8

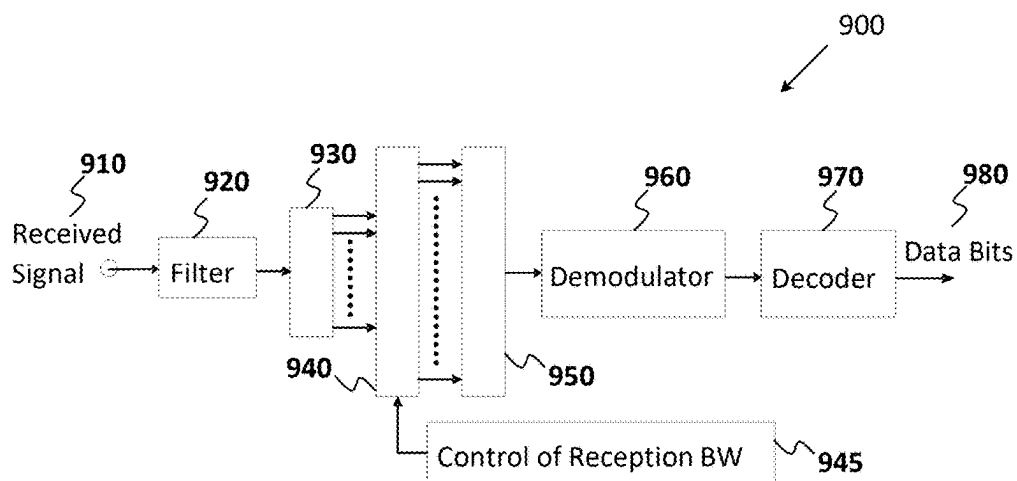


FIG. 9

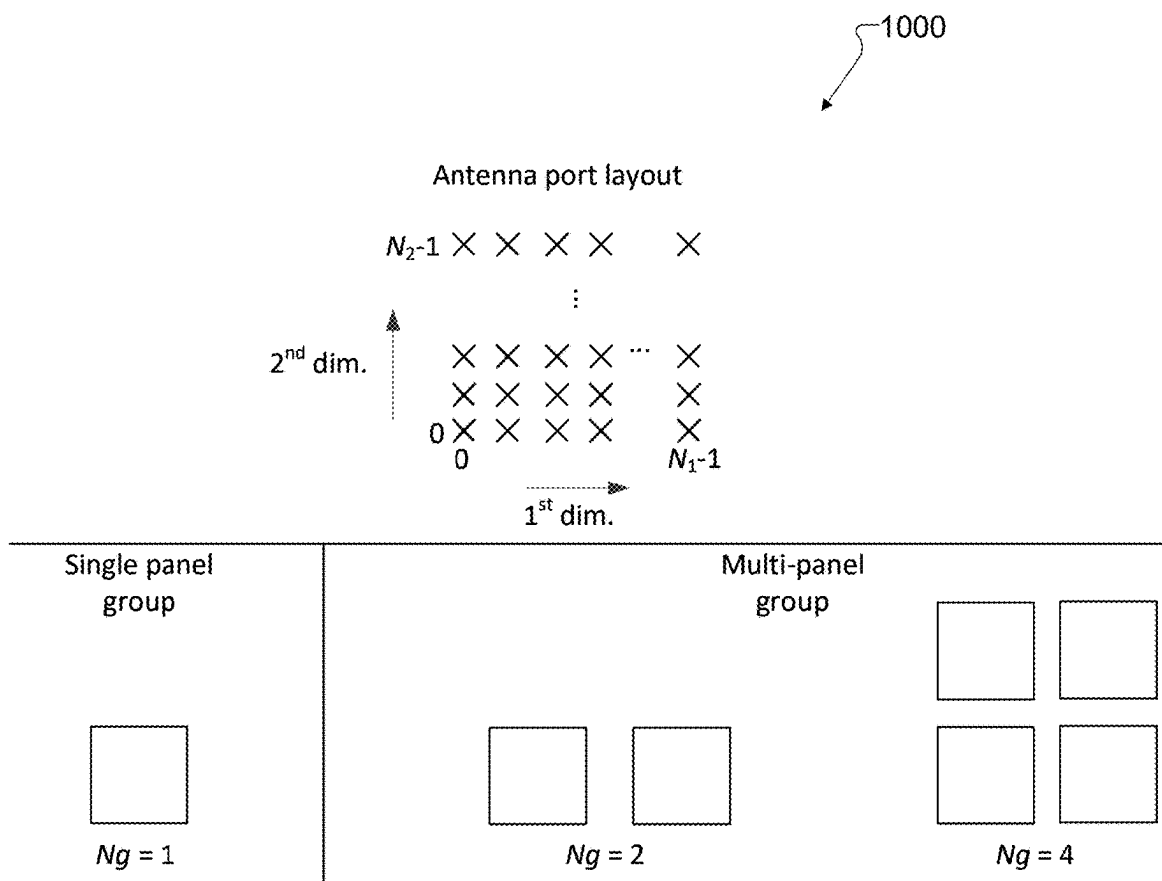


FIG. 10

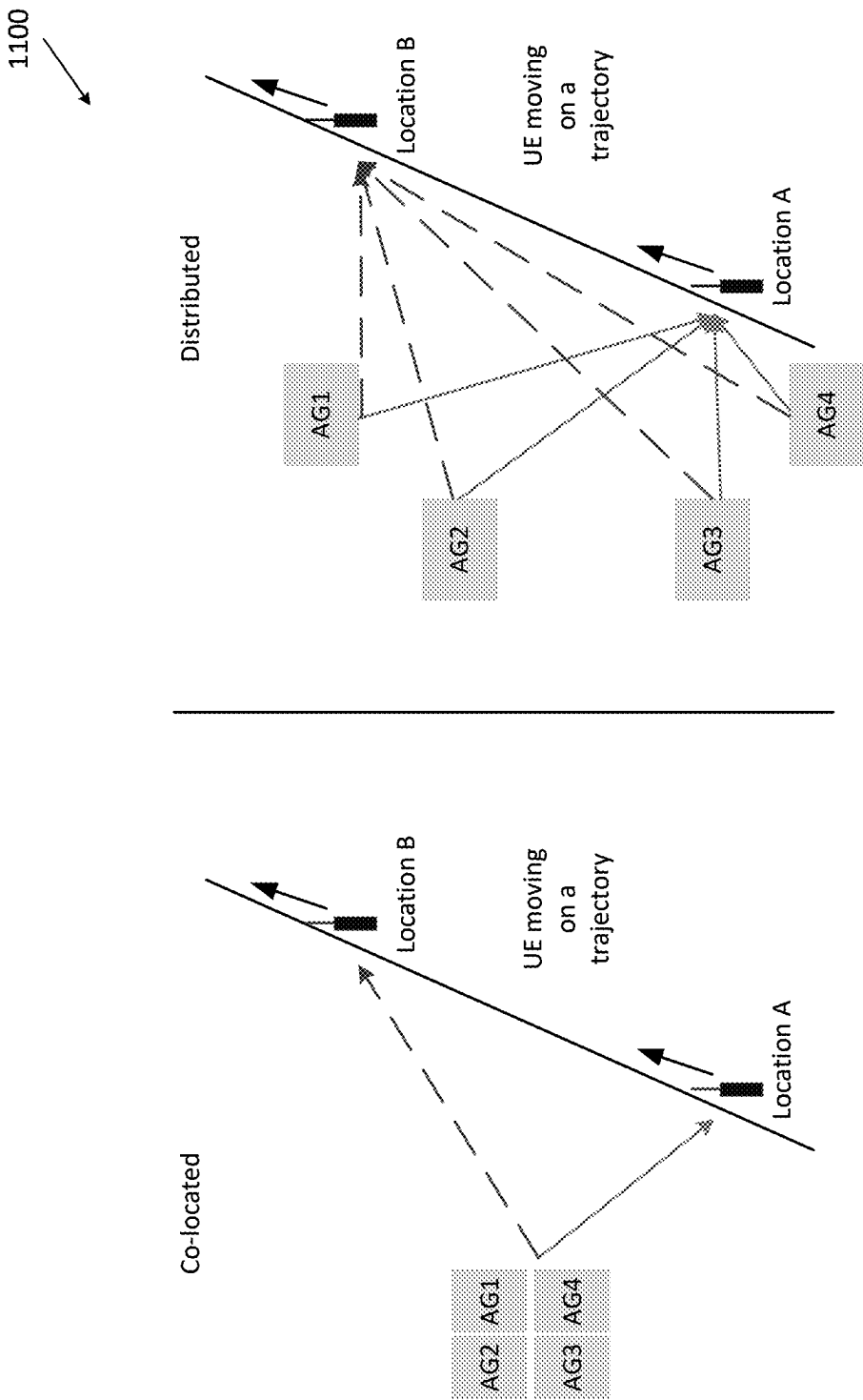


FIG. 11

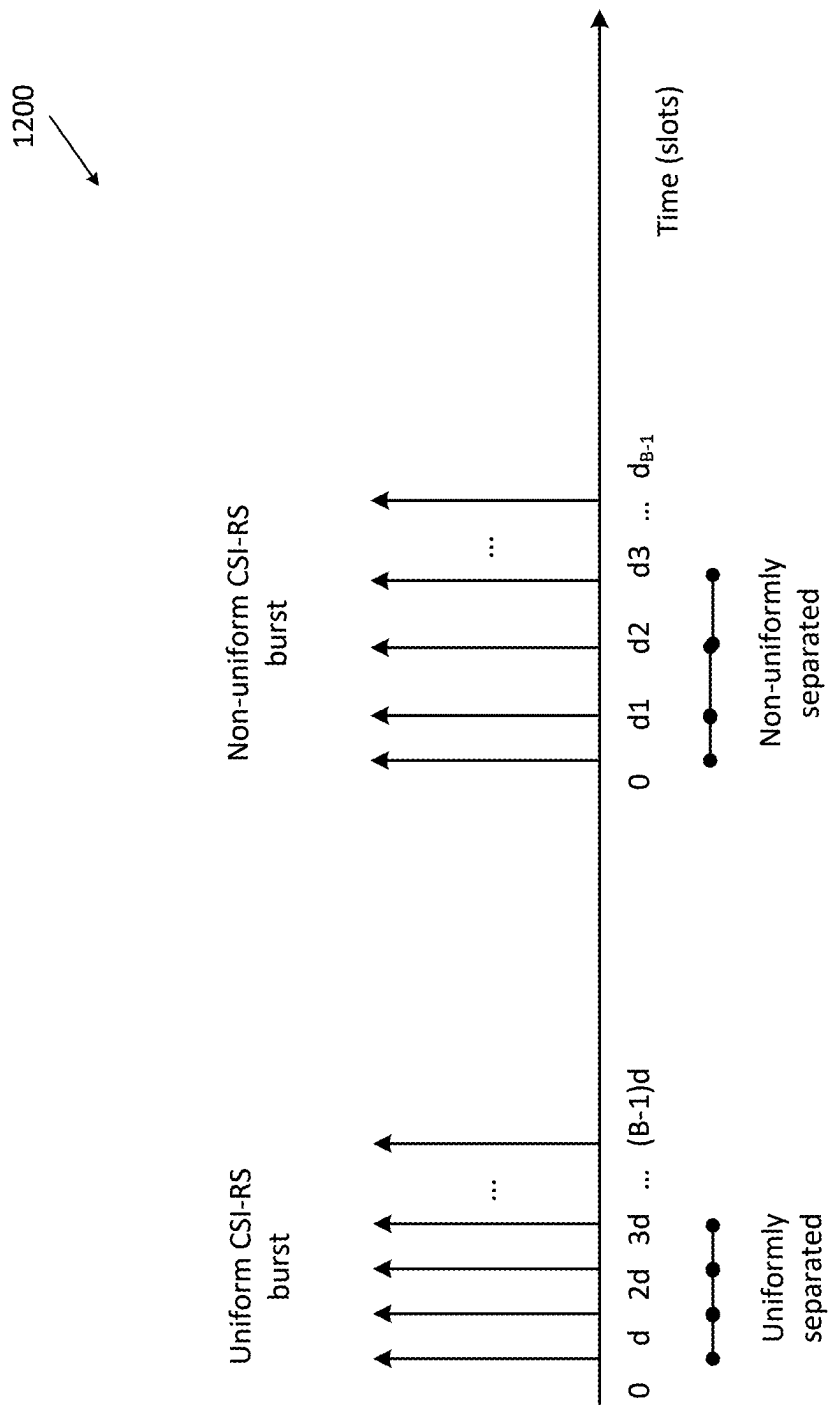


FIG. 12

-1300

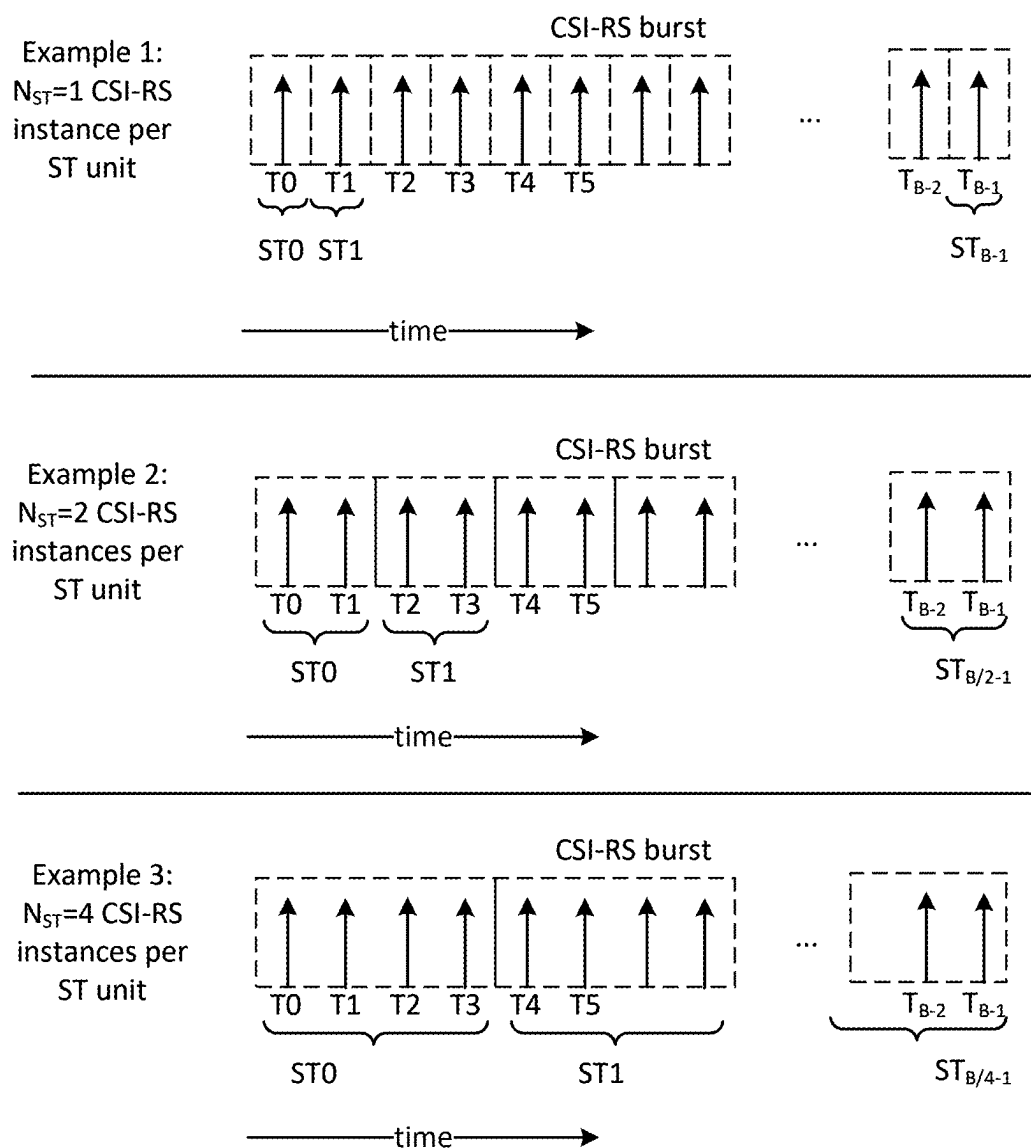


FIG. 13

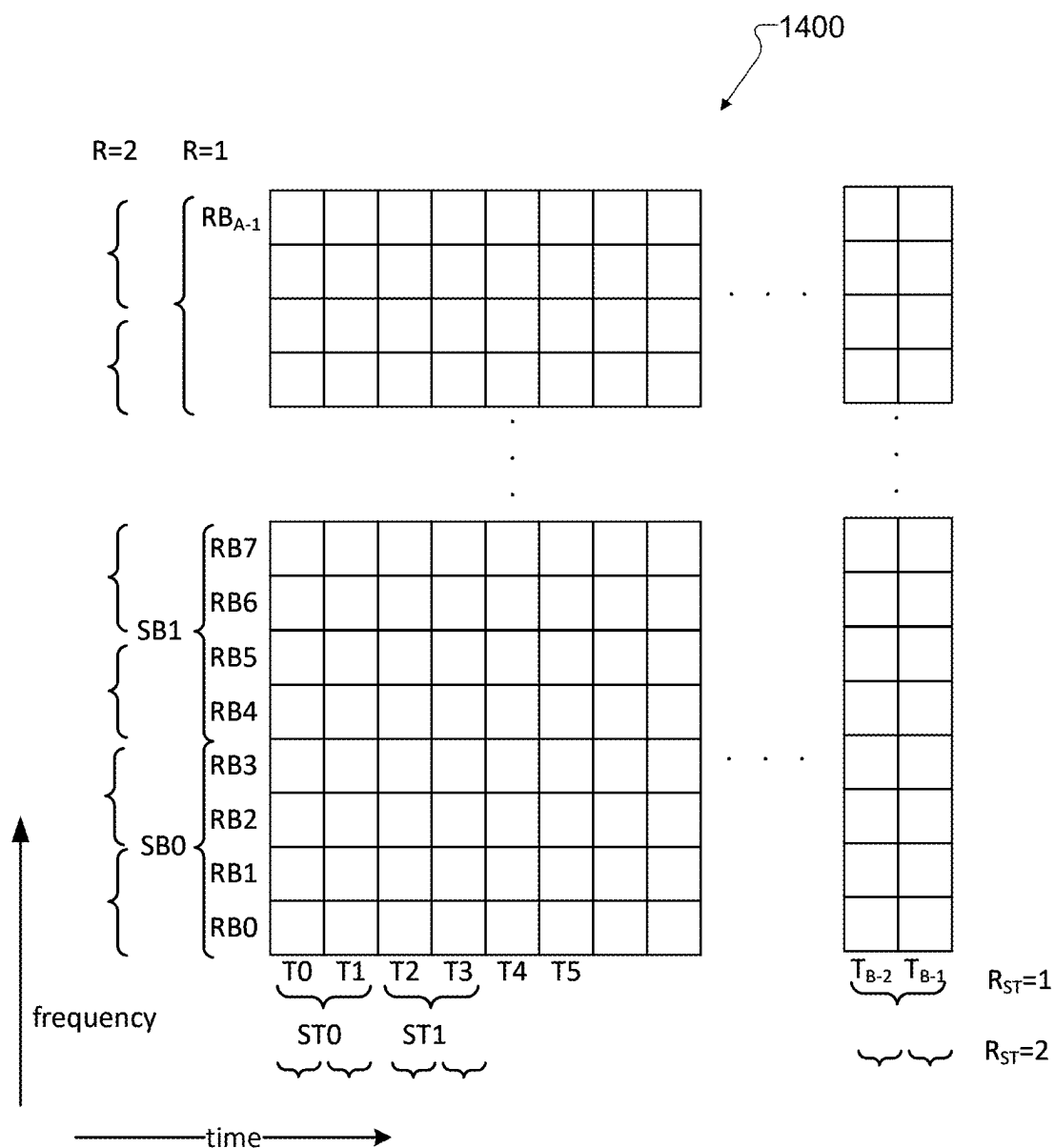


FIG. 14

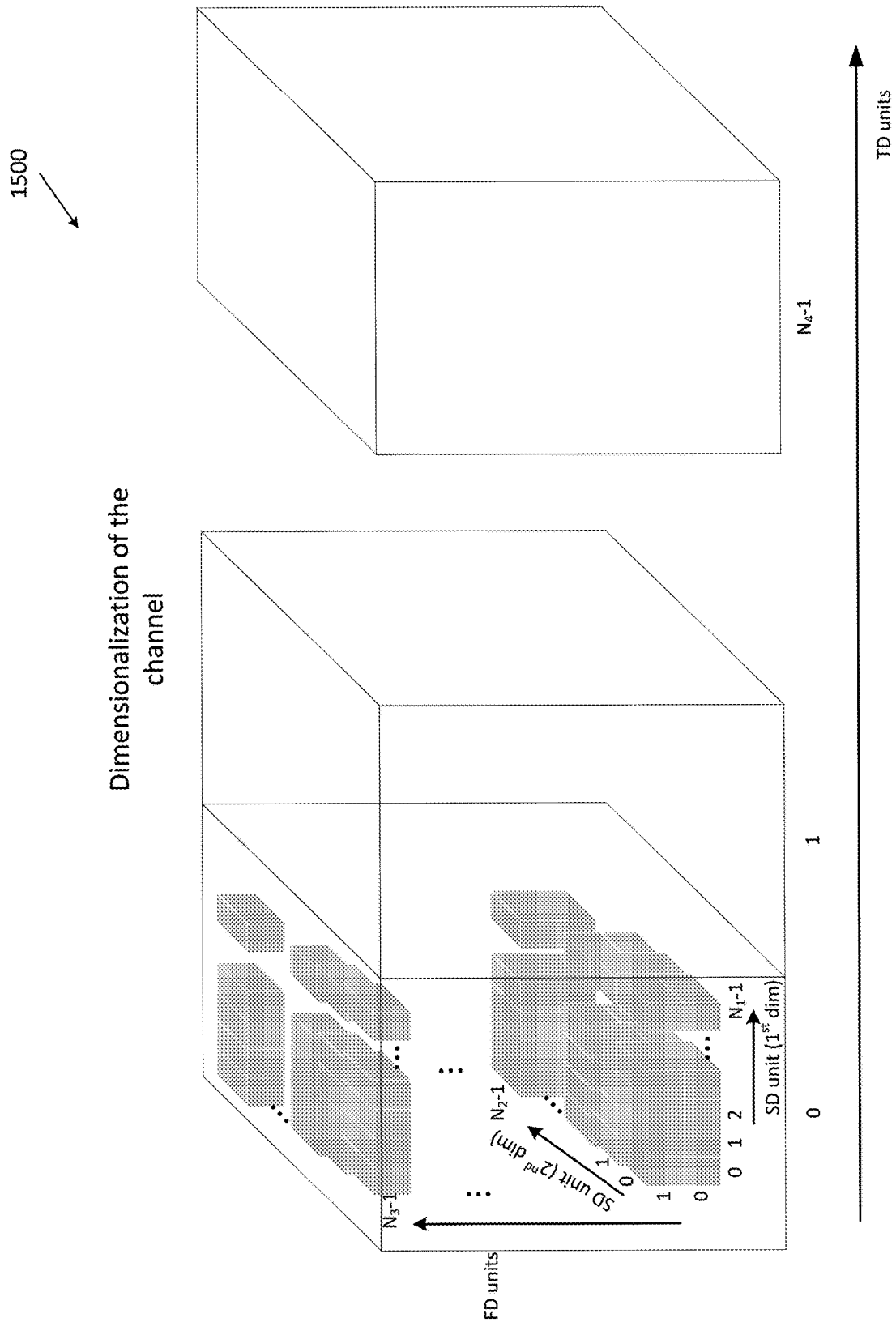


FIG. 15

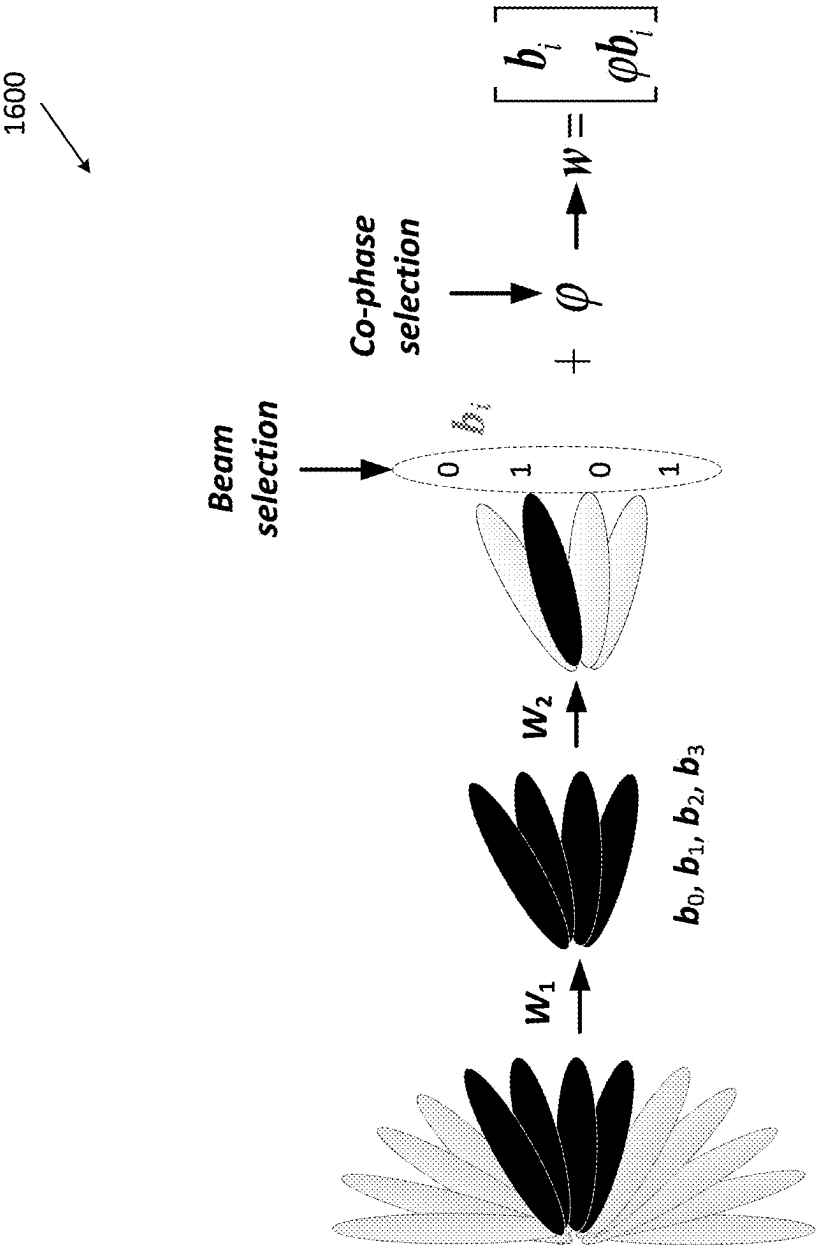


FIG. 16

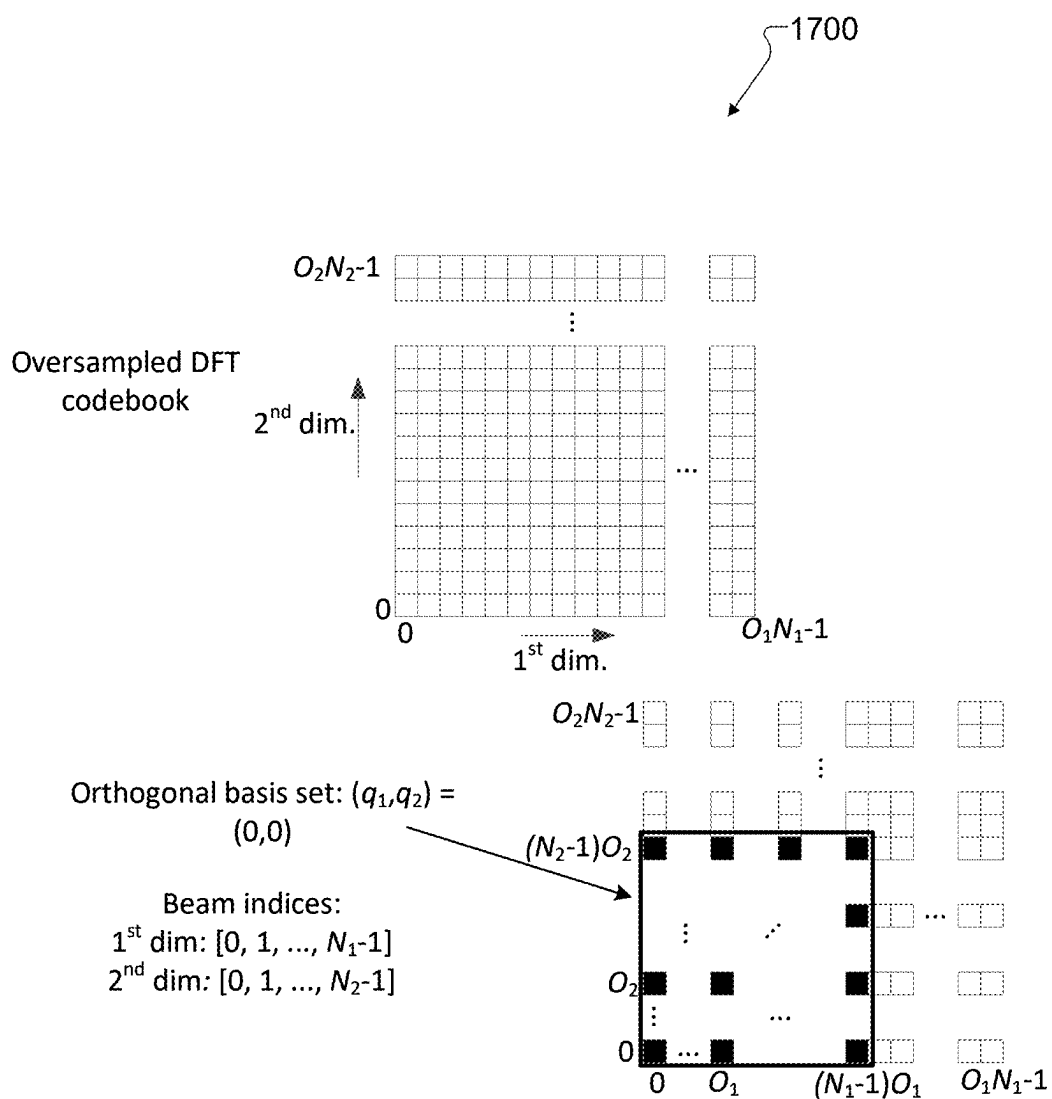


FIG. 17

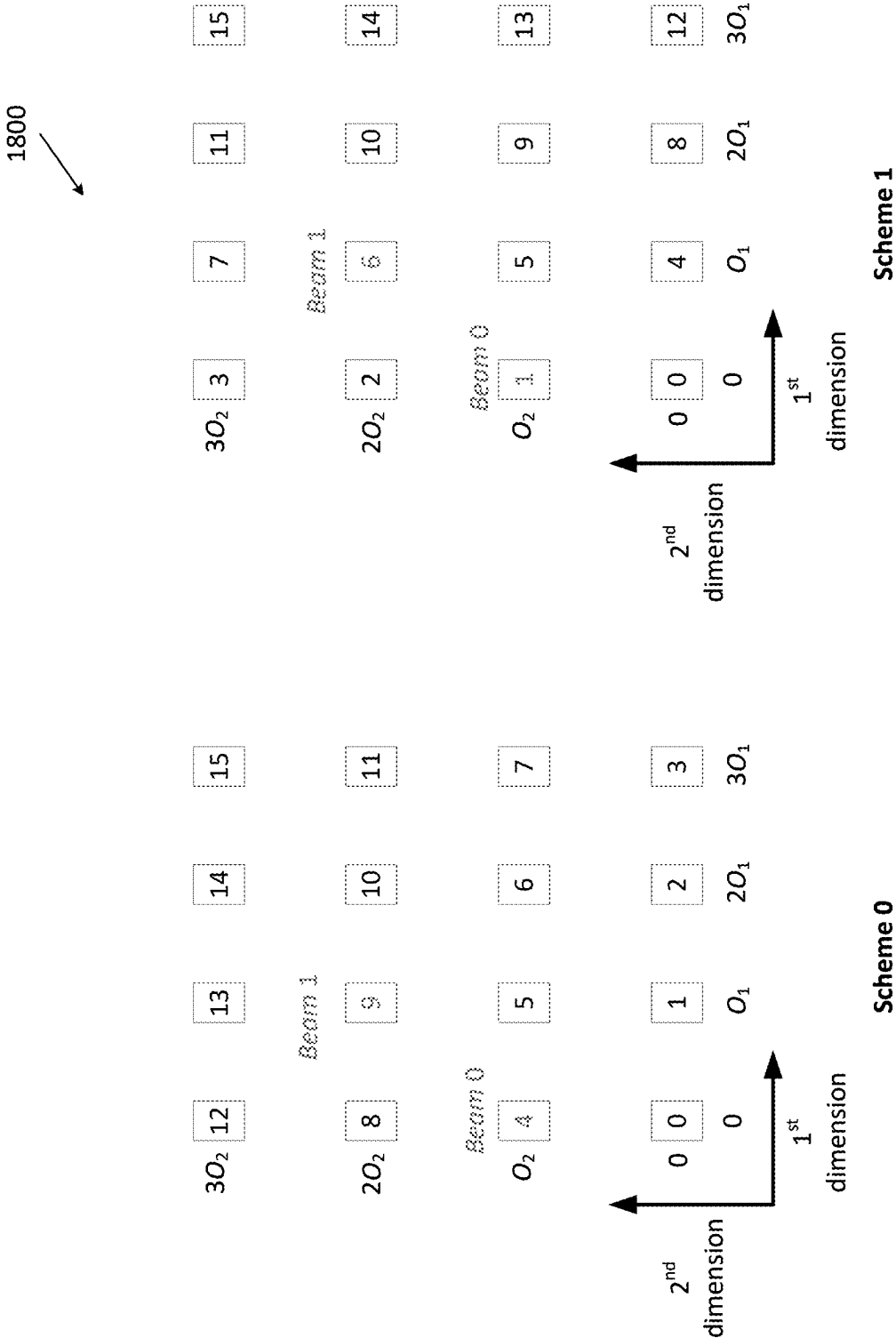


FIG. 18

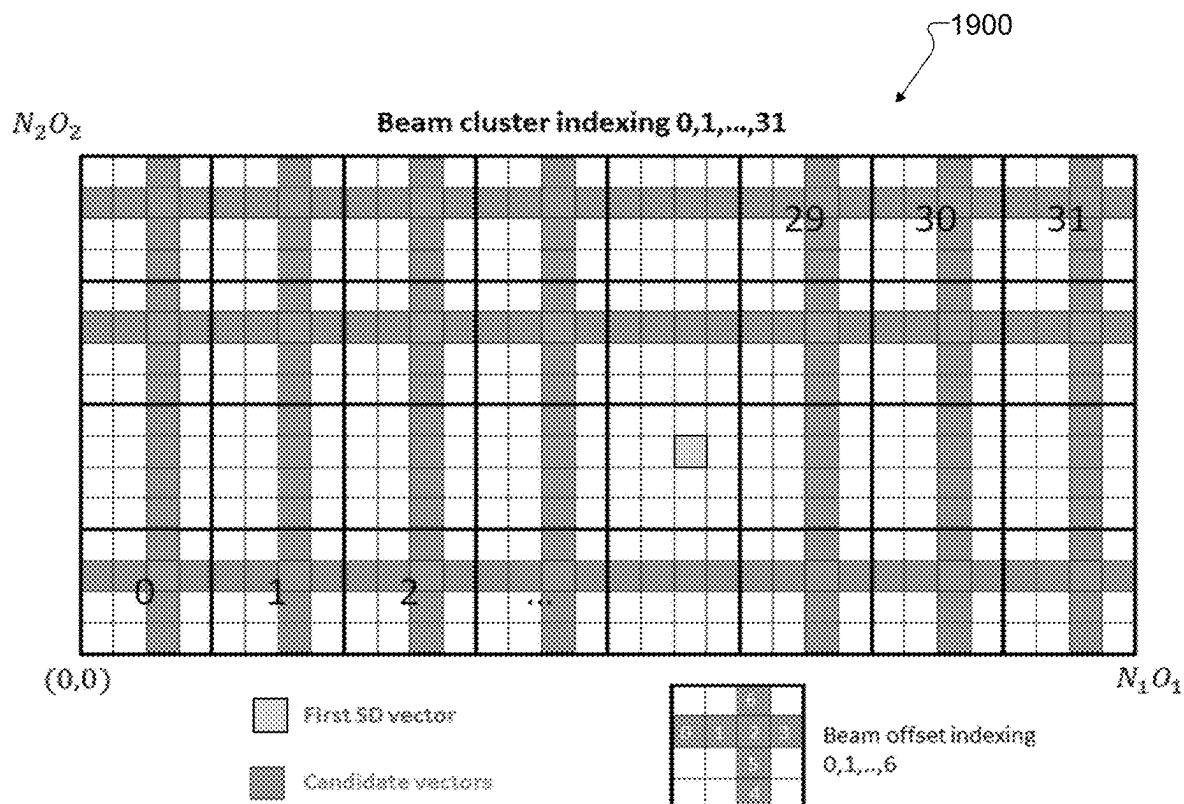


FIG. 19

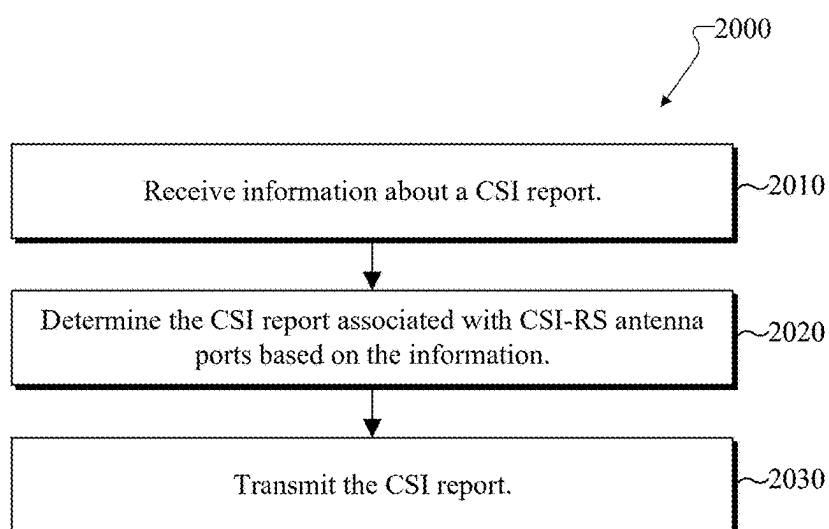


FIG. 20

CSI REPORTING FOR ENHANCED TYPE-I CODEBOOK

CROSS-REFERENCE TO RELATED AND CLAIM OF PRIORITY

[0001] The present application claims priority under 35 U.S.C. § 119(e) to:

[0002] U.S. Provisional Patent Application No. 63/557, 311 filed Feb. 23, 2024;

[0003] U.S. Provisional Patent Application No. 63/557, 424 filed Feb. 23, 2024;

[0004] U.S. Provisional Patent Application No. 63/565, 343 filed Mar. 14, 2024;

[0005] U.S. Provisional Patent Application No. 63/566, 840 filed Mar. 18, 2024;

[0006] U.S. Provisional Patent Application No. 63/645, 637 filed May 10, 2024;

[0007] U.S. Provisional Patent Application No. 63/679, 942 filed Aug. 6, 2024;

[0008] U.S. Provisional Patent Application No. 63/680, 425 filed Aug. 7, 2024;

[0009] U.S. Provisional Patent Application No. 63/681, 540 filed Aug. 9, 2024; and

[0010] U.S. Provisional Patent Application No. 63/684, 746 filed Aug. 19, 2024.

The above identified provisional patent applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0011] The present disclosure relates generally to wireless communication systems and, more specifically, the present disclosure is related to apparatuses and methods for channel state information (CSI) reporting for an enhanced Type-I codebook.

BACKGROUND

[0012] Wireless communication has been one of the most successful innovations in modern history. Recently, the number of subscribers to wireless communication services exceeded five billion and continues to grow quickly. The demand of wireless data traffic is rapidly increasing due to the growing popularity among consumers and businesses of smart phones and other mobile data devices, such as tablets, “note pad” computers, net books, eBook readers, and machine type of devices. In order to meet the high growth in mobile data traffic and support new applications and deployments, improvements in radio interface efficiency and coverage are of paramount importance. To meet the demand for wireless data traffic having increased since deployment of 4G communication systems, and to enable various vertical applications, 5G communication systems have been developed and are currently being deployed.

SUMMARY

[0013] The present disclosure relates to CSI reporting for an enhanced Type-I codebook.

[0014] In one embodiment, a user equipment (UE) is provided. The UE includes a transceiver configured to receive information about a CSI report. The information indicates codebookType set to ‘typeI-SinglePanel-r19’, codebookMode set to either ‘ModeA’ or ‘ModeB’, and P_{CSI-RS} CSI reference signal (CSI-RS) antenna ports, where $P_{CSI-RS} > 32$. The UE further includes a processor operably

coupled to the transceiver. The processor, based on the information, is configured to determine the CSI report associated with the CSI-RS antenna ports, the CSI report including a precoding matrix indicator (PMI). The transceiver is configured to transmit the CSI report.

[0015] In another embodiment, a base station (BS) is provided. The BS includes a transceiver operably coupled to the processor. The transceiver is configured to transmit information about a CSI report and receive the CSI report associated with CSI-RS antenna ports, the CSI report including a PMI. The information indicates codebookType set to ‘typeI-SinglePanel-r19’, codebookMode set to either ‘ModeA’ or ‘ModeB’, and P_{CSI-RS} CSI-RS antenna ports, where $P_{CSI-RS} > 32$.

[0016] In yet another embodiment, a method performed by a UE is provided. The method includes receiving information about a CSI report. The information indicates codebookType set to ‘typeI-SinglePanel-r19’, codebookMode set to either ‘ModeA’ or ‘ModeB’, and P_{CSI-RS} CSI-RS antenna ports, where $P_{CSI-RS} > 32$. The method further includes, based on the information, determining the CSI report associated with the CSI-RS antenna ports, the CSI report including a PMI, and transmitting the CSI report.

[0017] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

[0018] Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect communication between two or more elements, whether or not those elements are in physical contact with one another. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The term “controller” means any device, system, or part thereof that controls at least one operation. Such a controller may be implemented in hardware or a combination of hardware and software and/or firmware. The functionality associated with any particular controller may be centralized or distributed, whether locally or remotely. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

[0019] Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program

code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

[0020] Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

[0022] FIG. 1 illustrates an example wireless network according to embodiments of the present disclosure;

[0023] FIG. 2 illustrates an example gNodeB (gNB) according to embodiments of the present disclosure;

[0024] FIG. 3 illustrates an example UE according to embodiments of the present disclosure;

[0025] FIGS. 4A and 4B illustrate an example of a wireless transmit and receive paths according to embodiments of the present disclosure;

[0026] FIG. 5 illustrates an example of a transmitter structure for beamforming according to embodiments of the present disclosure;

[0027] FIG. 6 illustrates an example of a transmitter structure for physical downlink shared channel (PDSCH) in a subframe according to embodiments of the present disclosure;

[0028] FIG. 7 illustrates an example of a receiver structure for PDSCH in a subframe according to embodiments of the present disclosure;

[0029] FIG. 8 illustrates an example of a transmitter structure for physical uplink shared channel (PUSCH) in a subframe according to embodiments of the present disclosure;

[0030] FIG. 9 illustrates an example of a receiver structure for a PUSCH in a subframe according to embodiments of the present disclosure;

[0031] FIG. 10 illustrates a diagram of an antenna port layout according to embodiments of the present disclosure;

[0032] FIG. 11 illustrates examples co-located and distributed antenna groups (AGs)/port groups (PGs) serving a moving UE according to embodiments of the present disclosure;

[0033] FIG. 12 illustrates an example of a timeline for a UE to receive nonzero power (NZP) CSI reference signal (CSI-RS) resource(s) bursts according to embodiments of the present disclosure;

[0034] FIG. 13 illustrates examples of timelines for partitioned CSI-RS burst instances according to embodiments of the present disclosure;

[0035] FIG. 14 illustrates example of resource block (RB) and subband (SB) partitions according to embodiments of the present disclosure;

[0036] FIG. 15 illustrates an example of SD units, frequency-domain (FD) units, and time domain (TD) units according to embodiments of the present disclosure;

[0037] FIG. 16 illustrates a diagram of an example codebook for CSI reporting according to embodiments of the present disclosure;

[0038] FIG. 17 illustrates a diagram of an example orthogonal basis set for discrete Fourier transform (DFT) beams according to embodiments of the present disclosure;

[0039] FIG. 18 illustrates diagrams of example beam sorting schemes for beam selection according to embodiments of the present disclosure;

[0040] FIG. 19 illustrates an example of beam cluster indexing and beam offset indexing according to embodiments of the present disclosure; and

[0041] FIG. 20 illustrates an example method performed by a UE in a wireless communication system according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0042] FIGS. 1-20 discussed below, and the various, non-limiting embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged system or device.

[0043] To meet the demand for wireless data traffic having increased since deployment of 4G communication systems, and to enable various vertical applications, 5G/NR communication systems have been developed and are currently being deployed. The 5G/NR communication system is implemented in higher frequency (mmWave) bands, e.g., 28 GHz or 60 GHz bands, so as to accomplish higher data rates or in lower frequency bands, such as 6 GHz, to enable robust coverage and mobility support. To decrease propagation loss of the radio waves and increase the transmission distance, the beamforming, massive multiple-input multiple-output (MIMO), full dimensional MIMO (FD-MIMO), array antenna, an analog beam forming, large scale antenna techniques are discussed in 5G/NR communication systems.

[0044] In addition, in 5G/NR communication systems, development for system network improvement is under way based on advanced small cells, cloud radio access networks (RANs), ultra-dense networks, device-to-device (D2D) communication, wireless backhaul, moving network, cooperative communication, coordinated multi-points (CoMP), reception-end interference cancellation and the like.

[0045] The discussion of 5G systems and frequency bands associated therewith is for reference as certain embodiments of the present disclosure may be implemented in 5G systems. However, the present disclosure is not limited to 5G systems, or the frequency bands associated therewith, and embodiments of the present disclosure may be utilized in connection with any frequency band. For example, aspects of the present disclosure may also be applied to deployment of 5G communication systems, 6G, or even later releases which may use terahertz (THz) bands.

[0046] The following documents and standards descriptions are hereby incorporated by reference into the present disclosure as if fully set forth herein: [REF1]3GPP TS 36.211 v17.4.0, “E-UTRA, Physical channels and modulation;” [REF2]3GPP TS 36.212 v18.0.0, “E-UTRA, Multiplexing and Channel coding;” [REF3]3GPP TS 36.213 v18.1.0, “E-UTRA, Physical Layer Procedures;” [REF4] 3GPP TS 36.321 v18.0.0, “E-UTRA, Medium Access Control (MAC) protocol specification;” [REF5]3GPP TS 36.331 v18.0.0, “E-UTRA, Radio Resource Control (RRC) Protocol Specification;” [REF6]3GPP TR 22.891 v12.0; [REF7] 3GPP TS 38.212 v18.1.0, “E-UTRA, NR, Multiplexing and Channel coding;” [REF8]3GPP TS 38.214 v18.1.0, “E-UTRA, NR, Physical layer procedures for data;” [REF9] 3GPP TS 38.211 v18.1.0, “E-UTRA, NR, Physical channels and modulation;” and [REF 10]R1-2406645, “Views on Rel-19 CSI enhancements”, Samsung.”

[0047] FIGS. 1-3 below describe various embodiments implemented in wireless communications systems and with the use of orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA) communication techniques. The descriptions of FIGS. 1-3 are not meant to imply physical or architectural limitations to how different embodiments may be implemented. Different embodiments of the present disclosure may be implemented in any suitably arranged communications system.

[0048] FIG. 1 illustrates an example wireless network 100 according to embodiments of the present disclosure. The embodiment of the wireless network 100 shown in FIG. 1 is for illustration only. Other embodiments of the wireless network 100 could be used without departing from the scope of the present disclosure.

[0049] As shown in FIG. 1, the wireless network 100 includes a gNB 101 (e.g., base station, BS), a gNB 102, and a gNB 103. The gNB 101 communicates with the gNB 102 and the gNB 103. The gNB 101 also communicates with at least one network 130, such as the Internet, a proprietary Internet Protocol (IP) network, or other data network.

[0050] The gNB 102 provides wireless broadband access to the network 130 for a first plurality of user equipments (UEs) within a coverage area 120 of the gNB 102. The first plurality of UEs includes a UE 111, which may be located in a small business; a UE 112, which may be located in an enterprise; a UE 113, which may be a WiFi hotspot; a UE 114, which may be located in a first residence; a UE 115, which may be located in a second residence; and a UE 116, which may be a mobile device, such as a cell phone, a wireless laptop, a wireless PDA, or the like. The gNB 103 provides wireless broadband access to the network 130 for a second plurality of UEs within a coverage area 125 of the gNB 103. The second plurality of UEs includes the UE 115 and the UE 116. In some embodiments, one or more of the gNBs 101-103 may communicate with each other and with the UEs 111-116 using 5G/NR, long term evolution (LTE), long term evolution-advanced (LTE-A), WiMAX, WiFi, or other wireless communication techniques.

[0051] Depending on the network type, the term “base station” or “BS” can refer to any component (or collection of components) configured to provide wireless access to a network, such as transmit point (TP), transmit-receive point (TRP), an enhanced base station (eNodeB or eNB), a 5G/NR base station (gNB), a macrocell, a femtocell, a WiFi access point (AP), or other wirelessly enabled devices. Base sta-

tions may provide wireless access in accordance with one or more wireless communication protocols, e.g., 5G/NR 3rd generation partnership project (3GPP) NR, long term evolution (LTE), LTE advanced (LTE-A), high speed packet access (HSPA), Wi-Fi 802.11a/b/g/n/ac, etc. For the sake of convenience, the terms “BS” and “TRP” are used interchangeably in this patent document to refer to network infrastructure components that provide wireless access to remote terminals. Also, depending on the network type, the term “user equipment” or “UE” can refer to any component such as “mobile station,” “subscriber station,” “remote terminal,” “wireless terminal,” “receive point,” or “user device.” For the sake of convenience, the terms “user equipment” and “UE” are used in this patent document to refer to remote wireless equipment that wirelessly accesses a BS, whether the UE is a mobile device (such as a mobile telephone or smartphone) or is normally considered a stationary device (such as a desktop computer or vending machine).

[0052] The dotted lines show the approximate extents of the coverage areas 120 and 125, which are shown as approximately circular for the purposes of illustration and explanation only. It should be clearly understood that the coverage areas associated with gNBs, such as the coverage areas 120 and 125, may have other shapes, including irregular shapes, depending upon the configuration of the gNBs and variations in the radio environment associated with natural and man-made obstructions.

[0053] As described in more detail below, one or more of the UEs 111-116 include circuitry, programing, or a combination thereof for CSI reporting using an enhanced Type-I codebook. In certain embodiments, one or more of the BSs 101-103 include circuitry, programing, or a combination thereof to support CSI reporting for an enhanced Type-I codebook.

[0054] Although FIG. 1 illustrates one example of a wireless network, various changes may be made to FIG. 1. For example, the wireless network 100 could include any number of gNBs and any number of UEs in any suitable arrangement. Also, the gNB 101 could communicate directly with any number of UEs and provide those UEs with wireless broadband access to the network 130. Similarly, each gNB 102-103 could communicate directly with the network 130 and provide UEs with direct wireless broadband access to the network 130. Further, the gNBs 101, 102, and/or 103 could provide access to other or additional external networks, such as external telephone networks or other types of data networks.

[0055] FIG. 2 illustrates an example gNB 102 according to embodiments of the present disclosure. The embodiment of the gNB 102 illustrated in FIG. 2 is for illustration only, and the gNBs 101 and 103 of FIG. 1 could have the same or similar configuration. However, gNBs come in a wide variety of configurations, and FIG. 2 does not limit the scope of the present disclosure to any particular implementation of a gNB.

[0056] As shown in FIG. 2, the gNB 102 includes multiple antennas 205a-205n, multiple transceivers 210a-210n, a controller/processor 225, a memory 230, and a backhaul or network interface 235.

[0057] The transceivers 210a-210n receive, from the antennas 205a-205n, incoming radio frequency (RF) signals, such as signals transmitted by UEs in the wireless network 100. The transceivers 210a-210n down-convert the

incoming RF signals to generate IF or baseband signals. The IF or baseband signals are processed by receive (RX) processing circuitry in the transceivers **210a-210n** and/or controller/processor **225**, which generates processed baseband signals by filtering, decoding, and/or digitizing the baseband or IF signals. The controller/processor **225** may further process the baseband signals.

[0058] Transmit (TX) processing circuitry in the transceivers **210a-210n** and/or controller/processor **225** receives analog or digital data (such as voice data, web data, e-mail, or interactive video game data) from the controller/processor **225**. The TX processing circuitry encodes, multiplexes, and/or digitizes the outgoing baseband data to generate processed baseband or IF signals. The transceivers **210a-210n** up-converts the baseband or IF signals to RF signals that are transmitted via the antennas **205a-205n**.

[0059] The controller/processor **225** can include one or more processors or other processing devices that control the overall operation of the gNB **102**. For example, the controller/processor **225** could control the reception of uplink (UL) channel signals and the transmission of downlink (DL) channel signals by the transceivers **210a-210n** in accordance with well-known principles. The controller/processor **225** could support additional functions as well, such as more advanced wireless communication functions. For instance, the controller/processor **225** could support beam forming or directional routing operations in which outgoing/incoming signals from/to multiple antennas **205a-205n** are weighted differently to effectively steer the outgoing signals in a desired direction. As another example, the controller/processor **225** could support methods for CSI configurations in TDD scenarios. Any of a wide variety of other functions could be supported in the gNB **102** by the controller/processor **225**.

[0060] The controller/processor **225** is also capable of executing programs and other processes resident in the memory **230**, such as processes to support CSI reporting for an enhanced Type-I codebook. The controller/processor **225** can move data into or out of the memory **230** as required by an executing process.

[0061] The controller/processor **225** is also coupled to the backhaul or network interface **235**. The backhaul or network interface **235** allows the gNB **102** to communicate with other devices or systems over a backhaul connection or over a network. The interface **235** could support communications over any suitable wired or wireless connection(s). For example, when the gNB **102** is implemented as part of a cellular communication system (such as one supporting 5G/NR, LTE, or LTE-A), the interface **235** could allow the gNB **102** to communicate with other gNBs over a wired or wireless backhaul connection. When the gNB **102** is implemented as an access point, the interface **235** could allow the gNB **102** to communicate over a wired or wireless local area network or over a wired or wireless connection to a larger network (such as the Internet). The interface **235** includes any suitable structure supporting communications over a wired or wireless connection, such as an Ethernet or transceiver.

[0062] The memory **230** is coupled to the controller/processor **225**. Part of the memory **230** could include a RAM, and another part of the memory **230** could include a Flash memory or other ROM.

[0063] Although FIG. 2 illustrates one example of gNB **102**, various changes may be made to FIG. 2. For example,

the gNB **102** could include any number of each component shown in FIG. 2. Also, various components in FIG. 2 could be combined, further subdivided, or omitted and additional components could be added according to particular needs.

[0064] FIG. 3 illustrates an example UE **116** according to embodiments of the present disclosure. The embodiment of the UE **116** illustrated in FIG. 3 is for illustration only, and the UEs **111-115** of FIG. 1 could have the same or similar configuration. However, UEs come in a wide variety of configurations, and FIG. 3 does not limit the scope of the present disclosure to any particular implementation of a UE.

[0065] As shown in FIG. 3, the UE **116** includes antenna(s) **305**, a transceiver(s) **310**, and a microphone **320**. The UE **116** also includes a speaker **330**, a processor **340**, an input/output (I/O) interface (IF) **345**, an input **350**, a display **355**, and a memory **360**. The memory **360** includes an operating system (OS) **361** and one or more applications **362**.

[0066] The transceiver(s) **310** receives from the antenna(s) **305**, an incoming RF signal transmitted by a gNB of the wireless network **100**. The transceiver(s) **310** down-converts the incoming RF signal to generate an intermediate frequency (IF) or baseband signal. The IF or baseband signal is processed by RX processing circuitry in the transceiver(s) **310** and/or processor **340**, which generates a processed baseband signal by filtering, decoding, and/or digitizing the baseband or IF signal. The RX processing circuitry sends the processed baseband signal to the speaker **330** (such as for voice data) or is processed by the processor **340** (such as for web browsing data).

[0067] TX processing circuitry in the transceiver(s) **310** and/or processor **340** receives analog or digital voice data from the microphone **320** or other outgoing baseband data (such as web data, e-mail, or interactive video game data) from the processor **340**. The TX processing circuitry encodes, multiplexes, and/or digitizes the outgoing baseband data to generate a processed baseband or IF signal. The transceiver(s) **310** up-converts the baseband or IF signal to an RF signal that is transmitted via the antenna(s) **305**.

[0068] The processor **340** can include one or more processors or other processing devices and execute the OS **361** stored in the memory **360** in order to control the overall operation of the UE **116**. For example, the processor **340** could control the reception of DL channel signals and the transmission of UL channel signals by the transceiver(s) **310** in accordance with well-known principles. In some embodiments, the processor **340** includes at least one microprocessor or microcontroller.

[0069] The processor **340** is also capable of executing other processes and programs resident in the memory **360**. For example, the processor **340** may execute processes for CSI reporting using an enhanced Type-I codebook as described in embodiments of the present disclosure. The processor **340** can move data into or out of the memory **360** as required by an executing process. In some embodiments, the processor **340** is configured to execute the applications **362** based on the OS **361** or in response to signals received from gNBs or an operator. The processor **340** is also coupled to the I/O interface **345**, which provides the UE **116** with the ability to connect to other devices, such as laptop computers and handheld computers. The I/O interface **345** is the communication path between these accessories and the processor **340**.

[0070] The processor **340** is also coupled to the input **350**, which includes, for example, a touchscreen, keypad, etc.,

and the display 355. The operator of the UE 116 can use the input 350 to enter data into the UE 116. The display 355 may be a liquid crystal display, light emitting diode display, or other display capable of rendering text and/or at least limited graphics, such as from web sites.

[0071] The memory 360 is coupled to the processor 340. Part of the memory 360 could include a random-access memory (RAM), and another part of the memory 360 could include a Flash memory or other read-only memory (ROM).

[0072] Although FIG. 3 illustrates one example of UE 116, various changes may be made to FIG. 3. For example, various components in FIG. 3 could be combined, further subdivided, or omitted and additional components could be added according to particular needs. As a particular example, the processor 340 could be divided into multiple processors, such as one or more central processing units (CPUs) and one or more graphics processing units (GPUs). In another example, the transceiver(s) 310 may include any number of transceivers and signal processing chains and may be connected to any number of antennas. Also, while FIG. 3 illustrates the UE 116 configured as a mobile telephone or smartphone, UEs could be configured to operate as other types of mobile or stationary devices.

[0073] FIG. 4A and FIG. 4B illustrate an example of wireless transmit and receive paths 400 and 450, respectively, according to embodiments of the present disclosure. For example, a transmit path 400 may be described as being implemented in a gNB (such as gNB 102), while a receive path 450 may be described as being implemented in a UE (such as UE 116). However, it will be understood that the receive path 450 can be implemented in a gNB and that the transmit path 400 can be implemented in a UE. In some embodiments, the transmit path 400 is configured for a CSI reporting according to an enhanced Type-I codebook as described in embodiments of the present disclosure.

[0074] As illustrated in FIG. 4A, the transmit path 400 includes a channel coding and modulation block 405, a serial-to-parallel (S-to-P) block 410, a size N Inverse Fast Fourier Transform (IFFT) block 415, a parallel-to-serial (P-to-S) block 420, an add cyclic prefix block 425, and an up-converter (UC) 430. The receive path 450 includes a down-converter (DC) 455, a remove cyclic prefix block 460, a S-to-P block 465, a size N Fast Fourier Transform (FFT) block 470, a parallel-to-serial (P-to-S) block 475, and a channel decoding and demodulation block 480.

[0075] In the transmit path 400, the channel coding and modulation block 405 receives a set of information bits, applies coding (such as a low-density parity check (LDPC) coding), and modulates the input bits (such as with Quadrature Phase Shift Keying (QPSK) or Quadrature Amplitude Modulation (QAM)) to generate a sequence of frequency-domain modulation symbols. The serial-to-parallel block 410 converts (such as de-multiplexes) the serial modulated symbols to parallel data in order to generate N parallel symbol streams, where N is the IFFT/FFT size used in the gNB and the UE. The size N IFFT block 415 performs an IFFT operation on the N parallel symbol streams to generate time-domain output signals. The parallel-to-serial block 420 converts (such as multiplexes) the parallel time-domain output symbols from the size N IFFT block 415 in order to generate a serial time-domain signal. The add cyclic prefix block 425 inserts a cyclic prefix to the time-domain signal. The up-converter 430 modulates (such as up-converts) the output of the add cyclic prefix block 425 to a RF frequency

for transmission via a wireless channel. The signal may also be filtered at a baseband before conversion to the RF frequency.

[0076] As illustrated in FIG. 4B, the down-converter 455 down-converts the received signal to a baseband frequency, and the remove cyclic prefix block 460 removes the cyclic prefix to generate a serial time-domain baseband signal. The serial-to-parallel block 465 converts the time-domain baseband signal to parallel time-domain signals. The size N FFT block 470 performs an FFT algorithm to generate N parallel frequency-domain signals. The (P-to-S) block 475 converts the parallel frequency-domain signals to a sequence of modulated data symbols. The channel decoding and demodulation block 480 demodulates and decodes the modulated symbols to recover the original input data stream.

[0077] Each of the gNBs 101-103 may implement a transmit path 400 that is analogous to transmitting in the downlink to UEs 111-116 and may implement a receive path 450 that is analogous to receiving in the uplink from UEs 111-116. Similarly, each of UEs 111-116 may implement a transmit path 400 for transmitting in the uplink to gNBs 101-103 and may implement a receive path 450 for receiving in the downlink from gNBs 101-103.

[0078] Each of the components in FIGS. 4A and 4B can be implemented using only hardware or using a combination of hardware and software/firmware. As a particular example, at least some of the components in FIGS. 4A and 4B may be implemented in software, while other components may be implemented by configurable hardware or a mixture of software and configurable hardware. For instance, the FFT block 470 and the IFFT block 415 may be implemented as configurable software algorithms, where the value of size N may be modified according to the implementation.

[0079] Furthermore, although described as using FFT and IFFT, this is by way of illustration only and should not be construed to limit the scope of the present disclosure. Other types of transforms, such as Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT) functions, can be used. It will be appreciated that the value of the variable N may be any integer number (such as 1, 2, 3, 4, or the like) for DFT and IDFT functions, while the value of the variable N may be any integer number that is a power of two (such as 1, 2, 4, 8, 16, or the like) for FFT and IFFT functions.

[0080] Although FIGS. 4A and 4B illustrate examples of wireless transmit and receive paths 400 and 450, respectively, various changes may be made to FIGS. 4A and 4B. For example, various components in FIGS. 4A and 4B can be combined, further subdivided, or omitted and additional components can be added according to particular needs. Also, FIGS. 4A and 4B are meant to illustrate examples of the types of transmit and receive paths that can be used in a wireless network. Any other suitable architectures can be used to support wireless communications in a wireless network.

[0081] FIG. 5 illustrates an example of a transmitter structure 500 for beamforming according to embodiments of the present disclosure. In certain embodiments, one or more of gNB 102 or UE 116 includes the transmitter structure 500. For example, one or more of antenna 205 and its associated systems or antenna 305 and its associated systems can be included in transmitter structure 500. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0082] In a hybrid analog-digital beamforming, analog beamforming corresponds to a ‘dynamic/varying’ virtualization of multiple antenna elements to obtain one antenna port (or antenna panel). For mmWave bands, although a number of antenna elements can be larger for a given form factor, a number of CSI-RS ports, that can correspond to the number of digitally precoded ports, can be limited due to hardware constraints (such as the feasibility to install a large number of analog-to-digital converters (ADCs)/digital-to-analog converters (DACs) at mmWave frequencies) as illustrated in FIG. 5. Then, one CSI-RS port can be mapped onto a large number of antenna elements that can be controlled by a bank of analog phase shifters **501**. One CSI-RS port can then correspond to one sub-array which produces a narrow analog beam through analog beamforming **505**. This analog beam can be configured to sweep across a wider range of angles **520** by varying the phase shifter bank across symbols or slots/subframes. The number of sub-arrays (equal to the number of RF chains) is the same as the number of CSI-RS ports $N_{CSI-PORT}$. A digital beamforming unit **510** performs a linear combination across $N_{CSI-PORT}$ analog beams to further increase a precoding gain. While analog beams are wide-band (hence not frequency-selective), digital precoding can be varied across frequency sub-bands or resource blocks. Receiver operation can be conceived analogously.

[0083] Since the transmitter structure **500** of FIG. 5 utilizes multiple analog beams for transmission and reception (wherein one or a small number of analog beams are selected out of a large number, for instance, after a training duration that is occasionally or periodically performed), the term “multi-beam operation” is used to refer to the overall system aspect. This includes, for the purpose of illustration, indicating the assigned DL or UL TX beam (also termed “beam indication”), measuring at least one reference signal for calculating and performing beam reporting (also termed “beam measurement” and “beam reporting”, respectively), and receiving a DL or UL transmission via a selection of a corresponding RX beam. The system of FIG. 5 is also applicable to higher frequency bands such as >52.6 GHz (also termed frequency range 4 or FR4). In this case, the system can employ only analog beams. Due to the O_2 absorption loss around 60 GHz frequency (~10 dB additional loss per 100 m distance), a larger number and narrower analog beams (hence a larger number of radiators in the array) are essential to compensate for the additional path loss.

[0084] The present disclosure relates generally to wireless communication systems and, more specifically, to CSI reporting.

[0085] A communication system includes a downlink (DL) that conveys signals from transmission points such as Base Stations (BSs) or NodeBs to User Equipments (UEs) and an Uplink (UL) that conveys signals from UEs to reception points such as NodeBs. A UE, also commonly referred to as a terminal or a mobile station, may be fixed or mobile and may be a cellular phone, a personal computer device, or an automated device. An eNodeB, which is generally a fixed station, may also be referred to as an access point or other equivalent terminology. For LTE systems, a NodeB is often referred to as an eNodeB.

[0086] In a communication system, such as LTE, DL signals can include data signals conveying information content, control signals conveying DL Control Information (DCI), and Reference Signals (RS) that are also known as

pilot signals. An eNodeB transmits data information through a Physical DL Shared Channel (PDSCH). An eNodeB transmits DCI through a Physical DL Control Channel (PDCCH) or an Enhanced PDCCH (EPDCCH)—see also REF 3. An eNodeB transmits acknowledgement information in response to data Transport Block (TB) transmission from a UE in a Physical Hybrid ARQ Indicator Channel (PHICH). An eNodeB transmits one or more of multiple types of RS including a UE-Common RS (CRS), a Channel State Information RS (CSI-RS), or a DeModulation RS (DMRS). A CRS is transmitted over a DL system BandWidth (BW) and can be used by UEs to obtain a channel estimate to demodulate data or control information or to perform measurements. To reduce CRS overhead, an eNodeB may transmit a CSI-RS with a smaller density in the time and/or frequency domain than a CRS. DMRS can be transmitted only in the BW of a respective PDSCH or EPDCCH and a UE can use the DMRS to demodulate data or control information in a PDSCH or an EPDCCH, respectively. A transmission time interval for DL channels is referred to as a subframe (or slot) and can have, for example, duration of 1 millisecond.

[0087] DL signals also include transmission of a logical channel that carries system control information. A BCCH is mapped to either a transport channel referred to as a Broadcast Channel (BCH) when it conveys a Master Information Block (MIB) or to a DL Shared Channel (DL-SCH) when it conveys a System Information Block (SIB)—see also REF3 and REF 5. Most system information is included in different SIBs that are transmitted using DL-SCH. A presence of system information on a DL-SCH in a subframe (or slot) can be indicated by a transmission of a corresponding PDCCH conveying a codeword with a CRC scrambled with a special System Information RNTI (SI-RNTI). Alternatively, scheduling information for a SIB transmission can be provided in an earlier SIB and scheduling information for the first SIB (SIB-1) can be provided by the MIB.

[0088] DL resource allocation is performed in a unit of subframe (or slot) and a group of Physical resource blocks (PRBs). A transmission BW includes frequency resource units referred to as Resource Blocks (RBs). Each RB includes N_{sc}^{RB} sub-carriers, or Resource Elements (REs), such as 12 REs. A unit of one RB over one subframe (or slot) is referred to as a PRB. A UE can be allocated M_{PDSCH} RBs for a total of $M_{sc}^{PDSCH} = M_{PDSCH} \cdot N_{sc}^{RB}$ REs for the PDSCH transmission BW.

[0089] UL signals can include data signals conveying data information, control signals conveying UL Control Information (UCI), and UL RS. UL RS includes DMRS and Sounding RS (SRS). A UE transmits DMRS only in a BW of a respective PUSCH or PUCCH. An eNodeB can use a DMRS to demodulate data signals or UCI signals. A UE transmits SRS to provide an eNodeB with an UL CSI. A UE transmits data information or UCI through a respective Physical UL Shared Channel (PUSCH) or a Physical UL Control Channel (PUCCH). If a UE needs to transmit data information and UCI in a same UL subframe (or slot), it may multiplex both in a PUSCH. UCI includes Hybrid Automatic Repeat reQuest ACKnowledgement (HARQ-ACK) information, indicating correct (ACK) or incorrect (NACK) detection for a data TB in a PDSCH or absence of a PDCCH detection (DTX), Scheduling Request (SR) indicating whether a UE has data in its buffer, Rank Indicator (RI), and Channel State Information (CSI) enabling an eNodeB to perform link adaptation for PDSCH transmissions to a UE.

HARQ-ACK information is also transmitted by a UE in response to a detection of a PDCCH/EPDCCH indicating a release of semi-persistently scheduled PDSCH (see also REF 3).

[0090] An UL subframe (or slot) includes two slots. Each slot includes $N_{\text{ymb}}^{\text{UL}}$ symbols for transmitting data information, UCI, DMRS, or SRS. A frequency resource unit of an UL system BW is a RB. A UE is allocated N_{RB} RBs for a total of $N_{\text{RB}} \cdot N_{\text{sc}}^{\text{RB}}$ REs for a transmission BW. For a PUCCH, $N_{\text{RB}}=1$. A last subframe (or slot) symbol can be used to multiplex SRS transmissions from one or more UEs. A number of subframe (or slot) symbols that are available for data/UCI/DMRS transmission is $N_{\text{ymb}}=2 \cdot (N_{\text{ymb}}^{\text{UL}}-1)-N_{\text{SRS}}$, where $N_{\text{SRS}}=1$ if a last subframe (or slot) symbol is used to transmit SRS and $N_{\text{SRS}}=0$ otherwise.

[0091] FIG. 6 illustrates an example of a transmitter structure 600 for PDSCH in a subframe according to embodiments of the present disclosure. For example, transmitter structure 600 can be implemented in gNB 102 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0092] As illustrated in FIG. 6, information bits 610 are encoded by encoder 620, such as a turbo encoder, and modulated by modulator 630, for example using Quadrature Phase Shift Keying (QPSK) modulation. A Serial to Parallel (S/P) converter 640 generates M modulation symbols that are subsequently provided to a mapper 650 to be mapped to REs selected by a transmission BW selection unit 655 for an assigned PDSCH transmission BW, unit 660 applies an Inverse Fast Fourier Transform (IFFT), the output is then serialized by a Parallel to Serial (P/S) converter 670 to create a time domain signal, filtering is applied by filter 680, and a signal transmitted 690. Additional functionalities, such as data scrambling, cyclic prefix insertion, time windowing, interleaving, and others are well known in the art and are not shown for brevity.

[0093] FIG. 7 illustrates an example of a receiver structure 700 for PDSCH in a subframe according to embodiments of the present disclosure. For example, receiver structure 700 can be implemented by any of the UEs 111-116 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0094] With reference to FIG. 7, a received signal 710 is filtered by filter 720, REs 730 for an assigned reception BW are selected by BW selector 735, unit 740 applies a Fast Fourier Transform (FFT), and an output is serialized by a parallel-to-serial converter 750. Subsequently, a demodulator 760 coherently demodulates data symbols by applying a channel estimate obtained from a DMRS or a CRS (not shown), and a decoder 770, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits 780. Additional functionalities such as time-windowing, cyclic prefix removal, de-scrambling, channel estimation, and de-interleaving are not shown for brevity.

[0095] FIG. 8 illustrates an example of a transmitter structure 800 for PUSCH in a subframe according to embodiments of the present disclosure. For example, transmitter structure 800 can be implemented in gNB 103 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0096] As illustrated in FIG. 8, information data bits 810 are encoded by encoder 820, such as a turbo encoder, and modulated by modulator 830. A Discrete Fourier Transform (DFT) unit 840 applies a DFT on the modulated data bits, REs 850 corresponding to an assigned PUSCH transmission BW are selected by transmission BW selection unit 855, unit 860 applies an IFFT and, after a cyclic prefix insertion (not shown), filtering is applied by filter 870 and a signal transmitted 880.

[0097] FIG. 9 illustrates an example of a receiver structure 900 for a PUSCH in a subframe according to embodiments of the present disclosure; For example, receiver structure 900 can be implemented by the UE 116 of FIG. 3. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0098] As illustrated in FIG. 9, a received signal 910 is filtered by filter 920. Subsequently, after a cyclic prefix is removed (not shown), unit 930 applies a FFT, REs 940 corresponding to an assigned PUSCH reception BW are selected by a reception BW selector 945, unit 950 applies an Inverse DFT (IDFT), a demodulator 960 coherently demodulates data symbols by applying a channel estimate obtained from a DMRS (not shown), a decoder 970, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits 980.

[0099] There are two types of frequency range (FR) defined in 3GPP 5G NR specifications. The sub-6 GHz range is called frequency range 1 (FR1) and millimeter wave range is called frequency range 2 (FR2). An example of the frequency range for FR1 and FR2 is shown herein.

TABLE 0

Frequency range designation	Corresponding frequency range
FR1	450 MHz-600 MHz
FR2	24250 MHz-52600 MHz

[0100] For MIMO in FR1, up to 32 CSI-RS antenna ports is supported, and in FR2, up to 8 CSI-RS antenna ports is supported. In next generation cellular standards (e.g., 6G), in addition to FR1 and FR2, new carrier frequency bands can be evaluated, e.g., FR4 (>52.6 GHz), terahertz (>100 GHz) and upper mid-band (10-15 GHz). The number of CSI-RS ports that can be supported for these new bands is likely to be different from FR1 and FR2. In particular, for 10-15 GHz band, the max number of CSI-RS antenna ports is likely to be more than FR1, due to smaller antenna form factors, and feasibility of fully digital beamforming (as in FR1) at these frequencies. For instance, the number of CSI-RS antenna ports can grow up to 128. Besides, the NW (e.g., the network 130) deployment/topology at these frequencies is also expected to be denser/distributed, for example, antenna ports distributed at multiple (non-co-located, hence geographically separated) TRPs within a cellular region can be the main scenario of interest, due to which the number of CSI-RS antenna ports for MIMO can be even larger (e.g., up to 256).

[0101] A (spatial or digital) precoding/beamforming can be used across these large number of antenna ports in order to achieve MIMO gains. Depending on the carrier frequency, and the feasibility of radio RF/hardware (HW)-related components, the (spatial) precoding/beamforming can be fully digital or hybrid analog-digital. In fully digital

beamforming, there can be one-to-one mapping between an antenna port and an antenna element, or a ‘static/fixed’ virtualization of multiple antenna elements to one antenna port can be used. Each antenna port can be digitally controlled. Hence, a spatial multiplexing across antenna ports is provided.

[0102] In a wireless communication system, MIMO is often identified as key feature in order to achieve high system throughput requirements. One of the key components of a MIMO transmission scheme is the accurate CSI acquisition at the eNB (or gNB) (or TRP). For multi-user MIMO (MU-MIMO), in particular, the availability of accurate CSI is essential in order to guarantee high MU performance. For time division duplexing (TDD) systems, the CSI can be acquired using the SRS transmission relying on the channel reciprocity. For frequency division duplexing (FDD) systems, on the other hand, it can be acquired using the CSI-RS transmission from eNB (or gNB), and CSI acquisition and feedback from UE. In common FDD systems, the CSI feedback framework is ‘implicit’ in the form of channel quality indicator (CQI)/precoding matrix indicator (PMI)/rank indicator (RI) (also CSI reference signal identity (CRI) and layer identity (LI)) derived from a codebook implying SU transmission from eNB (or gNB).

[0103] In 5G or NR systems [REF7, REF8], the above-mentioned ‘implicit’ CSI reporting paradigm from LTE is also supported and referred to as Type I CSI reporting. In addition, a high-resolution CSI reporting, referred to as Type II CSI reporting, is also supported in Release 15 specification to provide more accurate CSI information to gNB for use cases such as high-order MU-MIMO. However, embodiments of the present disclosure recognize the overhead of Type II CSI reporting can be an issue in practical UE implementations. One approach to reduce Type II CSI overhead is based on frequency domain (FD) compression. In Rel. 16 NR, DFT-based FD compression of the Type II CSI has been supported (referred to as Rel. 16 enhanced Type II codebook in REF8).document and standard [8]). Some of the key components for this feature includes (a) spatial domain (SD) basis W_1 , (b) FD basis W_2 , and (c) coefficients W_N that linearly combine SD and FD basis. In a non-reciprocal FDD system, a complete CSI (comprising all components) needs to be reported by the UE. However, when reciprocity or partial reciprocity does exist between UL and DL, then some of the CSI components can be obtained based on the UL channel estimated using SRS transmission from the UE. In Rel. 16 NR, the DFT-based FD compression is extended to this partial reciprocity case (referred to as Rel. 16 enhanced Type II port selection codebook in REF8), wherein the DFT-based SD basis in W_1 is replaced with SD CSI-RS port selection, i.e., L out of $P_{\text{CSI-RS}}/2$ CSI-RS ports are selected (the selection is common for the two antenna polarizations or two halves of the CSI-RS ports). The CSI-RS ports in this case are beamformed in SD (assuming UL-DL channel reciprocity in angular domain), and the beamforming information can be obtained at the gNB based on UL channel estimated using SRS measurements.

[0104] In Rel. 17 NR, CSI reporting has been enhanced to support the following:

[0105] Further enhanced Type II port selection codebook: it has been known in the literature that UL-DL channel reciprocity can exist in both angular and delay domains if the UL-DL duplexing distance is small.

Since delay in time domain transforms (or closely related to) basis vectors in frequency domain (FD), the Rel. 16 enhanced Type II port selection can be further extended to both angular and delay domains (or SD and FD). In particular, the DFT-based SD basis in W_1 and DFT-based FD basis in W_2 can be replaced with SD and FD port selection, i.e., L CSI-RS ports are selected in SD or/and M ports are selected in FD. The CSI-RS ports in this case are beamformed in SD (assuming UL-DL channel reciprocity in angular domain) or/and FD (assuming UL-DL channel reciprocity in delay/frequency domain), and the corresponding SD or/and FD beamforming information can be obtained at the gNB based on UL channel estimated using SRS measurements. In Rel. 17, such a codebook is supported (which is referred to as Rel. 17 further enhanced Type II port selection codebook in REF8).

[0106] Non-coherent joint transmission (NCJT) CSI reporting: When the UE can communicate with multiple TRPs that are distributed at different locations in space (e.g. within a cell), the CSI reporting can correspond to a single TRP hypothesis (i.e. CSI reporting for one of the multiple TRPs), or multi-TRP hypothesis (i.e. CSI reporting for at least two of the multiple TRPs). The CSI reporting for both single TRP and multi-TRP hypotheses are supported in Rel. 17. However, the multi-TRP CSI reporting assume a non-coherent joint transmission (NCJT), i.e. a layer (and precoder) of the transmission is restricted to be transmitted from only one TRP.

[0107] In Rel. 18 NR MIMO, the following CSI enhancements are further provided targeting two use cases (coherent joint transmission from multiple TRPs, and high/medium velocity UEs):

[0108] Enhancements of CSI acquisition for Coherent-JT targeting FR1 and up to 4 TRPs, assuming ideal backhaul and synchronization as well as the same number of antenna ports across TRPs, as follows:

[0109] Rel-16/17 Type-II codebook refinement for CJT mTRP targeting FDD and its associated CSI reporting, taking into account throughput-overhead trade-off

[0110] CSI reporting enhancement for high/medium UE velocities by exploiting time-domain correlation/Doppler-domain information to assist DL precoding, targeting FR1, as follows:

[0111] Rel-16/17 Type-II codebook refinement, without modification to the spatial and frequency domain basis

[0112] UE reporting of time-domain channel properties measured via CSI-RS for tracking

[0113] Although Rel-18 CJT CSI can support up to 128 antenna ports by configuring 4 CSI-RS resources each with 32 antenna ports, there is another interest arising to support up to 128 antenna ports using Type-I CSI, which requires smaller feedback overhead than Rel-18 CJT CSI. Currently, a single CSI-RS resource can support up to 32 antenna ports for Type-I single-panel (SP) and multi-panel (MP) CSI. By using multiple CSI-RS resources for Type-I CSI, it is allowed to configure up to 64 antenna ports according to one of the following two schemes:

[0114] Two CSI-RS resources each with 32 antenna ports for Rel-17 NCJT (which is Type-I CSI-based)

[0115] 8 CSI-RS resources each with 8 antenna ports for Type-I SP CSI or Type-I MP CSI

[0116] However, both of the schemes do not offer CSI feedback associated with the entire channel of 64 antenna ports, but are designed for specific use cases, 1) NCJT from two TRP, and 2) one CSI-RS resource selection and reporting associated with the selected CSI-RS resource, respectively. Hence, embodiments of the present disclosure recognize Type-I CSI with more than 32 antenna ports is limited in terms of use cases, and needs some enhancement.

[0117] In next generation MIMO systems, the number of antenna ports is expected to increase further (e.g., up to 256), for example, for carrier frequencies in upper mid-band (10-15 GHz); the NW deployments are likely to be denser/more distributed (when compared with 5G NR); and the system is expected to work seamlessly even in challenging scenarios such as medium-high (e.g. 120 kmph) speed UEs, 'higher-order) multi-user MIMO.

[0118] Similar to common (Rel.15/18 NR) both low-resolution (aka Type I) and high-resolution (aka Type II) CSI reporting for the distributed systems mentioned herein are needed and beneficial depending on use cases and scenarios. Unlike the common, however, it is preferable to have a common framework or components between the two CSI reporting settings, in order to have a simple, future-proof, and scalable solution, thereby making it more feasible in real deployments.

[0119] Various embodiments of the present disclosure relate generally to wireless communication systems and, more specifically, to compression-based CSI reporting and/or framework supporting multiple codebooks.

[0120] Various embodiments of the present disclosure provide a framework having multiple codebooks (e.g., different codebook mode), where one codebook can be configured by NW based on its preference and/or UE capability, and the multiple codebooks include two codebooks: 1) Rel-15-based Type-I codebook with new (N_1, N_2) values for up to 128 CSI-RS ports, and 2) Rel-19 enhanced Type-I codebook designed based on a common framework with Type-I and Type-II described in this disclosure.

[0121] Various embodiments of the present disclosure relate to a CSI reporting framework in next generation MIMO systems. In particular, it relates to the CSI reporting based on a low-resolution (or Type I) codebook comprising spatial-, frequency- and time- (Doppler-) domain components for a mTRP CJT with distributed antenna structure (DMIMO). Two aspects include:

[0122] Multiple codebooks (via e.g., different codebook mode) for Type-I are supported and one of them can be configured by NW based on its preference and/or UE capability.

[0123] Multiple codebooks include 1) Rel-15-based Type-I codebook with new (N_1, N_2) values for up to 128 CSI-RS ports, and 2) Rel-19 enhanced Type-I codebook designed based on a common framework with Type-I and Type-II described herein.

[0124] Various embodiments of the present disclosure also relate generally to wireless communication systems and, more specifically, also to compression-based CSI reporting with low-resolution codebook.

[0125] Various embodiments of the present disclosure further provide a mixture codebook of two codebook design methods, method 1) an enhanced Type-I codebook extended based on Rel-15 Type-I codebook described herein, and

method 2) an enhanced Type-I codebook extended based on a common framework with Type-I and Type-II described herein.

[0126] Various embodiments of the present disclosure relate to a CSI reporting framework in next generation MIMO systems. In particular, various embodiments of the present disclosure relate to the CSI reporting based on a low-resolution (or Type I) codebook for STRP or comprising spatial-, frequency- and time- (Doppler-) domain components for a mTRP CJT with distributed antenna structure (DMIMO). Three aspects include:

[0127] With respect to rank values, a mixture codebook of method 1) an enhanced Type-I codebook extended based on Rel-15 Type-I codebook described herein, and method 2) an enhanced Type-I codebook extended based on a common framework with Type-I and Type-II as described herein.

[0128] The mixture codebook is provided with respect to (N_1, N_2) pair values.

[0129] The mixture codebook is provided with respect to (O_1, O_2) pair values.

[0130] Aspects, features, and advantages of the present disclosure are readily apparent from the following detailed description, simply by illustrating a number of particular embodiments and implementations, including the best mode contemplated for carrying out the present disclosure. Embodiments of the present disclosure also capable of other and different embodiments, and its several details can be modified in various obvious respects, all without departing from the spirit and scope of the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive. Embodiments of the present disclosure are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

[0131] In the following, for brevity, both FDD and TDD are considered as the duplex method for both DL and UL signaling.

[0132] Although exemplary descriptions and embodiments to follow assume orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA), the present disclosure can be extended to other OFDM-based transmission waveforms or multiple access schemes such as filtered OFDM (F-OFDM).

[0133] The present disclosure covers several components which can be used in conjunction or in combination with one another or can operate as standalone schemes.

[0134] Each of the following components and embodiments are applicable for UL transmission with CP-OFDM (cyclic prefix OFDM) waveform as well as DFT-SOFDM (DFT-spread OFDM) and SC-FDMA (single-carrier FDMA) waveforms. Furthermore, each of the following components and embodiments are applicable for UL transmission when the scheduling unit in time is either one subframe (which can include one or multiple slots) or one slot.

[0135] In the present disclosure, the frequency resolution (reporting granularity) and span (reporting bandwidth) of CSI reporting can be defined in terms of frequency "subbands" and "CSI reporting band" (CRB), respectively.

[0136] A subband for CSI reporting is defined as a set of contiguous PRBs which represents the smallest frequency unit for CSI reporting. The number of PRBs in a subband can be fixed for a given value of DL system bandwidth,

configured either semi-statically via higher-layer/RRC signaling, or dynamically via L1 DL control signaling or MAC control element (MAC CE). The number of PRBs in a subband can be included in CSI reporting setting.

[0137] “CSI reporting band” is defined as a set/collection of subbands, either contiguous or non-contiguous, wherein CSI reporting is performed. For example, CSI reporting band can include each of the subbands within the DL system bandwidth. This can also be termed “full-band”. Alternatively, CSI reporting band can include only a collection of subbands within the DL system bandwidth. This can also be termed “partial band”.

[0138] The term “CSI reporting band” is used only as an example for representing a function. Other terms such as “CSI reporting subband set” or “CSI reporting bandwidth” or bandwidth part (BWP) can also be used.

[0139] In terms of UE configuration, a UE can be configured with at least one CSI reporting band. This configuration can be semi-static (via higher-layer signaling or RRC) or dynamic (via MAC CE or L1 DL control signaling). When configured with multiple (N) CSI reporting bands (e.g. via RRC signaling), a UE can report CSI associated with $n \leq N$ CSI reporting bands. For instance, >6 GHz, large system bandwidth may require multiple CSI reporting bands. The value of n can either be configured semi-statically (via higher-layer signaling or RRC) or dynamically (via MAC CE or L1 DL control signaling). Alternatively, the UE can report a recommended value of n via an UL channel.

[0140] Therefore, CSI parameter frequency granularity can be defined per CSI reporting band as follows. A CSI parameter is configured with “single” reporting for the CSI reporting band with M_n subbands when one CSI parameter for each of the M_n subbands within the CSI reporting band. A CSI parameter is configured with “subband” for the CSI reporting band with M_n subbands when one CSI parameter is reported for each of the M_n subbands within the CSI reporting band.

[0141] FIG. 10 illustrates a diagram of an antenna port layout 1000 according to embodiments of the present disclosure. For example, antenna port layout 1000 of an antenna port layout can be implemented by the BS 102 of FIG. 2. This example is for illustration only and can be used without departing from the scope of the present disclosure.

[0142] With reference to FIG. 10, N_1 and N_2 are the number of antenna ports with the same polarization in the first and second dimensions, respectively. For 2D antenna port layouts, $N_1 > 1$, $N_2 > 1$, and for 1D antenna port layouts $N_1 > 1$ and $N_2 = 1$ (or $N_1 = 1$ and $N_2 > 1$). For a single-polarized (or co-polarized) antenna port layout, the total number of antenna ports is $P_{CSIRS} = N_1 N_2$. And, for a dual-polarized antenna port layout, the total number of antenna ports is $P_{CSIRS} = 2N_1 N_2$. “X” represents two antenna polarizations. In this disclosure, the term “polarization” refers to a group of antenna ports with the same polarization. For example, antenna ports

comprise a first antenna polarization, and antenna ports

$$j = X + \frac{P_{CSIRS}}{2}, X + \frac{P_{CSIRS}}{2} + 1, \dots, X + P_{CSIRS} - 1$$

comprise a second antenna polarization, where P_{CSIRS} is a number of CSI-RS antenna ports and X is a starting antenna port number (e.g. $X=3000$, then antenna ports are **3000**, **3001**, **3002**, . . .). Dual-polarized antenna payouts are implied in this disclosure. The embodiments (and examples) in this disclosure however are general and are applicable to single-polarized antenna layouts as well.

[0143] Let N_g be a number of antenna groups or port groups (AGs/PGs). With reference to FIG. 10, when there are multiple antenna groups ($N_g > 1$), each group ($g \in \{1, \dots, N_g\}$) comprises dual-polarized antenna ports with $N_{1,g}$ and $N_{2,g}$ ports in two dimensions. Note that the antenna port layouts may be the same ($N_{1,g} = N_1$ and $N_{2,g} = N_2$) in different antenna groups, or they can be different across antenna groups. For group g, the number of antenna ports is $P_{CSIRS,g} = N_{1,g} N_{2,g}$ or $2N_{1,g} N_{2,g}$ (for co-polarized or dual-polarized respectively).

[0144] In one example, an antenna group corresponds to an antenna panel. In one example, an antenna group corresponds to a TRP. In one example, an antenna group corresponds to a remote radio head (RRH). In one example, an antenna group corresponds to CSI-RS antenna ports of a NZP CSI-RS resource. In one example, an antenna group corresponds to a subset of CSI-RS antenna ports of a NZP CSI-RS resource (comprising multiple antenna groups). In one example, an antenna group corresponds to CSI-RS antenna ports of multiple NZP CSI-RS resources (e.g., comprising a CSI-RS resource set).

[0145] In one example, an antenna group corresponds to a reconfigurable intelligent surface (RIS) in which the antenna group can be (re-)configured more dynamically (e.g., via MAC CE or/and DCI). For example, the number of antenna ports associated with the antenna group can be changed dynamically.

[0146] FIG. 11 illustrates examples co-located and distributed antenna groups AGs/PGs **1100** serving a moving UE according to embodiments of the present disclosure. For example, the co-located and distributed AGs/PGs **1100** implemented by any of the BSs **101-103** of FIG. 1. This example is for illustration only and can be used without departing from the scope of the present disclosure.

[0147] In one example scenario, multiple AGs/PGs can be co-located or distributed, and can serve static (non-mobile) or moving UEs. With reference to FIG. 11, an illustration of AGs/PGs serving a moving UE is shown. While the UE moves from a location A to another location B, the UE measures the channel, e.g. via NZP CSI-RS resources, (may also measure the interference, e.g. via CSI interference measurement (CSI-IM) resources or CSI-RS resources for interference measurement), uses the measurement to determine/report CSI considering joint transmission from multiple AGs/PGs. The reported CSI can be based on a codebook. The codebook can include components considering multiple AGs/PGs, and frequency/delay-domain channel profile and time/Doppler-domain channel profile.

[0148] In one example, the antenna architecture of the MIMO system is structured. For example, the antenna structure at each AG/PG is dual-polarized (single or multi-panel as shown in FIG. 10). The antenna structure at each AG/PG can be the same. Or the antenna structure at an

$$j = X + 0, X + 1, \dots, X + \frac{P_{CSIRS}}{2} - 1$$

AG/PG can be different from another AG/PG. Likewise, the number of ports at each AG/PG can be the same. Or the number of ports at one AG/PG can be different from another AG/PG.

[0149] In another example, the antenna architecture of the MIMO system is unstructured. For example, the antenna structure at one AG/PG can be different from another AG/PG.

[0150] A structured antenna architecture in the rest of the disclosure. For simplicity, it is expected that each AG/PG is equivalent to a panel (cf. FIG. 10), although, an AG/PG can have multiple panels in practice. The disclosure however is not restrictive to a single panel expectation at each AG, and extends (covers) the case when an AG/PG has multiple antenna panels.

[0151] In various embodiments, an AG/PG constitutes (or corresponds to or is equivalent to) at least one of the following:

[0152] In one example A.1, an AG/PG corresponds to a TRP.

[0153] In one example A.2, an AG/PG corresponds to a CSI-RS resource. A UE is configured with $K=N_g>1$ non-zero-power (NZP) CSI-RS resources, and a CSI reporting is configured to be across multiple CSI-RS resources. This is similar to Class B, $K>1$ configuration in Rel. 14 LTE. The K NZP CSI-RS resources can belong to a CSI-RS resource set or multiple CSI-RS resource sets (e.g. K resource sets each comprising one CSI-RS resource). The details are as explained in this disclosure herein.

[0154] In one example, an AG/PG corresponds to a CSI-RS resource group, where a group comprises one or multiple NZP CSI-RS resources. A UE is configured with $K \geq N_g > 1$ non-zero-power (NZP) CSI-RS resources, and a CSI reporting is configured to be across multiple CSI-RS resources from resource groups. This is similar to Class B, $K>1$ configuration in Rel. 14 LTE. The K NZP CSI-RS resources can belong to a CSI-RS resource set or multiple CSI-RS resource sets (e.g., K resource sets each comprising one CSI-RS resource). The details are as explained in this disclosure herein. In particular, the K CSI-RS resources can be partitioned into N_g resource groups. The information about the resource grouping can be provided together with the CSI-RS resource setting/configuration, or with the CSI reporting setting/configuration, or with the CSI-RS resource configuration.

[0155] In one example, an AG/PG corresponds to a subset (or a group) of CSI-RS ports. A UE is configured with at least one NZP CSI-RS resource comprising (or associated with) CSI-RS ports that can be grouped (or partitioned) multiple subsets/groups/parts of antenna ports, each corresponding to (or constituting) an AG/PG. The information about the subsets of ports or grouping of ports can be provided together with the CSI-RS resource setting/configuration, or with the CSI reporting setting/configuration, or with the CSI-RS resource configuration.

[0156] In one example, an AG/PG corresponds to one or more examples described herein depending on a configuration. For example, this configuration can be explicit via a parameter (e.g. an RRC parameter). Or it can be implicit.

[0157] In one example, when implicit, it could be based on the value of K . For example, when $K>1$ CSI-RS resources, an AG/PG corresponds to one or more examples described herein. For example, when $K=1$ CSI-RS resource, an AG/PG corresponds to one or more examples described herein.

[0158] In another example, the configuration could be based on the configured codebook. For example, an AG/PG corresponds to a CSI-RS resource or resource group according to one or more examples described herein when the codebook corresponds to a decoupled codebook (modular or separate codebook for each AG/PG), and an AG/PG corresponds to a subset (or a group) of CSI-RS ports when codebook corresponds to a coupled (joint or coherent) codebook (one joint codebook across AGs/PGs).

[0159] In one example, when AG/PG maps (or corresponds to) a CSI-RS resource or resource group, and a UE can select a subset of AGs/PGs (resources or resource groups) and report the CSI for the selected AGs/PGs (resources or resource groups), the selected AGs/PGs can be reported via an indicator. For example, the indicator can be a CRI or a PMI (component) or a new indicator.

[0160] In one example, when AG/PG maps (or corresponds to) a CSI-RS port group (example A.4), and a UE can select a subset of AGs/PGs (port groups) and report the CSI for the selected AGs/PGs (port groups), the selected AGs/PGs can be reported via an indicator. For example, the indicator can be a CRI or a PMI (component) or a new indicator.

[0161] In one example, when multiple ($K>1$) CSI-RS resources are configured for N_g AGs/PGs, a decoupled (modular) codebook is used/configured, and when a single ($K=1$) CSI-RS resource for N_g AGs/PGs, a joint codebook is used/configured.

[0162] FIG. 12 illustrates an example of a timeline 1200 for a UE to receive NZP CSI-RS resource(s) bursts according to embodiments of the present disclosure. For example, timeline 1200 for a UE to receive NZP CSI-RS resource(s) bursts can be received by the UE 116 of FIG. 3. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0163] In one embodiment, with reference to FIG. 12, a UE is configured to receive a burst of non-zero power (NZP) CSI-RS resource(s), referred to as CSI-RS burst for brevity, within B time slots comprising a measurement window, where $B \geq 1$. The B time slots can be accordingly to at least one of the following examples.

[0164] In one example, the B time slots are evenly/uniformly spaced with an inter-slot spacing d .

[0165] In one example, the B time slots can be non-uniformly spaced with inter-slot spacing $e_1 d_1, e_2 d_2 - d_1, e_3 d_3 - d_2, \dots$, so on, where $e_i \neq e_j$ for at least one pair (i,j) with $i \neq j$.

[0166] The UE (e.g., the UE 116) receives the CSI-RS burst, estimates the B instances of the DL channel measurements, and uses the channel estimates to obtain the Doppler component(s) of the DL channel. The CSI-RS burst can be linked to (or associated with) a single CSI reporting setting (e.g., via higher layer parameter CSI-ReportConfig), wherein the corresponding CSI report includes an information about the Doppler component(s) of the DL channel.

[0167] Let h_t be the DL channel estimate based on the CSI-RS resource(s) received in time slot $t \in \{0, 1, \dots, B-1\}$. When the DL channel estimate in slot t is a matrix G_t of size $N_{Rx} \times N_{Tx} \times N_{Sc}$, then $h_t = \text{vec}(G_t)$, where N_{Rx} , N_{Tx} , and N_{Sc} are number of receive (Rx) antennae at the UE, number of CSI-RS ports measured by the UE, and number of subcarriers in frequency band of the CSI-RS burst, respectively. The notation $\text{vec}(X)$ is used to denote the vectorization operation wherein the matrix X is transformed into a vector by concatenating the elements of the matrix in an order, for example, $1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$ and so on, implying that the concatenation starts from the first dimension, then moves second dimension, and continues until the last dimension. Let $H_B = [h_0 \ h_1 \ \dots \ h_{B-1}]$ be a concatenated DL channel. The Doppler component(s) of the DL channel can be obtained based on H_B . For example, H_B can be represented as $C\Phi^H = \sum_{s=0}^{N-1} c_s \phi_s^H$ where $\Phi = [\phi_0 \phi_1 \ \dots \ \phi_{N-1}]$ is a Doppler domain (DD) basis matrix whose columns comprise basis vectors, $C = [c_0 \ c_1 \ \dots \ c_{N-1}]$ is a coefficient matrix whose columns comprise coefficient vectors, and $N \leq B$ is the number of DD basis vectors. Since the columns of H_B are likely to be correlated, a DD compression can be achieved when the value of N is small (compared to the value of B). In this example, the Doppler component(s) of the channel is represented by the DD basis matrix Φ and the coefficient matrix C .

[0168] When there are multiple TRPs/RRHs ($N_{TRP} > 1$), the UE can be configured to measure the CSI-RS burst(s) according to at least one of the following examples.

[0169] In one example, the UE is configured to measure N_{TRP} CSI-RS bursts, one from each TRP/RRH. The N_{TRP} CSI-RS bursts can be overlapping in time (i.e., measured in same time slots). Or they can be staggered in time (i.e., measured in different time slots). Whether the bursts are overlapping or staggered can be determined based on configuration. It can also depend on the total number of CSI-RS ports across RRHs/TRPs. When the total number of ports is small (e.g., ≤ 32), they can overlap, otherwise (> 32), they are staggered. The number of time instances B can be the same for each of the N_{TRP} bursts. Or the number B can be the same or different across bursts (or TRPs/RRHs).

[0170] In one example, each CSI-RS burst corresponds to a semi-persistent (SP) CSI-RS resource. The SP CSI-RS resource can be activated or/and deactivated based on a MAC CE or/and DCI based signaling. Additional details can be as described in the U.S. patent application Ser. No. 17/689,838 filed Mar. 8, 2022 (the '838 Application), which is incorporated by reference in its entirety.

[0171] In one example, each CSI-RS burst corresponds to a group of $B \geq 1$ aperiodic (Ap) CSI-RS resources. The Ap-CSI-RS resources can be triggered via a DCI with slot offsets such that they can be measured in B different time slots. The rest of the details can be as described in the '838 Application.

[0172] In one example, each CSI-RS burst corresponds to a periodic (P) CSI-RS resource. The P-CSI-RS resource can be configured via higher layer. The first measurement instance (time slot) and the measurement window of the CSI-RS burst (from the P-CSI-RS resource) can be fixed or configured. The rest of the details can be as described in the '838 Application.

[0173] In one example, a CSI-RS burst can either be a P-CSI-RS, or SP-CSI-RS or Ap-CSI-RS resource.

[0174] In one example, the time-domain behavior (P, SP, or Ap) of N_{TRP} CSI-RS bursts is the same.

[0175] In one example, the time-domain behavior of N_{TRP} CSI-RS bursts can be the same or different.

[0176] In one example, the UE (such as UE 116) is configured to measure $K \geq N_{TRP}$ CSI-RS bursts, where $K = \sum_{r=1}^{N_{TRP}} K_r$ and K_r is a number of CSI-RS bursts associated with RRH/TRP r , where $r \in \{1, \dots, N_{TRP}\}$. Each CSI-RS burst is according to one or more examples described herein. When $K_r > 1$, multiple CSI-RS bursts are linked to (or associated with) a CSI reporting setting, i.e., the UE receives the N_r CSI-RS bursts, estimates the DL channels, and obtains the Doppler component(s) of the channel using each of the N_r CSI-RS bursts. The rest of the details can be as described in the '838 Application.

[0177] In one example, the UE is configured to measure one CSI-RS burst across each of the N_{TRP} TRPs/RRHs. Let P be a number of CSI-RS ports associated with the NZP CSI-RS resource measured via the CSI-RS burst. The CSI-RS burst is according to one or more examples described herein. The total of P ports can be divided into N_{TRP} groups/subsets of ports and one group/subset of ports is associated with (or corresponds to) a TRP/RRH. Then, $P = \sum_{r=1}^{N_{TRP}} P_r$ and P_r is a number of CSI-RS ports in the group/subset of ports associated with RRH/TRP r .

[0178] In one example, in each of the B time instances, a UE is configured to measure each groups/subsets of ports, i.e., in each time instance within the burst, the UE measures each of P ports (or N_{TRP} groups/subsets of ports).

[0179] In one example, a UE is configured to measure subsets/groups of ports across multiple time instances, i.e., in each time instance within the burst, the UE measures a subset of P ports or a subset of groups of ports (RRHs/TRPs).

[0180] In one example, in each time instance, the UE measures only one group/subset of ports (1 TRP per time instance). In this case, $B = N_{TRP} \times C$ or $B \geq N_{TRP} \times C$, where C is a number of measurement instances for each TRP/RRH.

[0181] In one example, the UE is configured to measure one half of the port groups in a time instance, and the remaining half in another time instance.

[0182] In one example, the two time instances can be consecutive, for example, the UE measures one half of port groups in even-numbered time instances, and the remaining half in the odd-numbered time instances.

[0183] In one example, a first half of the time instances

$$\left(\text{e.g., } 0, 1, \dots, \frac{B}{2} - 1 \right)$$

is configured to measure one half of the port groups, and the second half of the time instances

$$\left(\text{e.g., } \frac{B}{2}, \dots, B-1 \right)$$

is configured to measure the remaining half of the port groups.

[0184] In one example, the UE is configured to measure multiple CSI-RS bursts, where each burst is according to one or more examples described herein. Multiple CSI-RS bursts are linked to (or associated with) a CSI reporting setting, i.e., the UE receives multiple CSI-RS bursts, estimates the DL channels, and obtains the Doppler component(s) of the channel using each of multiple CSI-RS bursts.

[0185] Let N_4 be the length of the DD basis vectors $\{\phi_s\}$, e.g., each basis vector is a length $N_4 \times 1$ column vector.

[0186] FIG. 13 illustrates examples of timelines 1300 for partitioned CSI-RS burst instances according to embodiments of the present disclosure. For example, timelines 1300 for partitioned CSI-RS burst instances can be received by the UE 113 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0187] In one embodiment, a UE is configured to determine a value of N_4 based on the value B (number of CSI-RS instances) in a CSI-RS burst and components across which the DD compression is performed, where each component corresponds to one or multiple time instances within the CSI-RS burst. In one example, N_4 is fixed (e.g., $N_4=B$) or configured (e.g., via RRC or MAC CE or DCI) or reported by the UE (e.g., the UE 116) (as part of the CSI report). In one example, the B CSI-RS instances can be partitioned into sub-time (ST) units (instances), where each ST unit is defined as (up to) N_{ST} contiguous time instances in the CSI-RS burst. In this example, a component for the DD compression corresponds to a ST unit. With reference to FIG. 13, three examples of the ST units are shown. In the first example, each ST unit comprises $N_{ST}=1$ time instance in the CSI-RS burst. In the second example, each ST unit comprises $N_{ST}=2$ contiguous time instances in the CSI-RS burst. In the third example, each ST unit comprises $N_{ST}=4$ contiguous time instances in the CSI-RS burst.

[0188] The value of N_{ST} can be fixed (e.g., $N_{ST}=1$ or 2 or 4) or indicated to the UE (e.g., via higher layer RRC or MAC CE or DCI based signaling) or reported by the UE (e.g., as part of the CSI report). The value of N_{ST} (fixed or indicated or reported) can be subject to a UE capability reporting. The value of N_{ST} can also be dependent on the value of B (e.g., one value for a range of values for B and another value for another range of values for B).

[0189] When there are multiple TRPs/RRHs ($N_{TRP}>1$), the UE can be configured to determine a value of N_4 according to at least one of the following examples.

[0190] In one example, a value of N_4 is the same for each TRPs/RRHs.

[0191] In one example, a value of N_4 can be the same or different across TRPs/RRHs.

[0192] FIG. 14 is an example of RB and SB partitions 1400 according to embodiments of the present disclosure. For example, the RB and SB partitions 1400 can be followed by the UE 116 of FIG. 3. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0193] In one embodiment, a UE is configured with $J \geq 1$ CSI-RS bursts (as illustrated herein) that occupy a frequency band and a time span (duration), wherein the frequency band comprises A RBs, and the time span comprises B time instances (of CSI-RS resource(s)). When $J \geq 1$, the A RBs or/and B time instances can be aggregated across J CSI-RS

bursts. In one example, the frequency band equals the CSI reporting band, and the time span equals the number of CSI-RS resource instances (across J CSI-RS bursts). Both can be configured to the UE (e.g., the UE 116) for a CSI reporting, which can be based on the DD compression.

[0194] The UE is further configured to partition (divide) the A RBs into subbands (SBs) or/and the B time instances into sub-times (STs). The partition of A RBs can be based on a SB size value N_{SB} , which can be configured to the UE (cf. Table 5.2.1.4-2 of REF8). The partition of B time instances can be based either a ST size value N_{ST} or an r value, as described in this disclosure. With reference to FIG. 14, RB0, RB1, ..., RB_{A-1} comprise A RBs, To, T₁, ..., T_{B-1} comprise B time instances, the SB size $N_{SB}=4$, and the ST size $N_{ST}=2$.

[0195] When there are multiple TRPs/RRHs ($N_{TRP}>1$), the UE can be configured to determine subbands (SBs) or/and sub-times (STs) according to at least one of the following examples.

[0196] In one example, both subbands (SBs) or/and sub-times (STs) are the same for each of the TRPs/RRHs.

[0197] In one example, subbands (SBs) are the same for each TRPs/RRHs, but sub-times (STs) can be the same or different across RRHs/TRPs.

[0198] In one example, sub-times (STs) are the same for each TRPs/RRHs, but subbands (SBs) can be the same or different across RRHs/TRPs.

[0199] In one example, both sub-times (STs) and subbands (SBs) can be the same or different across RRHs/TRPs.

[0200] For illustration, the example where both SBs or/and STs are the same for each of the TRPs/RRHs is used in the description below.

[0201] The CSI reporting is based on channel measurements (based on CSI-RS bursts) in three-dimensions (3D): the first dimension corresponds to SD comprising P_{CSIRS} CSI-RS antenna ports (in total across each of the N_{TRP} RRHs/TRPs), the second dimension corresponds to FD comprising N_3 FD units (e.g. SB), and the third dimension corresponds to DD comprising N_4 DD units (e.g. ST). The 3D channel measurements can be compressed using basis vectors (or matrices) similar to the Rel. 16 enhanced Type II codebook. Let W_1 , W_p and W_d respectively denote basis matrices whose columns comprise basis vectors for SD, FD, and DD.

[0202] In one embodiment, the DD compression (or DD component or W_d basis) can be turned OFF/ON from the codebook. When turned OFF, W_d can be fixed (hence not reported), e.g., $W_d=1$ (scalar 1) or $W_d=[1, \dots, 1]$ (all-one vector) or

$$W_d = \frac{1}{n} [1, \dots, 1]$$

(all-one vector) or

$$W_d = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(identity matrix), where n is a scaling factor (e.g. $n=N_4$) or $W_d=h_{d^*}=[\phi_0^{(d^*)} \phi_1^{(d^*)} \dots \phi_{N_4-1}^{(d^*)}]$, where d^* is an index of

a fixed DD basis vector h_d^* . In one example, $d^*=0$. In one example, when the DD basis vectors comprise an orthogonal DFT basis set, h_d^* is a DD basis vector which corresponds to the DC component. When turned ON, W_d (DD basis vectors) is reported.

[0203] In one example, W_d is turned OFF/ON via an explicit signaling, e.g., an explicit RRC parameter.

[0204] In one example, W_d is turned OFF/ON via a codebook parameter. For example, similar to $M=1$ in Rel.17, when $N=1$ is configured, W_d is turned OFF, and when a value $N>1$ is configured, W_d is turned ON. Here, N denotes a number of DD basis vectors comprising columns of W_d .

[0205] In one example, the UE reports whether the DD component is turned OFF (not reported) or ON (reported). This reporting can be via a dedicated parameter (e.g., new UCI/CSI parameter). Or this reporting can be via an existing parameter (e.g., PMI component). A two-part UCI (cf. Rel. 15 NR) can be reused wherein the information whether W_d is turned OFF/ON is included in UCI part 1.

[0206] In one example, W_d is turned OFF/ON depending on the codebookType. When the codebookType is regular Type II codebook (similar to Rel 16 Type II codebook), W_d is turned ON, and when the codebookType is Type II port selection codebook (similar to Rel 17 Type II codebook), W_d is turned ON/OFF.

[0207] FIG. 15 illustrates an example of SD units, FD units, and TD units 1500 according to embodiments of the present disclosure. For example, the example SD units, FD units, and TD units 1500 can be utilized by any of the UEs 111-116 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0208] In one embodiment, a UE (e.g., the UE 116) is configured (e.g., via a higher layer CSI configuration information) with a CSI report, where the CSI report is based on a channel measurement (and interference measurement) and a codebook. When the CSI report is configured to be aperiodic, it is reported when triggered via a DCI field (e.g., a CSI request field) in a DCI.

[0209] The channel measurement can be based on $K \geq 1$ channel measurement resources (CMRs) that are transmitted from a plurality of spatial-domain (SD) units (e.g., a SD unit=a CSI-RS antenna port), and are measured via a plurality of frequency-domain (FD) units (e.g., a FD unit=one or more PRBs/SBs) and via either a time-domain (TD) unit or a plurality of TD units (e.g., a TD unit=one or more time slots). In one example, a CMR can be a NZP-CSI-RS resource.

[0210] The CSI report can be associated with the plurality of FD units and the plurality of TD units associated with the channel measurement. Alternatively, the CSI report can be associated with a second set of FD units (different from the plurality of FD units associated with the channel measurement) or/and a second set of TD units (different from the plurality of TD units associated with the channel measurement). In this later case, the UE (e.g., the UE 116), based on the channel measurement, can perform prediction (interpolation or extrapolation) in the second set of FD units or/and the second set of TD units associated with the CSI report.

[0211] With reference to FIG. 15, an illustration of the SD units (in 1^{st} and 2^{nd} antenna dimensions), FD units, and TD units is shown:

[0212] The first dimension is associated with the 1st antenna port dimension and comprises N_1 units,

[0213] The second dimension is associated with the 2nd antenna port dimension and comprises N_2 units,

[0214] The third dimension is associated with the frequency dimension and comprises N_3 units, and

[0215] The fourth dimension is associated with the time/Doppler dimension and comprises N_4 units.

[0216] Regarding SD units, the plurality of SD units can be associated with antenna ports (e.g., co-located at one site or distributed across multiple sites) comprising one or multiple antenna groups (i.e., $N_g \geq 1$), and dimensionalizes the spatial-domain profile of the channel measurement.

[0217] When $K=1$, there is one CMR comprising P_{CSIRS} CSI-RS antenna ports.

[0218] When $N_g=1$, there is one AG/PG comprising P_{CSIRS} ports, and the CSI report is based on the channel measurement from the one AG/PG.

[0219] When $N_g>1$, there are multiple AGs/PGs, and the CSI report is based on the channel measurement from/across the multiple AGs/PGs.

[0220] When $K \geq 1$, there are multiple CMRs, and the CSI report is based on the channel measurement across the multiple CMRs. In one example, a CMR corresponds to an AG/PG (one-to-one mapping). In one example, multiple CMRs can correspond to an AG/PG (many-to-one mapping).

[0221] In one example, when the P_{CSIRS} antenna ports are co-located at one site, $N_g=1$. In one example, when the P_{CSIRS} antenna ports are distributed (non-co-located) across multiple sites, $N_g>1$.

[0222] In one example, when the P_{CSIRS} antenna ports are co-located at one site and within a single antenna panel, $N_g=1$. In one example, when the P_{CSIRS} antenna ports are distributed across multiple antenna panels (can be co-located or non-co-located), $N_g>1$.

[0223] The value of N_g can be configured, e.g., via higher layer RRC parameter. Or it can be indicated via a MAC CE. Or it can be provided via a DCI field.

[0224] Likewise, the value of K can be configured, e.g., via higher layer RRC parameter. Or it can be indicated via a MAC CE. Or it can be provided via a DCI field.

[0225] In one example, $K=N_g=X$. The value of X can be configured, e.g., via higher layer RRC parameter. Or it can be indicated via a MAC CE. Or it can be provided via a DCI field.

[0226] In one example, the value of K is determined based on the value of N_g . In one example, the value of N_g is determined based on the value of K .

[0227] Regarding FD units, the plurality of FD units can be associated with a frequency domain allocation of resources (e.g., one or multiple CSI reporting bands, each comprising multiple PRBs) and dimensionalizes the frequency (or delay)-domain profile of the channel measurement.

[0228] Regarding TD/DD units, the plurality of TD units can be associated with a time domain allocation of resources (e.g., one or multiple CSI reporting windows, each comprising multiple time slots) and dimensionalizes the time (or Doppler)-domain profile of the channel measurement.

[0229] The CSI report includes an information about a precoding matrix (e.g. the information is an indicator such as PMI). The information about the precoding matrix comprises/includes at least two components (W_1 and W_2). The

first component (W_1) includes a basis which corresponds to a set of basis entities. The second component (W_2) includes:

[0230] For low-resolution (Type I), selection of a basis entity from the basis entities (per layer) and co-phasing across two polarizations.

[0231] For high-resolution (Type II), combining coefficients which linearly combine the basis entities, i.e., the precoding matrix can be represented as a weighted summation over the basis entities, where the weights are the combining coefficients.

[0232] The first component W_1 is codebook-based. When the basis needs reporting (or configured to be reported), the codebook configured for the CSI report includes at least one component for reporting the basis W_1 . This component is similar to typical (e.g. Type I and II codebooks in 5G NR) codebooks. However, since W_1 is decoupled from W_2 , the framework allows more options and parameterization for the W_1 basis as future upgrades when newer antenna types become available. The basis can be dictated by (or associated with) at least one of the spatial-domain profile, frequency (or delay)-domain profile, or time (Doppler)-domain profile of the channel measurement. Even though the number of CSI-RS antenna ports can be large (e.g. 256), the antenna ports are expected to have some antenna structure (e.g. similar to 2D active antenna array), hence the SD channel profile can be represented using SD basis entities, where the SD basis entities have dimension depending on the number of SD units

$$\left(P_{CSIRS} \text{ or } \frac{P_{CSIRS}}{2} \text{ or } 2N_1N_2 \text{ or } N_1N_2 \right).$$

Likewise, the FD channel profile is likely to be correlated across FD units, and the DD/TD channel profile is also expected to have some correlation across DD/TD units (e.g. for low-medium speed UEs). Hence, FD and DD/TD channel profiles can be represented using FD and DD/TD basis entities, respectively, where their dimensions depend on the number of FD units (N_3) and the number of DD/TD units (N_4), respectively.

[0233] The second component (W_2) is also codebook-based and is derived based on the channel measurement and W_1 . For instance, the channel measurement can be projected on to the basis W_1 and projected channel can be used to derive the W_2 components (coefficients), e.g., based on Type I or Type II codebooks in 5G NR.

[0234] In one example, the number of antenna ports across K CSI-RS resources is the same. For example, each of the K CSI-RS resources can be associated with $2N_1N_2$ antenna ports. In this case, the total number of antenna ports is $P_{CSIRS, tot} = 2KN_1N_2$.

[0235] In one example, the number of antenna ports across K CSI-RS resources can be the same or different. For example, each of the K CSI-RS resources can be associated with $2N_{1,r}N_{2,r}$ antenna ports. In this case, the total number of antenna ports is $P_{CSIRS, tot} = \sum_{r=1}^K 2N_{1,r}N_{2,r}$.

[0236] In port numbering scheme 1, the CSI-RS ports are numbered according to the order of (polarization p, NZP CSI-RS resource r) as CSI-RS ports of (p=0, r=1) followed by CSI-RS ports of (p=1, r=1), followed by CSI-RS ports of (p=0, r=2), followed by CSI-RS ports of (p=1, r=2), followed by CSI-RS ports of (p=0, r=N) followed by CSI-RS ports of (p=1, r=N).

[0237] In port numbering scheme 2, the CSI-RS ports are numbered according to the order of (polarization p, NZP CSI-RS resource r) as

[0238] CSI-RS ports of (p=0, r=1) followed by CSI-RS ports of (p=0, r=1), , followed by CSI-RS ports of (p=0, r=N), and

[0239] then CSI-RS ports of (p=1, r=1) followed by CSI-RS ports of (p=1, r=1), , followed by CSI-RS ports of (p=1, r=N).

[0240] FIG. 16 illustrates a diagram of an example codebook ≤ 1600 for CSI reporting according to embodiments of the present disclosure. For example, codebook ≤ 1600 can be utilized by an of the UEs 111-116 of FIG. 1, such as the UE 116 of FIG. 3. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0241] The present disclosure provides embodiments on CB based on Type II W_1 and Type I W_2 . In another embodiment, a UE is configured with a CSI report associated with (or across) $N > 1$ NZP CSI-RS resources (or $N > 1$ subsets of CSI-RS antenna ports or antenna port groups within a NZP CSI-RS resource), the CSI report is determined based on a codebook comprising components corresponding to W_1 , and W_2 . In particular, the precoder for layer 1 is given by

$$W_l = \frac{1}{\gamma} W_1 W_2$$

[0242] Here,

[0243] W_l is a $P_{CSIRS} \times 1$ vector, where $P_{CSIRS} = \sum_{r=1}^K 2N_{1,r}N_{2,r}$ or $P_{CSIRS} = 2NN_1N_2$.

[0244] W_1 is a block diagonal matrix including 2 blocks, where two blocks are associated with two antenna polarizations (two halves or groups of CSI-RS antenna ports) of NZP CSI-RS resources and each block is a

$$\frac{P_{CSIRS}}{2} \times L$$

SD basis or port selection matrix (similar to Rel. 15/16/18 Type II or Rel-15 Type I codebook or Rel. 16/17/18 Type II port selection (PS) or CJT PS codebook)

[0245] W_2 is a $2L \times X$ coefficients matrix, where e.g., $X=1$ or $X>1$, and

[0246] γ is a normalization factor.

[0247] In one example, $N \leq K$ and K is a number of NZP CSI-RS resources (e.g. in a CSI resource set) configured for channel measurements. In one example, K is fixed (e.g. 2 or 3 or 4 or >4) or configured (e.g. via higher layer from {2, 3, 4} or {1, 2, 3, 4}), or reported by the UE (e.g. as part of UE capability). In one example, the value of N can be ≥ 1 . In one example, the value of N can be ≥ 2 . In one example, the value of N is configured (e.g. via higher layer). In one example, the value of N is reported by the UE (e.g. as part of the CSI report). In one example, the UE is configured with $N=K$ (i.e., no selection of NZP CSI-RS resources) or N; K (i.e., dynamic selection of NZP CSI-RS resources by the UE). When the UE performs dynamic selection, the selected N

NZP CSI-RS resources can be reported via part 1 of the two part CSI (or UCI). The reporting can be via a bitmap indicator of size K bits.

[0248] In one example, a codebook with W_1 of one or more embodiments described herein can be based on Rel-15 Type-I codebook (or low-resolution codebook, 5.2.2.2.1 [REF8]), where the codebook includes W_1 component in one or more embodiments described herein and W_2 component for basis vector selection and/or co-phase selection (e.g., referred to as Rel-19 Type-I CSI).

[0249] In one example, a codebook with W_1 of one or more embodiments described herein can be based on Rel-16 Type-II codebook (or high-resolution codebook, 5.2.2.2.5 [REF8]), where the codebook includes W_1 component in one or more embodiments described herein, W_f component for frequency-domain basis vector selection, and W_2 component for coefficient selection associated with (SD, FD) basis vector pairs (e.g., referred to as Rel-19 Type-II CSI).

[0250] In one example, a codebook with W_1 of one or more embodiments described herein can be based on Rel-18 Type-II codebook (or high-resolution codebook, 5.2.2.2.8 [REF8]), where the codebook includes W_1 component in one or more embodiments described herein, W_f component for frequency-domain basis vector selection, and W_2 component for coefficient selection associated with (SD, FD) basis vector pairs (e.g., referred to as Rel-19 Type-II CSI).

[0251] In one embodiment, Type-I and Type-II CSI reporting can be (implicitly) configured from a same codebook via configuring the value of L. The codebook is designed based on W_1 according to one or more embodiments described herein.

[0252] In one example, Type-I CSI reporting can be (implicitly) configured when $L=1$ is configured.

[0253] In one example, when $L=1$ is configured, FD compression component (i.e., W_f component, e.g., FD basis vector selection ($i_{1,5}$, $i_{1,6}$) and corresponding coefficient selection) is not applied in the codebook, i.e., $W=W_1W_2$.

[0254] In one example, when $L=1$ is configured, FD compression component (i.e., W_f component, e.g., FD basis vector selection ($i_{1,5}$, $i_{1,6}$) and corresponding coefficient selection) can be turned on or turned off by using a higher-layer parameter.

[0255] In one example, Type-II CSI reporting can be (implicitly) configured when $L>1$ is configured.

[0256] In one example, when $L>1$ is configured, FD compression component (i.e., W_f component, e.g., FD basis vector selection ($i_{1,5}$, $i_{1,6}$) and corresponding coefficient selection) can be turned on or turned off by using a higher-layer parameter.

[0257] In one example, when $L>1$ is configured, FD compression component (i.e., W_f component, e.g., FD basis vector selection ($i_{1,5}$, $i_{1,6}$) and corresponding coefficient selection) is turned on, i.e., $W=W_1W_2W_f$.

[0258] In one embodiment, Type-I and Type-II CSI reporting can be explicitly configured from a same codebook via a higher-layer parameter, e.g., codebookType, codebookMode, etc. The codebook is designed based on W_1 according to one or more embodiments described herein.

[0259] In one example, for Type-I CSI reporting, the candidate values of L can include 1 and other value(s) larger than 1 (e.g., 4), and one out of L basis vectors is selected.

[0260] In another example, for Type-I CSI reporting, $L=1$ is only allowed to configure.

[0261] In one example, for Type-II CSI reporting, the candidate values of L can include values larger than 1 (e.g., 2, 4, 6).

[0262] In one example, for Type-II CSI reporting, the candidate values of L can include 1 and other values larger than 1 (e.g., 2, 4, 6).

[0263] In the examples described in this disclosure, the terminology of Type-I/Type-II does not limit the scope of the present disclosure. They can be denoted by different terminologies such as low-resolution/high-resolution CSI codebook, low-resolution/high-resolution CSI reporting, etc.

[0264] With reference to FIG. 16, at least one of the following examples is used/configured regarding W_1 and W_2 .

[0265] FIG. 17 illustrates a diagram of an example orthogonal basis set 1700 for discrete Fourier transform (DFT) beams according to embodiments of the present disclosure. For example, orthogonal basis set 1700 can be utilized by any of the UEs 111-116 of FIG. 1, such as the UE 111. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0266] In one example, W_1 is a block diagonal matrix

$$\begin{bmatrix} W_{1,1} & 0 \\ 0 & W_{1,2} \end{bmatrix}$$

comprising 2 blocks, $W_{1,1}$ and $W_{1,2}$, are (spatial-domain, SD) basis matrices associated with two antenna polarizations (two halves or groups of CSI-RS antenna ports) of (all) N NZP CSI-RS resources, and W_2 can be

$$W_2 = W_{2,1}W_{2,2} = \begin{bmatrix} e_j^L & 0 \\ 0 & e_j^L \end{bmatrix} \begin{bmatrix} 1 \\ c \end{bmatrix} = \begin{bmatrix} e_j^L \\ ce_j^L \end{bmatrix},$$

e_j^L is a L-element column (selection) vector containing a value of 1 in element j or (j mod L) and zeros elsewhere, and c is a coefficient. Note that when $L=1$,

$$W_{2,1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

hence e_j^L does not need reporting when $L>1$.

[0267] In one example,

$$\begin{bmatrix} W_{1,1} & 0 \\ 0 & W_{1,2} \end{bmatrix} = \begin{bmatrix} B & 0 \\ 0 & B \end{bmatrix}$$

is a $P_{CSIRS} \times 2L$ SD basis matrix, where the L SD basis vectors comprising columns of B are determined the same way as in Rel. 15/16 Type II codebooks (cf. 5.2.2.2.3, REF 8), i.e., the SD basis vectors $v_{m_1(i), m_2(i)}$, $i=0, 1, \dots, L-1$ are identified by the indices q_1, q_2, n_1, n_2 , can be indicated by PMI components $i_{1,1}, i_{1,2}$, and are obtained as in 5.2.2.2.3 of [REF 8].

$$\begin{aligned}
i_{1,1} &= [q_1 q_2] \\
q_1 &\in \{0, 1, \dots, O_1 - 1\} \\
q_2 &\in \{0, 1, \dots, O_2 - 1\} \\
i_{1,2} &\in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L} - 1\right)\right\} \\
\text{Let} \\
n_1 &= [n_1^{(0)}, \dots, n_1^{(L-1)}] \\
n_2 &= [n_2^{(0)}, \dots, n_2^{(L-1)}] \\
n_1^{(i)} &\in \{0, 1, \dots, N_1 - 1\} \\
n_2^{(i)} &\in \{0, 1, \dots, N_2 - 1\} \\
\text{and} \\
C(x, y) &= \begin{cases} \begin{pmatrix} x \\ y \end{pmatrix} & x \geq y \\ 0 & x < y \end{cases}
\end{aligned}$$

where the values of $C(x, y)$ are given in Table 5.2.2.2.3-1 [REF8].

[0268] The quantities $m_1^{(i)}$, $m_2^{(i)}$ are given by

$$\begin{aligned}
m_1^{(i)} &= O_1 n_1^{(i)} + q_1 \\
m_2^{(i)} &= O_2 n_2^{(i)} + q_2
\end{aligned}$$

and correspond to the DFT beam (vector) indices in the oversampled DFT codebook.

[0269] The L DFT beams or DFT vectors are selected or identified by the components $i_{1,1}$ and $i_{1,2}$ of the codebook index i_1 , where (q_1, q_2) indicates the orthogonal basis set comprising of $N_1 N_2$ DFT beams. With reference to FIG. 17, an example is shown for $(q_1, q_2) = (0, 0)$ where beams are shown as black squares located in an (N_1, N_2) grid.

[0270] FIG. 18 illustrates diagrams of example beam sorting schemes 1800 for beam selection according to embodiments of the present disclosure. For example, beam sorting schemes 1800 for beam selection can be utilized by the UE 116 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0271] For the L out of $N_1 N_2$ beam selection, the $N_1 N_2$ beams in the orthogonal basis set, indicated by (q_1, q_2) , are sorted or numbered according to at least one of the following schemes:

[0272] Scheme 0: Starting from the leading beam (q_1, q_2) , $N_1 N_2$ beams in the orthogonal basis set are sorted or numbered sequentially 0 to $N_1 N_2 - 1$ first in the 1st dimension and then in the 2nd dimension. For a given beam $(n_1^{(i)}, n_2^{(i)})$ in the orthogonal basis set, the sorted beam index is then given by $n^{(i)} = N_1 n_2^{(i)} + n_1^{(i)}$ where the indices $i = 0, 1, \dots, L-1$ are assigned such that $n^{(i)}$ increases as i increases.

[0273] Scheme 1: Starting from the leading beam (q_1, q_2) , $N_1 N_2$ beams are numbered sequentially 0 to $N_1 N_2 - 1$ first in the 2nd dimension and then in the 1st dimension. For a given beam $(n_1^{(i)}, n_2^{(i)})$ in the orthogonal basis set, the sorted beam index is then given by $n^{(i)} = N_1 n_2^{(i)} + n_1^{(i)}$ where the indices $i = 0, 1, \dots, L-1$ are assigned such that $n^{(i)}$ increases as i increases.

[0274] The sorted beam indices $n^{(i)} \in \{0, 1, \dots, N_1 N_2 - 1\}$. With reference to FIG. 18, two beam sorting (numbering) schemes are shown. An example of $L=2$ out of $N_1 N_2=16$ beam selection is also shown according to the two schemes, where $(n_1^{(0)}, n_2^{(0)}) = (0, 1)$ for Beam 0 and $(n_1^{(1)}, n_2^{(1)}) = (1, 2)$ for Beam 1. According to Scheme 0, Beam 0 and Beam 1 are numbered as $n^{(0)}=4$ and $n^{(1)}=9$, respectively, and according to Scheme 1, they are numbered as $n^{(0)}=1$ and $n^{(1)}=6$, respectively.

[0275] For a given antenna port layout (N_1, N_2) and oversampling factors (O_1, O_2) for two dimensions, a DFT vector $v_{l,m}$ (the superscript N_1 and N_2 shall be used when needed in this disclosure) can be expressed as follows.

$$\begin{aligned}
v_{l,m}^{N_1, N_2} &= \left[u_m \ e^{j \frac{2\pi l}{O_1 N_1}} u_m \ \dots \ e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \right]^T \\
u_m^{N_2} &= \left[1 \ e^{j \frac{2\pi m}{O_2 N_2}} \ \dots \ e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \right]
\end{aligned}$$

where $l \in \{0, 1, \dots, O_1 N_1 - 1\}$ and $m \in \{0, 1, \dots, O_2 N_2 - 1\}$. Here, (O_1, O_2) can be fixed, e.g. (1,1), (2,2), (2,1), (2,2), (4,1), or (4,4), or configured. (O_1, O_2) can be different across resources. (O_1, O_2) can depend on (N_1, N_2) . For example, $O_i N_i = v$ or $\leq v$ where v can be fixed, e.g. 64, 128 or configured.

[0276] Let $P_{CSIRS,r} = 2N_{1,r} N_{2,r}$ be number of CSI-RS ports associated with CSI-RS resource r . Let $K = M_1 M_2$ be a total number of resources or port groups, where M_i is a number of resources in i -th dimension, and $i = 1, 2$. In one example, the UE is configured with one of the following:

[0277] In one example, the UE is configured with K or (M_1, M_2) , and $P_{CSIRS,r}$.

[0278] In one example, the UE is configured with K or (M_1, M_2) , and P_{CSIRS} .

[0279] In one example, the UE is configured with P_{CSIRS} , and $P_{CSIRS,r}$.

[0280] In one example, the UE is configured with K or (M_1, M_2) , $P_{CSIRS,r}$ and P_{CSIRS} .

[0281] In one example,

$$c = e^{j \frac{2\pi \phi}{N_{PSK}}}$$

where $\phi \in \{0, 1, \dots, N_{PSK} - 1\}$, and $N_{PSK} = 2$ (for BPSK), $N_{PSK} = 4$ (for QPSK), 8 (for 8PSK), or 16 (for 16PSK). In one example, N_{PSK} is fixed, e.g. $N_{PSK} = 4$, or $N_{PSK} = 2$. In one example, N_{PSK} is configured via higher layer, e.g. from {2,4}. In one example, UE determines/selects which N_{PSK} is used and reports it as a part of CSI. In one example, the range of ϕ is a subset of $\{0, 1, \dots, N_{PSK} - 1\}$. For example,

$$\phi \in \left\{0, 1, \dots, \frac{N_{PSK}}{2} - 1\right\}.$$

In another example,

$$\phi \in \left\{0, 1, \dots, \frac{N_{PSK}}{4} - 1\right\}.$$

[0282] In one example, N_1 and N_2 are numbers of CSI-RS antenna ports associated with N CSI-RS resources in a first dimension and a second dimension, respectively. In one example, N_1 and N_2 can be denoted by other notations, e.g., N_x , $N_{x,tot}$, $N_{x,sum}$, $N_{x,all}$, or other notations where $x \in \{1, 2\}$. In this disclosure, N_1 and N_2 are used for the entities. In one example, $P_{CSI-RS} = 2N_1N_2$.

[0283] In one example, the UE is configured with one of the above parameters from a set of supported combinations of values, which can be all of or a subset of the combinations shown in Table 1.

TABLE 1

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)
4	(2, 1)
8	(2, 2)
	(4, 1)
12	(3, 2)
	(6, 1)
16	(4, 2)
	(8, 1)
24	(4, 3)
	(6, 2)
32	(12, 1)
	(4, 4)
	(8, 2)
64	(16, 1)
	(8, 4)
	(16, 2)
96	(32, 1)
	(8, 6)
	(12, 4)
	(16, 3)
	(24, 2)
	(48, 1)
128	(8, 8)
	(16, 4)
	(32, 2)
	(64, 1)
36	(9, 2)
	(18, 1)
48	(6, 4)
	(8, 3)
	(12, 2)
	(24, 1)
72	(6, 6)
	(9, 4)
	(12, 3)
	(18, 2)
	(36, 1)

[0284] In one example, the UE is configured with one of the above parameters from a set of supported combinations of values, which can be all of or a subset of the combinations shown in the following table.

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)
4	(2, 1)
8	(2, 2)
	(4, 1)
12	(3, 2)
	(6, 1)
16	(4, 2)
	(8, 1)
24	(4, 3)
	(6, 2)
	(12, 1)
32	(4, 4)
	(8, 2)
	(16, 1)

-continued

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)
48	(6, 4)
	(8, 3)
64	(8, 4)
	(16, 2)
128	(8, 8)
	(16, 4)

[0285] In one example, the UE is configured with one of the above parameters from a set of supported combinations of values, which can be all of or a subset of the combinations shown in the following table.

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)
48	(6, 4)
	(8, 3)
64	(8, 4)
	(16, 2)
128	(8, 8)
	(16, 4)

[0286] In one example,

$$c = e^{\frac{j2\pi\phi}{N_{PSK}}}$$

where $\phi \in \{0, 1, \dots, N_{PSK}-1\}$, and $N_{PSK}=2$ (for BPSK), $N_{PSK}=4$ (for QPSK), 8 (for 8PSK), or 16 (for 16PSK). In one example, N_{PSK} is fixed, e.g. $N_{PSK}=4$, or $N_{PSK}=2$. In one example, N_{PSK} is configured via higher layer, e.g. from $\{2, 4\}$. In one example, UE determines/selects which N_{PSK} is used and reports it as a part of CSI. In one example, the range of ϕ is a subset of $\{0, 1, \dots, N_{PSK}-1\}$. For example,

$$\phi \in \left\{0, 1, \dots, \frac{N_{PSK}}{2} - 1\right\}.$$

In another example,

$$\phi \in \left\{0, 1, \dots, \frac{N_{PSK}}{4} - 1\right\}.$$

[0287] The selection vector e_j^L and coefficient c are identified by the indices $j \in \{0, 1, \dots, L-1\}$ and ϕ respectively, can be indicated by PMI components $i_{2,1}$, $i_{2,2}$, and are obtained as

$$i_2 = [i_{2,1}, i_{2,2}]$$

$$i_{2,1} = j$$

$$i_{2,2} = \phi$$

[0288] In one example,

$$c = pe^{\frac{j2\pi\phi}{N_{PSK}}}$$

where p is an amplitude or power level. The selection vector e and coefficient c for are identified by the indices $j \in \{0, 1, \dots, L-1\}$ and (θ, k) respectively, can be indicated by PMI components $i_{2,1}$, $i_{2,2}$, $i_{2,3}$, and are obtained as

$$i_2 = [i_{2,1}, i_{2,2}, i_{2,3}]$$

[0289] The selection vector indicator $i_{2,1}=j$.

[0290] The phase coefficient indicators $i_{2,2}=0$.

[0291] The amplitude coefficient indicators $i_{2,3}=k$.

[0292] In one example, the mapping from k to the amplitude coefficient p is given one of the examples in Table 2.

TABLE 2

Mapping of elements of $i_{2,3}$: k to p				
k	p (Ex1)	p (Ex2)	p (Ex3)	p (Ex4)
0	$\frac{1}{2\sqrt{2}}$	0	$\frac{1}{8\sqrt{2}}$	0
1	1/2	1/2	1/2	1/2
2	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{4\sqrt{2}}$	$\frac{1}{4\sqrt{2}}$
3	1	1	1/4	1/4
4			$\frac{1}{2\sqrt{2}}$	$\frac{1}{2\sqrt{2}}$
5			1/2	1/2
6			$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
7			1	1

[0293] In one example, the rank-1 (1-layer) precoder is then given by

$$W_{i_{1,1}, i_{1,2}, i_{2,2}}^{(1)} = \frac{1}{\sqrt{P_{CSIRS}}} \begin{bmatrix} v_{m_1^{(j)}, m_2^{(j)}} \\ c v_{m_1^{(j)}, m_2^{(j)}} \end{bmatrix}$$

[0294] In another variation (V1) of previous example,

$$W_2 = W_{2,1} W_{2,2} = \begin{bmatrix} e_j^L & 0 \\ 0 & e_j^L \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} c_0 e_j^L \\ c_1 e_j^L \end{bmatrix},$$

where c_0 and c_1 are coefficients (phase only or phase and amplitude) associated with two antenna polarizations (0 and 1).

$$W_2 = W_{2,1} W_{2,2} = \begin{bmatrix} e_j^L & 0 \\ 0 & e_j^L \end{bmatrix} \begin{bmatrix} 1 \\ c \end{bmatrix} = \begin{bmatrix} e_j^L \\ c e_j^L \end{bmatrix},$$

[0295] In another variation (V2) of previous example,

$$W_2 = W_{2,1} W_{2,2} = \begin{bmatrix} e_{j_0}^L & 0 \\ 0 & e_{j_1}^L \end{bmatrix} \begin{bmatrix} 1 \\ c \end{bmatrix} = \begin{bmatrix} e_{j_0}^L \\ c e_{j_1}^L \end{bmatrix},$$

where j_0 and j_1 are selected basis vectors associated with two antenna polarizations (0 and 1).

[0296] In another variation (V3) of previous example,

$$W_2 = W_{2,1} W_{2,2} = \begin{bmatrix} e_{j_0}^L & 0 \\ 0 & e_{j_1}^L \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} c_0 e_{j_0}^L \\ c_1 e_{j_1}^L \end{bmatrix},$$

where j_0 and j_1 are selected basis vectors associated with two antenna polarizations (0 and 1), and c_0 and c_1 are coefficients (phase only or phase and amplitude) associated with two antenna polarizations (0 and 1).

[0297] In general, as an example, the selection vector $e_{j_p}^L$ and coefficient c_p for $p=0, 1$ are identified by the indices $j_p \in \{0, 1, \dots, L-1\}$ and (ϕ_p, k_p) respectively, can be indicated by PMI components $i_{2,1}$, $i_{2,2}$, $i_{2,3}$, and are obtained as

$$i_2 = [i_{2,1}, i_{2,2}, i_{2,3}]$$

$$i_{2,1} = [i_0 i_1]$$

$$i_{2,2} = [\phi_0 \phi_1]$$

$$i_{2,3} = [k_0 k_1]$$

[0298] In general, as an example, the rank-1 (1-layer) precoder can be given by

$$W_{i_{1,1}, i_{1,2}, i_{2,2}}^{(1)} = \frac{1}{\sqrt{P_{CSIRS}}} \begin{bmatrix} c_0 v_{m_1^{(j_0)}, m_2^{(j_0)}} \\ c_1 v_{m_1^{(j_1)}, m_2^{(j_1)}} \end{bmatrix}$$

[0299] Where depending on the above-mentioned variations, either

[0300] j_0 and j_1 can be the same, i.e., $j_0=j_1$, or

[0301] j_0 and j_1 can be different, i.e., both $j_0=j_1$ and $j_0 \neq j_1$ are possible, or

[0302] k_0 and k_1 can be the same, i.e., $k_0=k_1$, or

[0303] k_0 and k_1 can be different, i.e., both $k_0=k_1$ and $k_0 \neq k_1$ are possible, or

[0304] c_0 can be fixed (e.g. 1) and $c_0=c_1$ or $c_0 \neq c_1$.

[0305] In one example, the W_1 has the following structure, i.e., SD basis vectors are the same for two polarizations (polarization common) but can be different across layers (layer-specific):

$$W_1^{(l)} = \begin{bmatrix} B_l & 0 \\ 0 & B_l \end{bmatrix},$$

for $l=1, \dots, v$, where $B_l = [b_{l,0}, b_{l,1}, \dots, b_{l,L-1}]$ including L SD basis vectors (as columns), and v is the number of layers (i.e., rank).

[0306] In one example (for layer-specific q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}}^{(i)},$$

$i=0, 1, \dots, L-1, l=1, \dots, V$ are indicated by PMI components $i_{1,1,l}, i_{1,2,l}$, where

$$\begin{aligned} i_{1,1,l} &= [q_{1,l} \ q_{2,l}], \\ q_{1,l} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,l} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2,l} &\in \left\{0, 1, \dots, \binom{N_1 N_2}{L} - 1\right\}. \end{aligned}$$

[0307] In another example (for layer-specific q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}}^{(i)},$$

$i=0, 1, \dots, L-1, l=1, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned} i_{1,1} &= [i_{1,1,1} \ \dots \ i_{1,1,v}], \\ i_{1,1,l} &= [q_{1,l} \ q_{2,l}], \\ q_{1,l} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,l} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2} &= [i_{1,2,1} \ \dots \ i_{1,2,v}], \\ i_{1,2,l} &\in \left\{0, 1, \dots, \binom{N_1 N_2}{L} - 1\right\}. \end{aligned}$$

[0308] In one example (for layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}}^{(i)},$$

$i=0, 1, \dots, L-1, l=1, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2,l}$, where

$$\begin{aligned} i_{1,1} &= [q_1 \ q_2], \\ q_1 &\in \{0, 1, \dots, O_1 - 1\}, \\ q_2 &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2,l} &\in \left\{0, 1, \dots, \binom{N_1 N_2}{L} - 1\right\}. \end{aligned}$$

[0309] In another example (for layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}}^{(i)},$$

$i=0, 1, \dots, L-1, l=1, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned} i_{1,1} &= [q_1 \ q_2], \\ q_1 &\in \{0, 1, \dots, O_1 - 1\}, \\ q_2 &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2} &= [i_{1,2,1} \ \dots \ i_{1,2,v}], \\ i_{1,2,l} &\in \left\{0, 1, \dots, \binom{N_1 N_2}{L} - 1\right\}. \end{aligned}$$

[0310] In one example, the W_1 has the following structure, i.e., SD basis vectors can be different across two polarizations (polarization-specific) but are the same for all layers (layer-common):

$$W_1 = \begin{bmatrix} B_0 & 0 \\ 0 & B_1 \end{bmatrix},$$

where $B_p = [b_{p,0}, b_{p,1}, \dots, b_{p,L-1}]$ including L SD basis vectors (as columns), and v is the number of

[0311] In one example (for layer-specific q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}}^{(i)},$$

$i=0, 1, \dots, L-1, p=0, 1$ are indicated by PMI components $i_{1,1,p}, i_{1,2,p}$, where

$$\begin{aligned} i_{1,1,p} &= [q_{1,p} \ q_{2,p}], \\ q_{1,p} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,p} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2,p} &\in \left\{0, 1, \dots, \binom{N_1 N_2}{L} - 1\right\}. \end{aligned}$$

[0312] In another example (for layer-specific q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p}^{(i)},$$

$i=0, 1, \dots, L-1, p=0, 1$, are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned} i_{1,1} &= [i_{1,1,0} i_{1,1,1}], \\ i_{1,1,p} &= [q_{1,p} \ q_{2,p}], \\ q_{1,p} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,p} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2} &= [i_{1,2,0} i_{1,2,1}], \end{aligned}$$

-continued

$$i_{1,2,p} \in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}.$$

[0313] In one example (for layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p},$$

$i=0, 1, \dots, L-1, p=0, 1$ are indicated by PMI components $i_{1,1}, i_{1,2,p}$, where

$$\begin{aligned} i_{1,1} &= [q_1 \quad q_2], \\ q_1 &\in \{0, 1, \dots, O_1 - 1\}, \\ q_2 &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2,p} &\in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}. \end{aligned}$$

[0314] In another example (for layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p},$$

$i=0, 1, \dots, L-1, p=0, 1$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned} i_{1,1} &= [q_1 \quad q_2], \\ q_1 &\in \{0, 1, \dots, O_1 - 1\}, \\ q_2 &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2} &= [i_{1,2,0} \quad i_{1,2,1}], \\ i_{1,2,p} &\in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}. \end{aligned}$$

[0315] In one example, the W_1 has the following structure, i.e., SD basis vectors can be different across two polarizations (polarization-specific) and can be different across layers (layer-specific):

$$W_1^{(i)} = \begin{bmatrix} B_{i,0} & 0 \\ 0 & B_{i,1} \end{bmatrix},$$

where $B_{i,p} = [b_{i,p,0}, b_{i,p,1}, \dots, b_{i,p,L-1}]$ including L SD basis vectors (as columns), and v is the number of layers (i.e., rank).

[0316] In one example (for layer-specific and polarization-specific q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1,l,p}, i_{1,2,l,p}$, where

$$\begin{aligned} i_{1,1,l,p} &= [q_{1,l,p} \quad q_{2,l,p}], \\ q_{1,l,p} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,l,p} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2,l,p} &\in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}. \end{aligned}$$

[0317] In another example (for layer-specific and polarization-specific q_1, q_2), the SD basis vectors

$$\begin{aligned} i_{1,1} &= [i_{1,1,1} \quad \dots \quad i_{1,1,v}], \\ i_{1,1,l} &= [i_{1,1,l,0} \quad i_{1,1,l,1}], \\ i_{1,1,l,p} &= [q_{1,l,p} \quad q_{2,l,p}], \\ q_{1,l,p} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,l,p} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2} &= [i_{1,2,1} \quad \dots \quad i_{1,2,v}], \\ i_{1,2,l} &= [i_{1,2,l,0} \quad i_{1,2,l,1}], \\ i_{1,2,l,p} &\in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}. \end{aligned}$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned} i_{1,1,l} &= [q_{1,l} \quad q_{2,l}], \\ q_{1,l} &\in \{0, 1, \dots, O_1 - 1\}, \\ q_{2,l} &\in \{0, 1, \dots, O_2 - 1\}, \\ i_{1,2,l,p} &\in \left\{0, 1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}. \end{aligned}$$

[0318] In one example (for layer-specific but polarization-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2,l,p}$, where

$$v_{m_1^{(i)}, m_2^{(i)}, p},$$

[0319] In another example (for layer-specific but polarization-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned}
i_{1,1} &= [i_{1,1,1} \dots i_{1,1,v}], \\
i_{1,l} &= [q_{1,l} \ q_{2,l}], \\
q_{1,l} &\in \{0,1, \dots, O_1 - 1\}, \\
q_{2,l} &\in \{0,1, \dots, O_2 - 1\}, \\
i_{1,2} &= [i_{1,2,1} \dots i_{1,2,v}], \\
i_{1,2,l} &= [i_{1,2,l,0} \ i_{1,2,l,1}], \\
i_{1,2,l,p} &\in \left\{0,1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}.
\end{aligned}$$

[0320] In one example (for polarization-specific but layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p}^{(i)},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1,p}, i_{1,2,1,p}$, where

$$\begin{aligned}
i_{1,1,p} &= [q_{1,p} \ q_{2,p}], \\
q_{1,p} &\in \{0,1, \dots, O_1 - 1\}, \\
q_{2,p} &\in \{0,1, \dots, O_2 - 1\}, \\
i_{1,2,l,p} &\in \left\{0,1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}.
\end{aligned}$$

[0321] In another example (for polarization-specific but layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p}^{(i)},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned}
i_{1,1} &= [i_{1,1,0} \ i_{1,1,1}], \\
i_{1,1,p} &= [q_{1,p} \ q_{2,p}], \\
q_{1,p} &\in \{0,1, \dots, O_1 - 1\}, \\
q_{2,p} &\in \{0,1, \dots, O_2 - 1\}, \\
i_{1,2} &= [i_{1,2,1} \dots i_{1,2,v}], \\
i_{1,2,l} &= [i_{1,2,l,0} \ i_{1,2,l,1}], \\
i_{1,2,l,p} &\in \left\{0,1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}.
\end{aligned}$$

[0322] In one example (for polarization-common and layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p}^{(i)},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2,1,p}$, where

$$\begin{aligned}
i_{1,1} &= [q_1 \ q_2], \\
q_1 &\in \{0,1, \dots, O_1 - 1\}, \\
q_2 &\in \{0,1, \dots, O_2 - 1\}, \\
i_{1,2,l,p} &\in \left\{0,1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}.
\end{aligned}$$

[0323] In another example (for polarization-common and layer-common q_1, q_2), the SD basis vectors

$$v_{m_1^{(i)}, m_2^{(i)}, p}^{(i)},$$

$i=0, 1, \dots, L-1, p=0, 1, l=1, 2, \dots, v$ are indicated by PMI components $i_{1,1}, i_{1,2}$, where

$$\begin{aligned}
i_{1,1} &= [q_1 \ q_2], \\
q_1 &\in \{0,1, \dots, O_1 - 1\}, \\
q_2 &\in \{0,1, \dots, O_2 - 1\}, \\
i_{1,2} &= [i_{1,2,1} \dots i_{1,2,v}], \\
i_{1,2,l} &= [i_{1,2,l,0} \ i_{1,2,l,1}], \\
i_{1,2,l,p} &\in \left\{0,1, \dots, \left(\frac{N_1 N_2}{L}\right) - 1\right\}.
\end{aligned}$$

[0324] In one example, the L SD basis vectors comprising columns of B are determined the same way as in Rel. 15 Type I codebooks (5.2.2.2.1, REF 8), i.e., the L SD basis vectors are identified by the indices $i_{1,1}, i_{1,2}$, and are obtained as in 5.2.2.2.1 of [REF 8].

[0325] In one example, $i_{1,1}$ is selected from $\{0, 1, \dots, N_1 O_1 - 1\}$ and $i_{1,2}$ is selected from $\{0, 1, \dots, N_2 O_2 - 1\}$ to indicate one SD basis vector among $N_1 N_2 O_1 O_2$ candidate basis vectors.

[0326] In one example, $i_{1,1,1}$ is selected from $\{0, 1, \dots, N_1 O_1 - 1\}$ and $i_{1,2,l}$ is selected from $\{0, 1, \dots, N_2 O_2 - 1\}$ to indicate one SD basis vector among $N_1 N_2 O_1 O_2$ candidate basis vectors, for each layer $l=1, \dots, v$.

[0327] In one example, $i_{1,1}$ is selected from

$$\{0,1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2}$ is selected from

$$\{0,1, \dots, \frac{N_2 O_2}{2} - 1\}$$

to indicate four SD basis vectors among $N_1 N_2 O_1 O_2$ candidate basis vectors.

[0328] In one example, $i_{1,1}, \dots$, is selected from

$$\{0,1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2,l}$ is selected from

$$\{0, 1, \dots, \frac{N_2 O_2}{2} - 1\}$$

to indicate four SD basis vectors among $N_1 N_2 O_1 O_2$ candidate basis vectors, for each layer $l=1, \dots, v$.

[0329] In one example, $i_{1,1}$ is selected from

$$\{0, 1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2}$ is selected from

$$\{0, 1, \dots, \frac{N_2 O_2}{2} - 1\}$$

to indicate four SD basis vectors among $N_1 N_2 O_1 O_2$ candidate basis vectors.

[0330] In one example, $i_{1,1,l}$ is selected from

$$\{0, 1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2,l}$ is selected from

$$\{0, 1, \dots, \frac{N_2 O_2}{2} - 1\}$$

to indicate four SD basis vectors among $N_1 N_2 O_1 O_2$ candidate basis vectors, for each layer $l=1, \dots, v$.

[0331] In one example, $i_{1,1}$ is selected from

$$\{0, 1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2}$ is selected from $\{0\}$ to indicate four SD basis vectors (hence no report) among $N_1 N_2 O_1 O_2$ candidate basis vectors.

[0332] In one example, $i_{1,1,1}$ is selected from

$$\{0, 1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2,l}$ is selected from $\{0\}$ to indicate four SD basis vectors (hence no report) among $N_1 N_2 O_1 O_2$ candidate basis vectors, for each layer $l=1, \dots, v$.

[0333] In one example, $i_{1,1}$ is selected from

$$\{0, 1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2}$ is selected from $\{0, 1, \dots, N_2 O_2 - 1\}$ to indicate four SD basis vectors among $N_1 N_2 O_1 O_2$ candidate basis vectors.

[0334] In one example, $i_{1,1}$ is selected from

$$\{0, 1, \dots, \frac{N_1 O_1}{2} - 1\}$$

and $i_{1,2,l}$ is selected from $\{0, 1, \dots, N_2 O_2 - 1\}$ to indicate four SD basis vectors among $N_1 N_2 O_1 O_2$ candidate basis vectors, for each layer $l=1, \dots, v$.

[0335] In one embodiment, for rank value $(v) > 1$, at least one of the following example is used/configured. Let $W_1^{(v)}$ and $W_2^{(v)}$ denote that W_1 and W_2 components for v layers.

[0336] In one example, all components of W_1 and W_2 are determined/reported according to each example described in embodiments herein for layers $l=1, \dots, v$.

[0337] In one example, $L=1$ or

[0338] In one example, $L > 1$ or

[0339] In one example, $L \in \{1, x\}$ where $x > 1$.

[0340] In one example, $L=1$ for each rank $v > 1$.

[0341] In one example, $L \in \{1, x\}$ for rank ≤ 1 and $L=1$ for each rank $v > 1$.

[0342] In one example, $L \in \{1, x\}$ for rank $\leq 1, 2$ and $L=1$ for each rank $v > 2$.

[0343] In one example, $L \in \{1, x\}$ for rank $\leq 1, 2, 3, 4$ and $L=1$ for each rank $v > 4$.

[0344] In one example, $x=2, x=4$, or $x=6$ in the above examples.

[0345] In one example, the rotation factor (q_1, q_2) are determined/reported common for all layers, and L SD basis vectors and all components of W_2 , as described above, are determined/reported independently for each layer $l=1, \dots, v$, and W_2 is determined/reported for each layer $l=1, \dots, v$.

[0346] In one example, all components of W_1 , as described above, are determined/reported common for all layers, and all components of W_2 , as described above, are determined/reported independently for each layer $l=1, \dots, v$, and W_2 is determined/reported for each layer $l=1, \dots, v$.

[0347] In one example, all components of W_1 , as described above, are determined/reported independently for each layer $l=1, \dots, v$, and one joint W_2 across v layers are determined/reported, where the columns of W_2 correspond to W_2 for v layers.

[0348] In one example, the rotation factor (q_1, q_2) are determined/reported common for all layers, L SD basis vectors, as described above, are determined/reported independently for each layer $l=1, \dots, v$, and one joint W_2 across v layers are determined/reported, where the columns of W_2 correspond to W_2 for v layers.

[0349] In one example, all components of W_1 , as described above, are determined/reported common for all layers, and one joint W_2 across v layers are determined/reported, where the columns of W_2 correspond to W_2 for v layers.

[0350] In one embodiment, when the number of layers (rank) $v=2$, the rank-2 (2-layer) precoding matrix is given by

$$W_{i_{1,1} i_{1,2} i_{2,1} i_{2,2}}^{(2)} = \frac{1}{\sqrt{2P_{CSIRS}}} [U^{(2)}].$$

[0351] At least one of the following examples is used/configured regarding $U^{(2)}$.

[0352] In one example,

$$U^{(2)} = \begin{bmatrix} b & b \\ cb & -cb \end{bmatrix}$$

where $b = v_{m_1(i), m_2(i)}$ (precoder structure A1).

[0353] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates c (in a subband (SB) manner), using an N_{PSK} -PSK scheme (as described in an example herein).

[0354] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example herein), and a second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).

[0355] In one example,

$$U^{(2)} = \begin{bmatrix} b & b \\ c_1 b & c_2 b \end{bmatrix} \text{ where } b = v_{m_1^{(j)}, m_2^{(j)}}$$

and c_l , $l=1, 2$ is a co-phase for layer l (precoder structure A2).

[0356] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates c_l , $l=1, 2$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0357] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l , $l=1, 2$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).

[0358] In one example,

$$U^{(2)} = \begin{bmatrix} b_1 & b_2 \\ cb_1 & -cb_2 \end{bmatrix} \text{ where } b_l = v_{m_1^{(j)}, m_2^{(j)}},$$

$l=1, 2$ is a DFT vector for layer l (precoder structure A3).

[0359] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b_l for $l=1, 2$ (as described in an example of embodiment 1.1) and i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0360] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2$.

[0361] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a

second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).

[0362] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B_l for each layer $l=1, 2$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example herein), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2$.

[0363] In one example,

$$U^{(2)} = \begin{bmatrix} b_1 & b_2 \\ c_1 b_1 & c_2 b_2 \end{bmatrix}$$

where b_l and c_l are as described above (precoder structure A4).

[0364] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b_l for $l=1, 2$ (as described in an example of embodiment 1.1) and i_2 indicates c_l for $l=1, 2$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of herein).

[0365] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2$.

[0366] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).

[0367] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B_l for each layer $l=1, 2$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2$.

[0368] In one example,

$$c_l = e^{\frac{j2\pi\phi_l}{N_{PSK}}}$$

where $\phi_l \in \{0, 1, \dots, N_{PSK}-1\}$, and $N_{PSK}=2$ (for BPSK), $N_{PSK}=4$ (for QPSK), 8 (for 8PSK), or 16 (for 16PSK). In one example, N_{PSK} is fixed, e.g. $N_{PSK}=4$, or $N_{PSK}=2$. In one example, N_{PSK} is configured via higher layer, e.g. from $\{2, 4\}$. In one example, UE determines/selects which N_{PSK} is used and reports it as a part of CSI. In one example, the range of ϕ_l is a subset of $\{0, 1, \dots, N_{PSK}-1\}$. For example,

$$\phi_l \in \{0, 1, \dots, \frac{N_{PSK}}{2} - 1\}.$$

In another example,

$$\phi_l \in \{0, 1, \dots, \frac{N_{PSK}}{4} - 1\}.$$

[0369] In one example,

$$c_l = p_l e^{\frac{j2\pi\phi_l}{N_{PSK}}}$$

where p_l is an amplitude or power level. In this disclosure, coefficient c_l can be reported in a WB manner or in a SB manner.

[0370] In one embodiment, for rank >2, when the number of layers (rank) $v=3$, the rank-3 (3-layer) precoding matrix is given by

$$W_{i_{1,1}, i_{1,2}, i_2}^{(3)} = \frac{1}{\sqrt{3P_{CSIRS}}} [U^{(3)}].$$

[0371] At least one of the following examples is used/configured regarding $U^{(2)}$.

[0372] In one example,

$$U^{(3)} = \begin{bmatrix} b & b & b \\ cb & -cb & -cb \end{bmatrix} \text{ or } U^{(3)} = \begin{bmatrix} b & b & b \\ cb & cb & -cb \end{bmatrix} \text{ where } b = v_{m_1^{(j)}, m_2^{(j)}}$$

(precoder structure A1).

[0373] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates c (in a subband (SB) manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0374] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).

[0375] In one example,

$$U^{(3)} = \begin{bmatrix} b & b & b \\ c_1 b & c_2 b & c_3 b \end{bmatrix} \text{ where } b = v_{m_1^{(j)}, m_2^{(j)}}$$

and c_l , $l=1, 2, 3$ is a co-phase for layer l (precoder structure A2).

[0376] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates c_l , $l=1, 2, 3$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0377] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l , $l=1, 2, 3$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a

second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).

[0378] In one example,

$$U^{(3)} = \begin{bmatrix} b_1 & b_2 & b_3 \\ cb_1 & -cb_2 & -cb_3 \end{bmatrix} \text{ or } U^{(3)} = \begin{bmatrix} b_1 & b_2 & b_3 \\ cb_1 & cb_2 & -cb_2 \end{bmatrix}$$

$$\text{where } b_l = v_{m_1^{(j)}, m_2^{(j)}},$$

$l=1, 2, 3$ is a DFT vector for layer l (precoder structure A3).

[0379] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b_l for $l=1, 2, 3$ (as described in an example of embodiment 1.1) and i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0380] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3$.

[0381] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).

[0382] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B_l for each layer $l=1, 2, 3$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3$.

[0383] In one example,

$$U^{(3)} = \begin{bmatrix} b_1 & b_2 & b_3 \\ c_1 b_1 & c_2 b_2 & c_3 b_3 \end{bmatrix}$$

where b_l and c_l are as described above (precoder structure A4).

[0384] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b_l for $l=1, 2, 3$ (as described in an example of embodiment 1.1) and i_2 indicates c_l for $l=1, 2, 3$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0385] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3$.

[0386] In one example, for $L>1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a

second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).

[0387] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B_l for each layer $l=1, 2, 3$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3$.

[0388] FIG. 18 illustrates diagrams of example beam sorting schemes 1800 for beam selection according to embodiments of the present disclosure. For example, beam sorting schemes 1800 for beam selection can be utilized by the UE 112 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0389] In one embodiment, when the number of layers (rank) $v=4$, the rank-4 (4-layer) precoding matrix is given by

$$W_{1,1,i_{1,2},i_2}^{(3)} = \frac{1}{\sqrt{4P_{CSIRS}}} [U^{(4)}].$$

[0390] At least one of the following examples is used/configured regarding $U^{(2)}$.

[0391] In one example,

$$U^{(4)} = \begin{bmatrix} b & b & b & b \\ cb & -cb & cb & -cb \end{bmatrix} \text{ or } U^{(4)} = \begin{bmatrix} b & b & b & b \\ cb & cb & -cb & -cb \end{bmatrix}$$

where $b = v_{m_1^{(j)}, m_2^{(j)}}$

(precoder structure A1).

[0392] In one example, for $L=1$, the indicator ($i_{1,1}, i_{1,2}$) indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates c (in a subband (SB) manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0393] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).

[0394] In one example,

$$U^{(4)} = \begin{bmatrix} b & b & b & b \\ c_1 b & c_2 b & c_3 b & c_4 b \end{bmatrix} \text{ where } b = v_{m_1^{(j)}, m_2^{(j)}}$$

c_l , $l=1, 2, 3, 4$ is a co-phase for layer l (precoder structure A2).

[0395] In one example, for $L=1$, the indicator ($i_{1,1}, i_{1,2}$) indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates c_l , $l=1, 2, 3, 4$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0396] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l ,

$l=1, 2, 3, 4$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).

[0397] In one example,

$$U^{(4)} = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ cb_1 & -cb_2 & cb_3 & -cb_4 \end{bmatrix} \text{ or } U^{(4)} = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ cb_1 & cb_2 & -cb_3 & -cb_4 \end{bmatrix}$$

where $b_l = v_{m_1^{(j)}, m_2^{(j)}}$

$l=1, 2, 3, 4$ is a DFT vector for layer l (precoder structure A3).

[0398] In one example, for $L=1$, the indicator ($i_{1,1}, i_{1,2}$) indicates an SD vector b_l for $l=1, 2, 3, 4$ (as described in an example of embodiment 1.1) and i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0399] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3, 4$.

[0400] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).

[0401] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B_l for each layer $l=1, 2, 3, 4$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3, 4$.

[0402] In one example,

$$U^{(4)} = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ c_1 b_1 & c_2 b_2 & c_3 b_3 & c_4 b_4 \end{bmatrix}$$

where b_l and c_l are as described above (precoder structure A4).

[0403] In one example, for $L=1$, the indicator ($i_{1,1}, i_{1,2}$) indicates an SD vector b_l for $l=1, 2, 3, 4$ (as described in an example of embodiment 1.1) and i_2 indicates c_l for $l=1, 2, 3, 4$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).

[0404] In one example, for $L>1$, the indicator ($i_{1,1}, i_{1,2}$) indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3, 4$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3, 4$.

- [0405] In one example, for $L > 1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3, 4$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).
- [0406] In one example, for $L > 1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B_l for each layer $l=1, 2, 3, 4$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3, 4$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3, 4$.
- [0407] In one embodiment, when the number of layers (rank) $v > 4$, the rank- v (v -layer) precoding matrix is given by

$$W_{i_{1,1}, i_{1,2}, i_2}^{(v)} = \frac{1}{\sqrt{v P_{CSIRS}}} [U^{(v)}].$$

- [0408] In one example,

$$U^{(v)} = \begin{bmatrix} b & b & \dots & b \\ c_1 b & c_2 b & \dots & c_v b \end{bmatrix} \text{ where } b = v_{m_1^{(j)}, m_2^{(j)}}^{(j)}$$

and $c_l, l=1, 2, 3, \dots, v$ is a co-phase for layer l (precoder structure A2).

- [0409] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b (as described in an example of embodiment 1.1) and i_2 indicates $c_l, l=1, 2, 3, \dots, v$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).
- [0410] In one example, for $L > 1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates $c_l, l=1, 2, 3, \dots, v$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vector b out of the L SD vectors (in a SB manner).
- [0411] In one example,

$$U^{(v)} = \begin{bmatrix} b_1 & b_2 & \dots & b_v \\ c_1 b_1 & c_2 b_2 & \dots & c_v b_v \end{bmatrix}$$

where b_l and c_l are as described above $c_l b_l, c_2 b_2, \dots, c_v b_v$ (precoder structure A4).

- [0412] In one example, for $L=1$, the indicator $(i_{1,1}, i_{1,2})$ indicates an SD vector b_l for $l=1, 2, 3, \dots, v$ (as described in an example of embodiment 1.1) and i_2 indicates c_l for $l=1, 2, 3, \dots, v$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1).
- [0413] In one example, for $L > 1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3, \dots, v$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment

1.1), and a second part of i_2 indicate an SD vector b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3, \dots, v$.

- [0414] In one example, for $L > 1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B_l (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3, \dots, v$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate two SD vectors b_l and b_2 out of the L SD vectors (in a SB manner).
- [0415] In one example, for $L > 1$, the indicator $(i_{1,1}, i_{1,2})$ indicates L SD vectors B_l for each layer $l=1, 2, 3, \dots, v$ (as described in an example of embodiment 1.1) and a first part of i_2 indicates c_l for $l=1, 2, 3, \dots, v$ (in a SB manner), using an N_{PSK} -PSK scheme (as described in an example of embodiment 1.1), and a second part of i_2 indicate an SD vectors b_l out of the L SD vectors (in a SB manner) for each layer $l=1, 2, 3, \dots, v$.

- [0416] In one embodiment, a UE can be configured with CSI reporting with a codebook described in this disclosure under a restriction.

- [0417] In one example, the UE can be configured with CSI reporting with a codebook described in this disclosure when a total number of CSI-RS ports $P_{CSI-RS, total} \leq t$ across CSI-RS resources is greater than (or equal to) t . In one example, $t=32$. In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. For example, for $P_{CSI-RS, total} \leq t$, the UE can be configured with CSI reporting with a legacy codebook.

- [0418] Various embodiments of the present disclosure provide a framework having multiple codebooks where one codebook can be configured by NW based on its preference and/or UE capability.

- [0419] In one embodiment, a UE can be configured with a codebook via higher-layer signaling (e.g., codebookConfig-r19) for Type-I (single-panel) CSI reporting. The configured codebook can be a Rel-15-based Type-I codebook with supporting up to $2N_1 N_2 = x$ CSI-RS ports (total number of CSI-RS ports across aggregated NZP-CSI-RS resources), where $x=64$ or 128 , or an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure).

- [0420] In one example, for an enhanced Type-I codebook, new pair values of (N_1, N_2) specified in Rel-19 (i.e., (N_1, N_2) pairs s.t. $2N_1 N_2 > 32$) are supported.

- [0421] In one example, for an enhanced Type-I codebook, new pair values of (N_1, N_2) specified in Rel-19 (i.e., (N_1, N_2) pairs s.t. $2N_1 N_2 > 32$) and legacy pair values of (N_1, N_2) specified in Rel-15 (i.e., (N_1, N_2) pairs s.t. $2N_1 N_2 \leq 32$) are supported.

- [0422] In an embodiment, there are multiple codebook modes for Rel-19 eType-I codebook, and one of the codebook modes can be configured via higher-layer signaling (e.g., codebookMode for Rel-19).

- [0423] In an example, multiple codebook modes include two schemes: 1) Rel-15 Type-I codebook with codebookMode=1 (i.e., beam selection scheme), which supports $2N_1 N_2 = x$ CSI-RS ports, where $x=64$ or 128 , and 2) an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure).

- [0424] In one example, scheme 1) corresponds to codebook mode 1 and/or scheme 2) corresponds to codebook mode 2.

[0425] In one example, scheme 1) corresponds to codebook mode 2 and/or scheme 2) corresponds to codebook mode 1.

[0426] In one example, for a rank value=1, a Rel-15 Type-I codebook with codebookMode=1 is designed based on the following codebook [REF8]:

codebookMode = 1			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1O_1 - 1$	$0, \dots, N_2O_2 - 1$	$0, 1, 2, 3$	$W_{i_{1,1}i_{1,2}i_2}^{(1)}$
where $W_{l,m,n}^{(1)} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \end{bmatrix}$.			

[0427] In one example, for a rank value=2, a Rel-15 Type-I codebook with codebookMode=1 is designed based on the following codebook [REF8]:

codebookMode = 1			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1O_1 - 1$	$0, \dots, N_2O_2 - 1$	$0, 1$	$W_{i_{1,1}i_{1,2}i_2}^{(2)}$
where $W_{l,l',m,m',n}^{(2)} = \frac{1}{\sqrt{2P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$.			
and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1-3 [REF8].			

[0428] In one example, a mapping from $i_{1,3}$ to k_1 and k_2 is as follows:

$i_{1,3}$	k_1	k_2
0	0	0
1	O_1	0
2	0	O_2
3	$2O_1$	0

[0429] In another example, multiple codebook modes include two schemes: 1) Rel-15 Type-I codebook with codebookMode=2 (i.e., beam group selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , and 2) an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure)

[0430] In one example, scheme 1) corresponds to codebook mode 1 and/or scheme 2) corresponds to codebook mode 2.

[0431] In one example, scheme 1) corresponds to codebook mode 2 and/or scheme 2) corresponds to codebook mode 1.

[0432] In another example, multiple codebook modes include three schemes: 1) Rel-15 Type-I codebook with codebookMode=1 (i.e., beam selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , 2) Rel-15 Type-I codebook with codebookMode=2 (i.e., beam group selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , and 3) an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure).

[0433] In one example, scheme 1) corresponds to codebook mode 1 and/or scheme 2) corresponds to codebook mode 2 and/or scheme 3) corresponds to codebook mode 3.

[0434] In one example, scheme 1) corresponds to codebook mode 1 and/or scheme 3) corresponds to codebook mode 2 and/or scheme 1) corresponds to codebook mode 3.

[0435] In one example, scheme 2) corresponds to codebook mode 1 and/or scheme 1) corresponds to codebook mode 2 and/or scheme 3) corresponds to codebook mode 3.

[0436] In one example, scheme 2) corresponds to codebook mode 1 and/or scheme 3) corresponds to codebook mode 2 and/or scheme 1) corresponds to codebook mode 3.

[0437] In one example, scheme 3) corresponds to codebook mode 1 and/or scheme 1) corresponds to codebook mode 2 and/or scheme 2) corresponds to codebook mode 3.

[0438] In one example, scheme 3) corresponds to codebook mode 1 and/or scheme 2) corresponds to codebook mode 2 and/or scheme 1) corresponds to codebook mode 3.

[0439] In another example, multiple codebook modes include two schemes: 1) Rel-15 Type-I codebook, which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , and 2) an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure). Also, the codebook mode corresponding to the Rel-15 Type-I codebook includes two sub-modes: a) a codebook corresponding to Rel-15 Type-I codebook with codebookMode=1 and b) a codebook corresponding to Rel-15 Type-I codebook with codebookMode=2.

[0440] In one example, scheme 1) corresponds to codebook mode 1 and/or scheme 2) corresponds to codebook mode 2.

[0441] In one example, scheme a) corresponds to sub-mode 1 and/or Scheme b) corresponds to sub-mode 2

[0442] In one example, Scheme a) corresponds to sub-mode 2 and/or Scheme b) corresponds to sub-mode 1

[0443] In one example, scheme 1) corresponds to codebook mode 2 and/or scheme 2) corresponds to codebook mode 1.

[0444] In one example, Scheme a) corresponds to sub-mode 1 and/or Scheme b) corresponds to sub-mode 2

[0445] In one example, Scheme a) corresponds to sub-mode 2 and/or Scheme b) corresponds to sub-mode 1

[0446] In another embodiment, there are multiple codebook types for Rel-19 eType-I codebook, and one of the codebook types can be configured via higher-layer signaling (e.g., typeI-SinglePanel, typeI-SinglePanel-r19, or eTypeI-SinglePanel-r19).

[0447] In an example, multiple codebook types include two schemes: 1) Rel-15 Type-I codebook with codebookMode=1 (i.e., beam selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , and 2) an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure).

[0448] In one example, scheme 1) corresponds to codebookType='typeI-SinglePanel' or codebookType='typeI-SinglePanel-r19'

[0449] In one example, scheme 2) corresponds to codebookType='typeI-SinglePanel-r19' or codebookType='eTypeI-SinglePanel-r19'.

[0450] In another example, multiple codebook types include two schemes: 1) Rel-15 Type-I codebook with codebookMode=2 (i.e., beam group selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , and 2) an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure).

[0451] In one example, scheme 1) corresponds to codebookType='typeI-SinglePanel' or codebookType='typeI-SinglePanel-r19'

[0452] In one example, scheme 2) corresponds to codebookType='typeI-SinglePanel-r19' or codebookType='eTypeI-SinglePanel-r19'.

[0453] In another example, multiple codebook types include two schemes: 1) Rel-15 Type-I codebook, which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , and 2) an enhanced Type-codebook (e.g., designed based on one of the frameworks described in this disclosure). Also, the codebook type corresponding to the Rel-15 Type-I codebook includes two codebook modes: a) a codebook corresponding to Rel-15 Type-I codebook with codebookMode=1 and b) a codebook corresponding to Rel-15 Type-I codebook with codebookMode=2.

[0454] In one example, scheme 1) corresponds to codebookType='typeI-SinglePanel' or codebookType='typeI-SinglePanel-r19'

[0455] In one example, scheme a) correspond to codebook mode 1 and/or scheme b) correspond to codebook mode 2.

[0456] In one example, scheme a) correspond to codebook mode 2 and/or scheme b) correspond to codebook mode 1.

[0457] In one example, scheme 2) corresponds to codebookType='typeI-SinglePanel-r19' or codebookType='eTypeI-SinglePanel-r19'.

[0458] In another embodiment, configuring a codebook for Rel-19 eType-I CSI reporting is subject to UE capability. Depending on UE capability whether the UE supports a codebook for Rel-19 eType-I CSI reporting or not, NW can configure the codebook subject to UE capability. The UE capability can be according to at least one of the following examples.

[0459] In an example, a codebook corresponding to Rel-15 Type-I codebook with codebookMode=1 (i.e., beam selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , is a basic UE feature that the UE has to support if the UE is a Rel-19 feature supporting UE.

[0460] In another example, a codebook corresponding to Rel-15 Type-I codebook with codebookMode=1 (i.e., beam selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , is an optional UE feature that the UE may not support even if the UE is a Rel-19 feature supporting UE.

[0461] In another example, a codebook corresponding to Rel-15 Type-I codebook with codebookMode=2 (i.e., beam selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , is a basic UE feature that the UE has to support if the UE is a Rel-19 feature supporting UE.

[0462] In another example, a codebook corresponding to Rel-15 Type-I codebook with codebookMode=2 (i.e., beam selection scheme), which supports $2N_1N_2=x$ CSI-RS ports, where $x=64$ or 128 , is an optional UE feature that the UE may not support even if the UE is a Rel-19 feature supporting UE.

[0463] In another example, a codebook corresponding to an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure), is a basic UE feature that the UE has to support if the UE is a Rel-19 feature supporting UE.

[0464] In another example, a codebook corresponding to an enhanced Type-I codebook (e.g., designed based on one of the frameworks described in this disclosure), is an optional UE feature that the UE may not support even if the UE is a Rel-19 feature supporting UE.

[0465] In another embodiment, for a Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports, for rank value $v=3$ or 4 , (i.e., $RI=3-4$), (at least) one of the following examples can be supported.

[0466] In one example, regardless of number of CSI-RS ports, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows (only) the Rel-15 Type-I scheme for $P_{CSI-RS} < 16$ (Scheme X).

[0467] In one example, regardless of number of CSI-RS ports, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows (only) the Rel-15 Type-I scheme for $P_{CSI-RS} > 16$ (Scheme Y).

[0468] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows the same as the Rel-15 Type-I schemes for $P_{CSI-RS} < 16$ and for $P_{CSI-RS} > 16$.

[0469] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows the Scheme X for $P_{CSI-RS} < t$ (or $P_{CSI-RS} < t$) and Scheme Y for $P_{CSI-RS} \geq t$ (or $P_{CSI-RS} \geq t$).

[0470] In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. In one example, $t=32$. In one example, $t=48$. In one example, $t=64$. In one example, $t=128$.

[0471] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows the Scheme X for $P_{CSI-RS} > t$ and Scheme Y for $P_{CSI-RS} \leq t$.

[0472] In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. In one example, $t=32$. In one example, $t=48$. In one example, $t=64$. In one example, $t=128$.

Rel-15 Type-I Codebook for 3-Layer CSI Reporting Using Antenna Ports 3000 to 2999

codebookMode = 1-2, $P_{CSI-RS} < 16$ (e.g., Scheme X)			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1O_1 - 1 \quad 0, 1, \dots, N_2O_2 - 1 \quad 0, 1 \quad W_{i_{1,1}i_{1,2}+k_1i_{1,2}+k_2i_2}^{(3)}$			
where $W_{i_{1,1}i_{1,2}+k_1i_{1,2}+k_2i_2}^{(3)} = \frac{1}{\sqrt{3P_{CSI-RS}}} \begin{bmatrix} v_{1,m} & v_{1,m'} & v_{1,m} \\ \varphi_n v_{1,m} & \varphi_n v_{1,m'} & -\varphi_n v_{1,m} \end{bmatrix}$			
and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1-4.			

codebookMode = 1-2, $P_{CSI-RS} \geq 16$ (e.g., Scheme Y)				
$i_{1,1}$	$i_{1,2}$	$i_{1,3}$	i_2	
$0, \dots, \frac{N_1 O_1}{2} - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1, 2, 3$	$0, 1$	$W_{i_{1,1}, i_{1,2}, i_{1,3}, i_2}^{(3)}$
where $W_{i_{1,1}, i_{1,2}, i_{1,3}, i_2}^{(3)} = \frac{1}{\sqrt{3P_{CSI-RS}}} \begin{bmatrix} \tilde{v}_{l,m} & \tilde{v}_{l,m} & \tilde{v}_{l,m} \\ \theta_p \tilde{v}_{l,m} & -\theta_p \tilde{v}_{l,m} & \theta_p \tilde{v}_{l,m} \\ \varphi_n \tilde{v}_{l,m} & \varphi_n \tilde{v}_{l,m} & -\varphi_n \tilde{v}_{l,m} \\ \varphi_n \theta_p \tilde{v}_{l,m} & -\varphi_n \theta_p \tilde{v}_{l,m} & -\varphi_n \theta_p \tilde{v}_{l,m} \end{bmatrix}$				

Rel-15 Type-I Codebook for 4-Layer CSI Reporting Using Antenna Ports 3000 to 2999+

codebookMode = 1-2, $P_{CSI-RS} < 16$ (e.g., Scheme X)				
$i_{1,1}$	$i_{1,2}$	i_2		
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(4)}$	
where $W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(4)} = \frac{1}{\sqrt{4P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} & v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$				
and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1-4.				

codebookMode = 1-2, $P_{CSI-RS} \geq 16$ (e.g., Scheme Y)				
$i_{1,1}$	$i_{1,2}$	$i_{1,3}$	i_2	
$0, \dots, \frac{N_1 O_1}{2} - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1, 2, 3$	$0, 1$	$W_{i_{1,1}, i_{1,2}, i_{1,3}, i_2}^{(4)}$
where $W_{i_{1,1}, i_{1,2}, i_{1,3}, i_2}^{(4)} = \frac{1}{\sqrt{4P_{CSI-RS}}} \begin{bmatrix} \tilde{v}_{l,m} & \tilde{v}_{l,m} & \tilde{v}_{l,m} & \tilde{v}_{l,m} \\ \theta_p \tilde{v}_{l,m} & -\theta_p \tilde{v}_{l,m} & \theta_p \tilde{v}_{l,m} & -\theta_p \tilde{v}_{l,m} \\ \varphi_n \tilde{v}_{l,m} & \varphi_n \tilde{v}_{l,m} & -\varphi_n \tilde{v}_{l,m} & -\varphi_n \tilde{v}_{l,m} \\ \varphi_n \theta_p \tilde{v}_{l,m} & -\varphi_n \theta_p \tilde{v}_{l,m} & -\varphi_n \theta_p \tilde{v}_{l,m} & \varphi_n \theta_p \tilde{v}_{l,m} \end{bmatrix}$				

In one example, a mapping from $i_{1,3}$ to k_1 and k_2 is as follows:

$i_{1,3}$	k_1	k_2
0	O_1	0
1	0	O_2
2	O_1	O_2
3	$2O_1$	0

[0473] In another embodiment, (mixture of 0, 1 and 1.4) for a Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports, (at least) one of the following examples can be supported.

[0474] In one example, regardless of number of CSI-RS ports, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=1, where the Rel-15 Type-I scheme with codebookMode=1 uses (only) the codebook associated with $P_{CSI-RS} < 16$ in the case of rank value=3 or 4.

[0475] In one example, regardless of number of CSI-RS ports, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with

codebookMode=1, where the Rel-15 Type-I scheme with codebookMode=1 uses (only) the codebook associated with $P_{CSI-RS} > 16$ in the case of rank value=3 or 4.

[0476] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows the same Rel-15 Type-I scheme with codebookMode=1 (i.e., the Rel-15 Type-I scheme with codebookMode=1 follows both the codebooks associated with $P_{CSI-RS} < 16$ and $P_{CSI-RS} > 16$ in the case of rank value=3 or 4, depending on the configured number of CSI-RS ports.

[0477] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=1, where the Rel-15 Type-I scheme with codebookMode=1 uses Scheme A for $P_{CSI-RS} < t$ (or $P_{CSI-RS} < t$) and Scheme Y for $P_{CSI-RS} \geq t$ (or $P_{CSI-RS} \geq t$).

[0478] In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. In one example, $t=32$. In one example, $t=48$. In one example, $t=64$. In one example, $t=128$.

[0479] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=1, where the Rel-15 Type-I scheme with codebookMode=1 uses Scheme A for $P_{CSI-RS} \geq t$ (or $P_{CSI-RS} > t$) and Scheme Y for $P_{CSI-RS} < t$ (or $P_{CSI-RS} \leq t$).

[0480] In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. In one example, $t=32$. In one example, $t=48$. In one example, $t=64$. In one example, $t=128$.

[0481] In one example, regardless of number of CSI-RS ports, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=2, where the Rel-15 Type-I scheme with codebookMode=2 uses (only) the codebook associated with $P_{CSI-RS} < 16$ in the case of rank value=3 or 4.

[0482] In one example, regardless of number of CSI-RS ports, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=2, where the Rel-15 Type-I scheme with codebookMode=2 uses (only) the codebook associated with $P_{CSI-RS} > 16$ in the case of rank value=3 or 4.

[0483] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows the same Rel-15 Type-I scheme with codebookMode=2 (i.e., the Rel-15 Type-I scheme with codebookMode=2 follows both the codebooks associated with $P_{CSI-RS} < 16$ and $P_{CSI-RS} \geq 16$ in the case of rank value=3 or 4, depending on the configured number of CSI-RS ports.).

[0484] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=2, where the Rel-15 Type-I scheme with codebookMode=2 uses Scheme A for $P_{CSI-RS} < t$ (or $P_{CSI-RS} < t$) and Scheme Y for $P_{CSI-RS} \geq t$ (or $P_{CSI-RS} \geq t$).

[0485] In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. In one example, $t=32$. In one example, $t=48$. In one example, $t=64$. In one example, $t=128$.

[0486] In one example, the Rel-15-based Type-I codebook with supporting up to 128 CSI-RS ports follows a Rel-15 Type-I scheme with codebookMode=2, where the Rel-15 Type-I scheme with codebookMode=2 uses Scheme A for $P_{CSI-RS} \geq t$ (or $P_{CSI-RS} > t$) and Scheme Y for $P_{CSI-RS} < t$ (or $P_{CSI-RS} \leq t$).

[0487] In one example, $t=16$. In one example, $t=8$. In one example, $t=4$. In one example, $t=32$. In one example, $t=48$. In one example, $t=64$. In one example, $t=128$.

[0488] In one example, for $\text{rank} > 4$, (i.e., $RI=5-8$), $\text{ceil}(v/2)$ SD basis vectors are selected/reported for $RI=v$, where each SD basis vector is applied to two respective layers, and if v is odd, one of the SD basis vectors is applied to an orphan layer (i.e., the layer that doesn't belong to any $2(\text{ceil}(v/2)-1)$ layers). This can be called layer-pair specific SD basis selection. For inter-polarization co-phasing values, A codepoints are used for co-phasing for two layers sharing a same SD basis vector, and B codepoints are used for an orphan layer. For example, $A=B$. In one example, $A=B/2$. In one example, $A=2$, and $B=4$.

[0489] In one example, for $v=5$,

$$U^{(v)} = \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & b_3 \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & c_3 b_3 \end{bmatrix}.$$

At least one of the following examples can be used to indicate $U^{(v)}$.

[0490] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate each of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.

[0491] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate all of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.

[0492] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate all of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$, and an additional indicator is used to indicate which SD basis vector corresponds to the orphan layer.

[0493] In one example, for a coefficient/cophase value associated with two respective layers, (i.e., c_1 and c_2) it is selected from a A-PSK codebook, where $A=N_{PSK,1}$. In one example, $A=2$ or $A=4$. In one example, when $A=2$, A-PSK codebook= $\{1, j\}$. In another example, when $A=4$, A-PSK codebook= $\{1, j, -1, -j\}$.

[0494] In one example, for a coefficient/cophase value associated with an orphan layer, (i.e., c_3) it is selected

from a B-PSK codebook, where $B=N_{PSK,2}$. In one example, $B=4$. In one example, when $B=4$, B-PSK codebook= $\{1, j, -1, -j\}$.

[0495] In one example, for $v=6$,

$$U^{(v)} = \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & b_3 & b_3 \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & c_3 b_3 & -c_3 b_3 \end{bmatrix}.$$

At least one of the following examples can be used to indicate U .

[0496] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate each of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.

[0497] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate all of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.

[0498] In one example, for a coefficient/cophase value associated with two respective layers, (i.e., c_1, c_2 , and c_3) it is selected from a A-PSK codebook, where $A=N_{PSK,1}$. In one example, $A=2$ or $A=4$. In one example, when $A=2$, A-PSK codebook= $\{1, j\}$. In another example, when $A=4$, A-PSK codebook= $\{1, j, -1, -j\}$.

[0499] In one example, for $v=7$,

$$U^{(v)} = \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & b_3 & b_3 & b_4 \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & c_3 b_3 & -c_3 b_3 & c_4 b_4 \end{bmatrix}.$$

At least one of the following examples can be used to indicate $U^{(v)}$.

[0500] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate each of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.

[0501] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate all of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.

[0502] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate all of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$, and an additional indicator is used to indicate which SD basis vector corresponds to the orphan layer.

[0503] In one example, for a coefficient/cophase value associated with two respective layers, (i.e., c_1, c_2 , and c_3) it is selected from a A-PSK codebook, where $A=N_{PSK,1}$. In one example, $A=2$ or $A=4$. In one example, when $A=2$, A-PSK codebook= $\{1, j\}$. In another example, when $A=4$, A-PSK codebook= $\{1, j, -1, -j\}$.

[0504] In one example, for a coefficient/cophase value associated with an orphan layer, (i.e., c_4) it is selected from a B-PSK codebook, where $B=N_{PSK,2}$. In one example, $B=4$. In one example, when $B=4$, B-PSK codebook= $\{1, j, -1, -j\}$.

[0505] In one example, for $v=8$,

$$U^{(v)} = \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & b_3 & b_3 & b_4 & b_4 \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & c_3 b_3 & -c_3 b_3 & c_4 b_4 & -c_4 b_4 \end{bmatrix}.$$

At least one of the following examples can be used to indicate $U^{(v)}$.

- [0506] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate each of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.
- [0507] In one example, an indicator (e.g., $i_{1,1}$) is used to indicate (q_1, q_2) common for all $\text{ceil}(v/2)$ layers and a combinatorial indicator (e.g., $i_{1,2}$) is used to indicate all of the $\text{ceil}(v/2)$ SD basis vectors $l=1, \dots, \text{ceil}(v/2)$.
- [0508] In one example, for a coefficient/cophase value associated with two respective layers, (i.e., c_p, c_2 , and c_3, c_4) it is selected from a A-PSK codebook, where $A=N_{PS}K, 1$. In one example, $A=2$ or $A=4$. In one example, when $A=2$, A-PSK codebook= $\{1, j\}$. In another example, when $A=4$, A-PSK codebook= $\{1, j, -1, -j\}$.
- [0509] In one embodiment, a UE can be configured with CSI reporting with a codebook described in this disclosure under a restriction.
- [0510] In one example, the UE can be configured with CSI reporting with a codebook described in this disclosure when a total number of CSI-RS ports $P_{CSI-RS, total}$ across CSI-RS resources is greater than t . In one example, $t=32$. For example, for $P_{CSI-RS, total}$; t , the UE can be configured with CSI reporting with a legacy codebook.
- [0511] Various embodiments of the present disclosure provide an enhanced Type-I codebook designed based on Rel-15 Type-I codebook.
- [0512] In various embodiments, an enhanced Type-I codebook for e.g., Rel-19 type-I CSI reporting can be designed (or extended) based on Rel-15 Type-I codebook. An enhanced Type-I codebook can be according to at least one of the following examples.
- [0513] In one example, a first SD vector (2D-DFT) is selected from a set of $N_1 N_2 O_1 O_2$ 2D-DFT vectors, and a second SD vector is selected from a (sub-)set of X orthogonal vectors of the first SD vector, where, e.g., $X=(N_1-1)N_2 O_2 + (N_2-1)N_1 O_1 - (N_1-1)(N_2-1)$. In one example, an indicator with size of $\lceil \log_2 X \rceil$ -bit is used to indicate a second SD vector in a CSI report.
- [0514] Note that for a given 2D-DFT SD vector, there are $(N_1-1)N_2 O_2 + (N_2-1)N_1 O_1 - (N_1-1)(N_2-1)$ vectors, which are orthogonal to the given 2D-DFT SD vector, among $N_1 N_2 O_1 O_2$ 2D-DFT vectors. The $(N_1-1)N_2 O_2 + (N_2-1)N_1 O_1 - (N_1-1)(N_2-1)$ vectors can be called extended set of orthogonal vectors.
- [0515] Let denote a set of $(N_1-1)N_2 O_2 + (N_2-1)N_1 O_1 - (N_1-1)(N_2-1)$ vectors by $\mathcal{X}$.
- [0516] In one example, a second SD vector is selected from a subset of $\mathcal{X}$. In one example, an indicator with size of $\lceil \log_2 |\mathcal{X}| \rceil$ -bit is used to indicate a second SD vector in a CSI report, where $|\mathcal{X}|$ is the cardinality of $\mathcal{X}$.
- [0517] In one example, an indicator (of $i_{1,1}$) with size of $\lceil \log_2 N_1 O_1 \rceil$ -bit is used to indicate a vector index in a first dimension of antenna ports for a first SD vector (2D-DFT), and another indicator (of $i_{1,1}$) with size of $\lceil \log_2 N_2 O_2 \rceil$ -bit is

used to indicate a vector index in a second dimension of antenna ports for the first SD vector. (Rel-15-based indicator)

[0518] In one example, a joint indicator (of $i_{1,1}$ and $i_{1,2}$) with size of $\lceil \log_2 N_1 O_1 N_2 O_2 \rceil$ -bit is used to indicate a vector index in first and second dimensions of antenna ports for a first SD vector (2D-DFT).

[0519] In one example, an indicator (or multiple indicators) with size of $\lceil \log_2 Y \rceil$ -bit, where $Y \geq |\mathcal{X}|$, is used to indicate a second SD vector.

[0520] In one example, a joint indicator is used to indicate first and second vectors similar to Rel-16-based Type-II SD basis indicator. For example, an indicator (of $i_{1,1}$) with size of $\lceil \log_2 O_1 O_2 \rceil$ -bit is used to indicate (O_1, O_2) and another indicator (of $i_{1,2}$) with size of

$$\left\lceil \log_2 \binom{N_1 N_2}{L} \right\rceil$$

-bit is used to indicate two vectors (e.g., $L=2$) and another indicator is used to indicate rotated values or offset values for a second vector.

[0521] In one example, for co-phase (and/or co-amp) reporting, it is similar to or exactly the same as Rel-15 SP Type-I codebook.

[0522] In another example, for co-phase reporting, it is similar to an example/embodiment shown in embodiments herein, (e.g., layer-specific co-phase (and/or co-amp) reporting).

[0523] In one example, an enhanced Type-I codebook includes indicators $i_{1,1}, i_{1,2}, i_{1,3}$ of Rel-15 type-I codebook with codebookMode=1 which indicate two 2D DFT basis vectors, and a new indicator $i_{1,4}$ to shift the oversampled offset value for a second vector in direction either O_1 or O_2 . In one example, the payload of $i_{1,4}$ is 2 bits. In one example, the payload of $i_{1,4}$ is 3 bits. In one example, the payload of $i_{1,4}$ is 4 bits.

[0524] In another example, an enhanced Type-I codebook described in examples herein includes a joint indicator $i_{1,3, joint}=(1,3, i_{1,4})$, (instead of $i_{1,3}$ and $i_{1,4}$ indicators separately) where $i_{1,3}$ and $i_{1,4}$ are described in examples herein.

[0525] In another example, an enhanced Type-I codebook includes indicators $i_{1,1}, i_{1,2}$ of Rel-16 eType-II codebook to indicate two 2D DFT basis vectors, and a new indicator $i_{1,3}$ to shift the oversampled offset value for a second vector in direction O_1 or in direction O_2 . In one example, the payload of $i_{1,3}$ is 3 bits. In one example, the payload of $i_{1,3}$ is 4 bits.

[0526] In another example, an enhanced Type-I codebook includes indicators $i_{1,1}, i_{1,2}, i_{1,3}$ of Rel-15 type-I codebook with codebookMode=2 which indicate two 2D DFT vector groups, and a new indicator $i_{1,4}$ to shift the oversampled offset value for a second vector group in direction either O_1 or O_2 . In one example, the payload of $i_{1,4}$ is 1 bit. In one example, the payload of $i_{1,4}$ is 2 bits.

[0527] In another example, an enhanced Type-I codebook described in examples herein includes a joint indicator $i_{1,3, joint}=(1,3, i_{1,4})$, (instead of $i_{1,3}$ and $i_{1,4}$ indicators separately) where $i_{1,3}$ and $i_{1,4}$ are described in examples herein.

[0528] In one example, an enhanced Type-I codebook described in examples herein supports up to rank=4.

[0529] In another example, for RI=5-6 (i.e., when the rank value is selected/indicated as 5 or 6), three SD basis vectors are selected and indicated via indicator(s) and, for RI=7-8

(i.e., when the rank value is selected/indicated as 7 or 8), four SD basis vectors are selected and indicated via indicator (s), where a first SD basis vector is selected/indicated via indicator described in one of the examples in this disclosure (e.g., examples/embodiments herein). In one example, first, second, third, and fourth SD basis vectors are 2D-DFT vectors and mutually orthogonal, where any two of the four SD basis vectors are orthogonal at least one of the two-dimensional directions for the 2D-DFT vectors.

[0530] In another example, the payloads of all indicators associated with SD basis vectors are fixed and the indicators are included in CSI part 2.

[0531] In one example, a first SD vector (2D-DFT) is (freely) selected from a set of $N_1 N_2 O_1 O_2$ 2D-DFT vectors, and indicated via an indicator with size of $\lceil \log_2 N_1 O_1 \rceil$ bits or via two indicators with sizes of $\lceil \log_2 N_1 O_1 \rceil$ bits and $\lceil \log_2 N_2 O_2 \rceil$ bits.

[0532] In one example, second, third, and fourth SD vectors are selected from an extended orthogonal SD vector set from a first SD vector, and indicated via respective indicators each with size of $\lceil \log_2 X \rceil$ bits, where X is a number of an extended orthogonal SD vector set from a first SD vector, i.e., $X = (N_1 - 1)N_2 O_2 + (N_2 - 1)N_1 O_1 - (N_1 - 1)(N_2 - 1)$. In this case, since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two/three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is $L(\lceil \log_2 X \rceil)$ bits, where $L=2$ for RI=5-6 and $L=3$ for RI=7-8.

[0533] In one example, second, third, and fourth SD vectors are selected from an extended orthogonal SD vector set from a first SD vector, and indicated via a combinatorial indicator with size of

$$\left\lceil \log_2 \binom{X}{L} \right\rceil \text{ bits,}$$

where X is a number of an extended orthogonal SD vector set from a first SD vector, i.e., $X = (N_1 - 1)N_2 O_2 + (N_2 - 1)N_1 O_1 - (N_1 - 1)(N_2 - 1)$. In this case, since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two/three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is

$$\left\lceil \log_2 \binom{X}{L} \right\rceil \text{ bits,}$$

where $L=2$ for RI=5-6 and $L=3$ for RI=7-8.

[0534] In one example, each of second, third, and fourth SD vectors is selected from an extended orthogonal SD vector set from a first SD vector, and indicated via two indicators with size of $\lceil \log_2 X_{1,1} \rceil$ bits and size of $\lceil \log_2 X_2 \rceil$ bits, where X_1 is a number of critical SD basis vectors without considering oversampling factors (i.e., $X_1 \leq N_1 N_2$) and X_2 is a (possible) number of SD vectors that can be

shifted by offsets with oversampling factors of O_1 and O_2 for a critical SD vector (i.e., $X_2 \leq O_1 O_2$). In this case, since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two/three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is $L(\lceil \log_2 X_1 \rceil + \lceil \log_2 X_2 \rceil)$ bits, where $L=2$ for RI=5-6 and $L=3$ for RI=7-8.

[0535] In one example, $X_1 = N_1 N_2$ and $X_2 = O_1 + O_2 - 1$.

[0536] In one example, $X_1 = N_1 N_2$ and $X_2 = O_1 + O_2$.

[0537] In one example, $X_1 = N_1 N_2$ and $X_2 = O_1 O_2$.

[0538] In one example, $X_1 = N_1 N_2 - 1$ and $X_2 = O_1 + O_2 - 1$.

[0539] In one example, $X_1 = N_1 N_2 - 1$ and $X_2 = O_1 + O_2$.

[0540] In one example, $X_1 = N_1 N_2 - 1$ and $X_2 = O_1 O_2$.

[0541] In one example, each of second, third, and fourth SD vectors (if RI=7-8) is selected from an extended orthogonal SD vector set from a first SD vector, and indicated via three indicators with size of $\lceil \log_2 N_1 \rceil$ bits, size of $\lceil \log_2 N_2 \rceil$ bits, and size of $\lceil \log_2 X_2 \rceil$, and additional 1-bit indicator is (commonly) used for all the L SD basis vectors, where the 1-bit indicator indicates which dimensional direction (either N_1 or N_2 direction) is the direction where the L SD basis vectors are orthogonal to the first SD basis vector, and X_2 is a (possible) number of SD vectors that can be shifted by offsets with either oversampling factors of O_1 or O_2 for a critical SD vector. Here, each critical SD vector index is indicated via the two indicators of $\lceil \log_2 N_1 \rceil$ and $\lceil \log_2 N_2 \rceil$ bits. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is $L(\lceil \log_2 N_1 \rceil + \lceil \log_2 N_2 \rceil + \lceil \log_2 X_2 \rceil + 1)$ bits, where $L=2$ for RI=5-6 and $L=3$ for RI=7-8.

[0542] In one example, $X_2 = O_1$ for the case where one codepoint of the 1-bit indicator is indicated, and $X_2 = O_2$ for the case where the other code point of the 1-bit indicator is indicated.

[0543] In one example, $X_2 = O_1 - 1$ for the case where one codepoint of the 1-bit indicator is indicated, and $X_2 = O_2 - 1$ for the case where the other code point of the 1-bit indicator is indicated.

[0544] In one example, $X_2 = 4$.

[0545] In one example, $X_2 = 3$.

[0546] In one example, $X_2 = O_1 O_2 = O_1$.

[0547] In one example, $X_2 = O_1 - 1 = O_2 - 1 = O_1 - 1$.

[0548] In one example, in the above example, the additional 1-bit indicator is the same but the three indicators are replaced by two indicators, where a first indicator indicates an index of N_1 direction and a second indicator indicates an index of N_2 direction. The first indicator is an indicator with size of $\lceil \log_2 N_1 \rceil$ or $\lceil \log_2 N_1 O_1 \rceil$ depending on the indication of the 1-bit indicator, and the second indicator is an indicator with size of $\lceil \log_2 O_2 N_2 \rceil$ or $\lceil \log_2 N_2 \rceil$ depending on the indication of the 1-bit indicator. For example, for a codepoint of the 1-bit indicator, the first indicator is with size of $\lceil \log_2 N_1 \rceil$ and the second indicator is with size of $\lceil \log_2 O_2 N_2 \rceil$. For another codepoint of the 1-bit indicator, the first indicator is with size of $\lceil \log_2 O_1 N_1 \rceil$ and the second indicator is with size of $\lceil \log_2 N_2 \rceil$.

[0549] In one example, i-th SD vector is selected from an extended orthogonal SD vector set from a first SD vector for $i=2, 3, 4$, and indicated via two indicators with size of $\lceil \log_2 X_{1-(i-1)} \rceil$ bits and size of $\lceil \log_2 X_2 \rceil$ bits, where X_1 is a number of critical SD basis vectors without considering

oversampling factors (i.e., $X_1 < N_1 N_2$) and X_2 is a (possible) number of SD vectors that can be shifted by offsets with oversampling factors of ϕ_i and O_2 for a critical SD vector (i.e., $X_2 < O_1 O_2$). In this case, since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two/three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is $(\sum_i \lceil \log_2 X_{1-(i-1)} \rceil + L \lceil \log_2 X_2 \rceil)$ bits, where $L=2$ for $RI=5-6$ and $L=3$ for $RI=7-8$.

[0550] In one example, $X_1 = N_1 N_2$ and $X_2 = O_1 + O_2 - 1$.

[0551] In one example, $X_1 = N_1 N_2$ and $X_2 = O_1 + O_2$.

[0552] In one example, $X_1 = N_1 N_2$ and $X_2 = O_1 O_2$.

[0553] In one example, second, third, and fourth SD vectors are selected from an extended orthogonal SD vector set from a first SD vector, and indicated via a combinatorial indicator with size of $\lceil \log_2 X_3 \rceil$ indicating their associated critical SD basis vector indexes without considering oversampling factors and respective indicators each with size of $\lceil \log_2 X_2 \rceil$ bits, where X_3 is a possible number of indicating L SD vectors (e.g., $L=2$ for $RI=5-6$ or 3 for $RI=7-8$) via a combinatorial indicator and X_2 is a (possible) number of SD vectors that can be shifted by offsets with oversampling factors of O_1 and O_2 for a critical SD vector (i.e., $X_2; O_1 O_2$). In this case, since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two/three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is $(\lceil \log_2 X_3 \rceil + L \lceil \log_2 X_2 \rceil)$ bits, where $L=2$ for $RI=5-6$ and $L=3$ for $RI=7-8$.

[0554] In one example,

$$X_3 = \binom{N_1 N_2}{L}$$

and $X_2 = O_1 + O_2 - 1$.

[0555] In one example,

$$X_3 = \binom{N_1 N_2}{L}$$

and $X_2 = O_1 + O_2$.

[0556] In one example,

$$X_3 = \binom{N_1 N_2}{L}$$

and $X_2 = O_1 O_2$.

[0557] In one example,

$$X_3 = \binom{N_1 N_2 - 1}{L}$$

and $X_2 = O_1 + O_2 - 1$.

[0558] In one example,

$$X_3 = \binom{N_1 N_2 - 1}{L}$$

and $X_2 = O_1 + O_2$.

[0559] In one example,

$$X_3 = \binom{N_1 N_2 - 1}{L}$$

and $X_2 = O_1 O_2$.

[0560] In one example, for the above examples, the indicators with size of $L \lceil \log_2 X_2 \rceil$ bits can be replaced by a joint indicator with size of $\lceil \log_2 X_2 \rceil$.

[0561] Here, in this disclosure,

$$\binom{N}{K} = \frac{N!}{K!(N-K)!}$$

is a number of combinations.

[0562] In one example, second, third, and fourth SD vectors (if $RI=7-8$) are selected from an extended orthogonal SD vector set from a first SD vector, and indicated via 1) a joint indicator with size of $\lceil \log_2 X_3 \rceil$ bits and additional 1-bit indicator is (commonly) used for all the L SD basis vectors, where the 1-bit indicator indicates which dimensional direction (either N_1 or N_2 direction) is the direction where the L SD basis vectors are orthogonal to the first SD basis vector. For each of the L SD basis vectors, another indicator with size of $\lceil \log_2 X_2 \rceil$ is used to indicate a (beam) offset, where X_2 is a (possible) number of SD vectors that can be shifted by offsets with either oversampling factors of O_1 or O_2 for a critical SD vector. Hence, the total payload for indicating the L SD vectors (excluding the first SD vector) is $(\lceil \log_2 X_3 \rceil + L \lceil \log_2 X_2 \rceil) + 1$ bits, where $L=2$ for $RI=5-6$ and $L=3$ for $RI=7-8$.

[0563] In one example,

$$X_3 = \binom{N_1 N_2}{L}$$

[0564] In one example,

$$X_3 = \binom{N_1 N_2 - 1}{L}$$

[0565] In one example, $X_2 = O_1$ for the case where one codepoint of the 1-bit indicator is indicated, and $X_2 = O_2$ for the case where the other code point of the 1-bit indicator is indicated. In one example, $X_2 = O_1 - 1$ for the case where one codepoint of the 1-bit indicator is indicated, and $X_2 = O_2 - 1$ for the case where the other code point of the 1-bit indicator is indicated.

[0566] In one example, $X_2 = 4$.

[0567] In one example, $X_2 = 3$.

[0568] In one example, $X_2 = O = O_2 = O_1$.

[0569] In one example, $X_2 = O - 1 = O_2 - 1 = O_1 - 1$.

[0570] In one example, second, third, and fourth SD vectors (if RI=7-8) are selected from an extended orthogonal SD vector set from a first SD vector, and indicated via 1) a joint indicator with size of $\lceil \log_2 X_3 \rceil$ bits jointly with the first SD basis vector and additional 1-bit indicator is (commonly) used for all the L SD basis vectors, where the 1-bit indicator indicates which dimensional direction (either N_1 or N_2 direction) is the direction where the L SD basis vectors are orthogonal to the first SD basis vector. For each of the L SD basis vectors, another indicator with size of $\lceil \log_2 X_2 \rceil$ is used to indicate a (beam) offset, where X_2 is a (possible) number of SD vectors that can be shifted by offsets with either oversampling factors of O_1 or O_2 for a critical SD vector. Hence, the total payload for indicating the L+1 SD vectors including the first SD vector is $\lceil \log_2 X_3 \rceil + L \lceil \log_2 X_2 \rceil + 1 + \lceil \log_2 X_4 \rceil$ bits, where L=2 for RI=5-6 and L=3 for RI=7-8.

[0571] In one example,

$$X_3 = \binom{N_1 N_2}{L+1}.$$

[0572] In one example, $X_2=O_1$ for the case where one codepoint of the 1-bit indicator is indicated, and $X_2=O_2$ for the case where the other code point of the 1-bit indicator is indicated.

[0573] In one example, $X_2=O_1-1$ for the case where one codepoint of the 1-bit indicator is indicated, and $X_2=O_2-1$ for the case where the other code point of the 1-bit indicator is indicated.

[0574] In one example, $X_2=4$.

[0575] In one example, $X_2=3$.

[0576] In one example, $X_2=O=O_2=O_1$.

[0577] In one example, $X_2=O-1=O_2-1=O_1-1$.

[0578] In one example, $X_4=O_1 O_2$.

[0579] In another example, the payloads of at least one indicator associated with SD basis vectors are variable and information associated with the at least one indicator is included in CSI part 1 (or UCI part 1).

[0580] In another example, in CSI part 1, there is an indicator A to indicate a number of SD basis vectors, where the 2D-DFT SD basis vectors are orthogonal in both of the two-dimensional directions.

[0581] In one example, the indicator A has |A| codepoints (e.g., 2-bits when |A|=4). For example, codepoint 1 corresponds to 3, codepoint 2 corresponds to 2, codepoint 3 corresponds to 1, and codepoint 4 corresponds to 0.

[0582] In another example, the indicator A has four codepoints, where the codepoints (bit-field) correspond to the cases that are shown in Table 3.

[0583] Depending on the information that the indicator A indicates, the payload(s) of indicator(s) associated with the SD basis vectors are determined according to at least one of the following examples.

[0584] In one example, when the indicator A indicates k SD vectors are orthogonal in both directions for RI=7-8 (which is also indicated via RI in CSI part 1), the 3 SD vectors (excluding the first SD vector) are indicated via a combinatorial indicator with size of

$$\left\lceil \log_2 \binom{N_1 N_2}{3} \right\rceil \text{ bits or } \left\lceil \log_2 \binom{N_1 N_2 - 1}{3} \right\rceil \text{ bits,}$$

and offsets for the 3-k SD vectors are indicated via respective indicators each with size of $\lceil \log_2 X_2 \rceil$ bits. Since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the 3 SD vectors (excluding the first SD vector) is

$$\left(\left\lceil \log_2 \binom{N_1 N_2}{3} \right\rceil + (3-k) \lceil \log_2 X_2 \rceil \right) \text{ bits or } \left\lceil \log_2 \binom{N_1 N_2 - 1}{3} \right\rceil + (3-k) \lceil \log_2 X_2 \rceil \text{ bits.}$$

[0585] In one example, $X_2=O_1+O_2-1$.

[0586] In one example, $X_2=O_1+O_2$.

[0587] In one example, $X_2=O_1 O_2$.

[0588] In one example, $X_2=O_1+O_2-2$.

[0589] In one example, the same indicating ways of the above example are utilized except that the offsets for the 3-k SD vectors (where $k \leq 2$) are indicated via respective indicators, where a first indicator payload is $\lceil \log_2 X_{2,1} \rceil$ bits and the payload of each of the remaining indicators (i.e., associated with the offsets for the 2-k SD vectors) is $\lceil \log_2 X_{2,2} \rceil$ bits. In this case, the total payload for indicating the 3 SD vectors (excluding the first SD vector) for $k \leq 2$ is

$$\left(\left\lceil \log_2 \binom{N_1 N_2}{3} \right\rceil + \lceil \log_2 X_{2,1} \rceil + (2-k) \lceil \log_2 X_{2,2} \rceil \right) \text{ bits}$$

$$\text{or } \left\lceil \log_2 \binom{N_1 N_2 - 1}{3} \right\rceil + \lceil \log_2 X_{2,1} \rceil + (2-k) \lceil \log_2 X_{2,2} \rceil \text{ bits.}$$

[0590] In one example, $X_{2,1}=O_1+O_2-1$ and $X_{2,2}=O$, where $O=4$.

[0591] In one example, $X_{2,1}=O_1+O_2-1$ and $X_{2,2}=O-1$, where $O=4$.

[0592] In one example, $X_{2,1}=O_1+O_2-2$ and $X_{2,2}=O$, where $O=4$.

[0593] In one example, $X_{2,1}=O_1+O_2-2$ and $X_{2,2}=O-1$, where $O=4$.

[0594] In one example, $X_{2,1}=O_1+O_2$ and $X_{2,2}=O$, where $O=4$.

[0595] In one example, $X_{2,1}=O_1+O_2$ and $X_{2,2}=O-1$, where $O=4$.

[0596] In one example, the same indicating ways of the above example are utilized except that the offsets for the 3-k SD vectors (where $k \leq 2$) are indicated via a joint indicator, where the joint indicator's payload is $\lceil \log_2 X_4 \rceil$ bits. In this case, the total payload for indicating the 3 SD vectors (excluding the first SD vector) for $k \leq 2$ is

$$\left(\left\lceil \log_2 \binom{N_1 N_2}{3} \right\rceil + \lceil \log_2 X_4 \rceil \right) \text{ bits or } \left\lceil \log_2 \binom{N_1 N_2 - 1}{3} \right\rceil + \lceil \log_2 X_4 \rceil \text{ bits.}$$

[0597] In one example, $X_4=(O_1+O_{2,2})(O-1)^{2-k}$, where $O=O_1$ or $O=O_2$ or $O=4$.

[0598] In one example, $X_4=(O-1)^{3-k}$, where $O=O_1$ or $O=O_2$ or $O=4$ (when $k=0$).

[0599] In one example, $X_4=6 \cdot 3^{2-k}$.

[0600] In one example, $X_4=3^{3-k}$ (when $k=0$).

[0601] In one example, when the indicator A indicates k SD vectors are orthogonal in both directions for RI=7-8 (which is also indicated via RI in CSI part 1), the 3 SD vectors (excluding the first SD vector) are indicated via respective indicators each with size of $\lceil \log_2(N_1N_2) \rceil$ bits or $\lceil \log_2(N_1N_2-1) \rceil$ bits or the i-th SD vector of the 3 SD vectors is indicated via an indicator with size of $\lceil \log_2(N_1N_2-i) \rceil$ bits for $i=1, 2, 3$, and offsets for the 3-k SD vectors are indicated via respective indicators each with size of $\lceil \log_2 X_2 \rceil$ bits. Since there is a constraint that the first, second, third, and fourth SD vectors are mutually orthogonal and any two of the four SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the three vectors (second to fourth) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the 3 SD vectors (excluding the first SD vector) is $(3\lceil \log_2(N_1N_2) \rceil + (3-k)\lceil \log_2 X_2 \rceil)$ bits or $(3\lceil \log_2(N_1N_2-1) \rceil + (3-k)\lceil \log_2 X_2 \rceil)$ bits or $(\sum_i \lceil \log_2(N_1N_2-i) \rceil + (3-k)\lceil \log_2 X_2 \rceil)$ bits.

[0602] In one example, $X_2=O_1+O_2-1$.

[0603] In one example, $X_2=O_1+O_2$.

[0604] In one example, $X_2=O_1O_2$.

[0605] In one example, $X_2=O_1+O_{2,2}$.

[0606] In one example, the same indicating ways of the above example are utilized except that the offsets for the 3-k SD vectors (where $k \leq 2$) are indicated via respective indicators, where a first indicator payload is $\lceil \log_2 X_{2,1} \rceil$ bits and the payload of each of the remaining indicators (i.e., associated with the offsets for the 2-k SD vectors) is $\lceil \log_2 X_{2,2} \rceil$ bits. In this case, the total payload for indicating the 3 SD vectors (excluding the first SD vector) for $k \leq 2$ is $(3\lceil \log_2(N_1N_2) \rceil + \lceil \log_2 X_{2,1} \rceil + (2-k)\lceil \log_2 X_{2,2} \rceil)$ bits or $(3\lceil \log_2(N_1N_2-1) \rceil + \lceil \log_2 X_{2,1} \rceil + (2-k)\lceil \log_2 X_{2,2} \rceil)$ bits or $(3\lceil \log_2(N_1N_2-i) \rceil + \lceil \log_2 X_{2,1} \rceil + (2-k)\lceil \log_2 X_{2,2} \rceil)$ bits.

[0607] In one example, $X_{2,1}=O_1+O_2-1$ and $X_{2,2}=O$, where $O=4$.

[0608] In one example, $X_{2,1}=O_1+O_2-1$ and $X_{2,2}=O-1$, where $O=4$.

[0609] In one example, $X_{2,1}=O_1+O_{2,2}$ and $X_{2,2}=O$, where $O=4$.

[0610] In one example, $X_{2,1}=O_1+O_{2,2}$ and $X_{2,2}=O-1$, where $O=4$.

[0611] In one example, $X_{2,1}=O_1+O_2$ and $X_{2,2}=O$, where $O=4$.

[0612] In one example, $X_{2,1}=O_1+O_2$ and $X_{2,2}=O-1$, where $O=4$.

[0613] In one example, when the indicator A indicates k SD vectors (where $k \leq 2$) are orthogonal in both directions for RI=5-6 (which is also indicated via RI in CSI part 1), the 2 SD vectors (excluding the first SD vector) are indicated via a combinatorial indicator with size of

$$\left\lceil \log_2 \binom{N_1N_2}{3} \right\rceil \text{ bits or } \left\lceil \log_2 \binom{N_1N_2-1}{2} \right\rceil \text{ bits,}$$

and offsets for the 2-k SD vectors are indicated via respective indicators each with size of $\lceil \log_2 X_2 \rceil$ bits. Since there

is a constraint that the first, second, and third SD vectors are mutually orthogonal and any two of the three SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two vectors (second and third) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the 2 SD vectors (excluding the first SD vector) is

$$\left(\left\lceil \log_2 \binom{N_1N_2}{2} \right\rceil + (2-k)\lceil \log_2 X_2 \rceil \right) \text{ bits or } \left\lceil \log_2 \binom{N_1N_2-1}{2} \right\rceil + (2-\lceil \log_2 X_2 \rceil) \text{ bits.}$$

[0614] In one example, $X_2=O_1+O_2-1$.

[0615] In one example, $X_2=O_1+O_2$.

[0616] In one example, $X_2=O_1O_2$.

[0617] In one example, $X_2=O_1+O_{2,2}$.

[0618] In one example, the same indicating ways of the above example are utilized except that the offsets for the 2-k SD vectors (where $k \leq 1$) are indicated via respective indicators, where a first indicator payload is $\lceil \log_2 X_{2,1} \rceil$ bits and the payload of each of the remaining indicator (i.e., associated with the offset for the 1-k SD vectors) is $\lceil \log_2 X_{2,2} \rceil$ bits. In this case, the total payload for indicating the 3 SD vectors (excluding the first SD vector) for $k \leq 1$ is

$$\left(\left\lceil \log_2 \binom{N_1N_2}{2} \right\rceil + \lceil \log_2 X_{2,1} \rceil + (1-k)\lceil \log_2 X_{2,2} \rceil \right) \text{ bits or } \left\lceil \log_2 \binom{N_1N_2-1}{2} \right\rceil + \lceil \log_2 X_{2,1} \rceil + (1-k)\lceil \log_2 X_{2,2} \rceil \text{ bits.}$$

[0619] In one example, $X_{2,1}=O_1+O_2-1$ and $X_{2,2}=O$, where $O=4$.

[0620] In one example, $X_{2,1}=O_1+O_2-1$ and $X_{2,2}=O-1$, where $O=4$.

[0621] In one example, $X_{2,1}=O_1+O_{2,2}$ and $X_{2,2}=O$, where $O=4$.

[0622] In one example, $X_{2,1}=O_1+O_{2,2}$ and $X_{2,2}=O-1$, where $O=4$.

[0623] In one example, $X_{2,1}=O_1+O_2$ and $X_{2,2}=O$, where $O=4$.

[0624] In one example, $X_{2,1}=O_1+O_2$ and $X_{2,2}=O-1$, where $O=4$.

[0625] In one example, the same indicating ways of the above example are utilized except that the offsets for the 2-k SD vectors (where $k \leq 1$) are indicated via a joint indicator, where the joint indicator's payload is $\lceil \log_2 X_4 \rceil$ bits. In this case, the total payload for indicating the 3 SD vectors (excluding the first SD vector) for $k \leq 1$ is

$$\left(\left\lceil \log_2 \binom{N_1N_2}{2} \right\rceil + \lceil \log_2 X_4 \rceil \right) \text{ bits or } \left\lceil \log_2 \binom{N_1N_2-1}{2} \right\rceil + \lceil \log_2 X_4 \rceil \text{ bits.}$$

[0626] In one example, $X_4=(O_1+O_{2,2})(O-1)1-k$, where $O=O_1$ or $O=O_2$ or $O=4$.

[0627] In one example, $X_4=(O-1)^{2-k}$, where $O=O_1$ or $O=O_2$ or $O=4$ (when $k=0$).

[0628] In one example, $X_4=6 \cdot 3^{1-k}$.

[0629] In one example, $X_4=3^{2-k}$ (when $k=0$).

[0630] In one example, when the indicator A indicates k SD vectors (where $k \leq 2$) are orthogonal in both directions for

RI=5-6 (which is also indicated via RI in CSI part 1), the 2 SD vectors (excluding the first SD vector) are indicated via respective indicators each with size of $\lceil \log_2(N_1 N_2) \rceil$ bits or $\lceil \log_2(N_1 N_2 - 1) \rceil$ bits or the i -th SD vector of the 2 SD vectors is indicated via an indicator with size of $\lceil \log_2(N_1 N_2 - i) \rceil$ bits for $i=1, 2$, and offsets for the 2- k SD vectors are indicated via respective indicators each with size of $\lceil \log_2 X_2 \rceil$ bits. Since there is a constraint that the first, second, and third SD vectors are mutually orthogonal and any two of the three SD vectors are orthogonal at least one of the two-dimensional directions, the UE needs to indicate the two vectors (second and third) via the respective indicators satisfying the constraint. Hence, the total payload for indicating the 2 SD vectors (excluding the first SD vector) is $(2\lceil \log_2(N_1 N_2) \rceil + (2-k)\lceil \log_2 X_2 \rceil)$ bits or $(2\lceil \log_2(N_1 N_2 - 1) \rceil + (2-k)\lceil \log_2 X_2 \rceil)$ bits or $(\sum_i \lceil \log_2(N_1 N_2 - i) \rceil + (2-k)\lceil \log_2 X_2 \rceil)$ bits.

[0631] In one example, $X_2 = O_1 + O_2 - 1$.

[0632] In one example, $X_2 = O_1 + O_2$.

[0633] In one example, $X_2 = O_1 O_2$.

[0634] In one example, $X_2 = O_1 + O_{2,2}$.

[0635] In one example, the same indicating ways of the above example are utilized except that the offsets for the 2- k SD vectors (where $k \leq 1$) are indicated via respective indicators, where a first indicator payload is $\lceil \log_2 X_{2,1} \rceil$ bits and the payload of each of the remaining indicator (i.e., associated with the offset for the 1- k SD vectors) is $\lceil \log_2 X_{2,2} \rceil$ bits. In this case, the total payload for indicating the 3 SD vectors (excluding the first SD vector) for $k \leq 1$ is $(2\lceil \log_2(N_1 N_2) \rceil + \lceil \log_2 X_{2,1} \rceil + (1-k)\lceil \log_2 X_{2,2} \rceil)$ bits or $(2\lceil \log_2(N_1 N_2 - 1) \rceil + \lceil \log_2 X_{2,1} \rceil + (1-k)\lceil \log_2 X_{2,2} \rceil)$ bits or $(\sum_i \lceil \log_2(N_1 N_2 - i) \rceil + \lceil \log_2 X_{2,1} \rceil + (1-k)\lceil \log_2 X_{2,2} \rceil)$ bits.

[0636] In one example, $X_{2,1} = O_1 + O_2 - 1$ and $X_{2,2} = O$, where $O=4$.

[0637] In one example, $X_{2,1} = O_1 + O_2 - 1$ and $X_{2,2} = O - 1$, where $O=4$.

[0638] In one example, $X_{2,1} = O_1 + O_{2,2}$ and $X_{2,2} = O$, where $O=4$.

[0639] In one example, $X_{2,1} = O_1 + O_{2,2}$ and $X_{2,2} = O - 1$, where $O=4$.

[0640] In one example, $X_{2,1} = O_1 + O_2$ and $X_{2,2} = O$, where $O=4$.

[0641] In one example, $X_{2,1} = O_1 + O_2$ and $X_{2,2} = O - 1$, where $O=4$.

[0642] There is an open issue on how to indicate subsequent $L=2$ (for RI=5-6) or $L=3$ SD basis vectors (for RI=7-8), once the 1st SD basis vector is freely selected and indicated via indicators with size of $\lceil \log_2 N_1 O_1 \rceil$ bits and $\lceil \log_2 N_2 O_2 \rceil$ bits (assuming the using legacy scheme for the first SD basis vector).

[0643] There could be two simple methods to indicate the L SD basis vectors from the candidate SD basis vector set having $X = (N_1 - 1)N_2 O_2 + (N_2 - 1)N_1 O_1 - (N_1 - 1)(N_2 - 1)$ candidate vectors.

[0644] Method 1: Each of the L SD basis vectors is independently selected from the candidate SD basis vector set via an indicator with size of $\lceil \log_2 X \rceil$ bits.

[0645] Method 2: The L SD basis vectors are selected from the candidate SD basis vector set via a combinatorial indicator with size of

$$\left\lceil \log_2 \binom{X}{L} \right\rceil \text{ bits.}$$

[0646] Note that for both of Method 1 and Method 2, the UE needs to choose the L SD basis vectors that are mutually orthogonal in at least one dimension as described in the previous agreement, although the indicators of the both methods can allow non-orthogonal SD basis selections.

[0647] Method 1 and Method 2 can be a simple solution but they could not be efficient in terms of overhead since they have redundant combinations that are not allowed to select.

[0648] FIG. 19 illustrates an example of beam cluster indexing and beam offset indexing 1900 according to embodiments of the present disclosure. For example, the beam cluster indexing and beam offset indexing 1900 for beam selection can be utilized by the UE 116 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0649] Another method (Method 3) is to indicate the L SD basis vectors via 1) beam-cluster indicator and 2) beam offset indicator, where the beam-cluster indicator indicates beam cluster indexes for the L SD basis vectors and the beam-offset indicator beam offset indexes for the L SD basis vectors. Here, a beam cluster refers to a SD vector group containing $O_1 \times O_2$ adjacent SD vectors in N_1 and N_2 dimensions, and a beam offset refers to a particular SD vector of a beam cluster. FIG. 19 illustrates an example of beam cluster indexing and beam offset indexing when $(N_1, N_2) = (8, 4)$ with $(O_1, O_2) = (4, 4)$. Regarding the method, there are two properties with the candidate SD basis vectors that:

[0650] the L SD basis vectors (including the first SD basis vector) have to be located in different clusters, and

[0651] the number of beam offsets in a beam cluster is at most $O_1 + O_2 - 1$.

[0652] Based on the two properties, Method 3 can indicate the L SD basis vectors via the beam-cluster/beam-offset indicators with the payload size of

$$\left\lceil \log_2 \binom{N_1 N_2 - 1}{L} \right\rceil + L \lceil \log_2 (O_1 + O_2 - 1) \rceil$$

bits.

[0653] Method 3 has a good structure to indicate the L SD basis vectors, but it is still not efficient in terms of overhead since the beam offset indicator is designed in considering the worst-case number of beam offsets in a beam cluster.

[0654] To design a further efficient method (called Method 4) on top of Method 3 in terms of overhead, additional indicator in CSI part 1 is to provide useful information regarding relation among the first SD basis vector and the L SD basis vectors, based on the following two properties:

[0655] if an SD basis vector is orthogonal to the first SD basis vector in both the horizontal and vertical (or N_1 and N_2) dimensions, indicating only a beam-cluster index for the SD basis vector is enough since the beam offset index is the same as that of the first SD basis vector; and

[0656] if an SD basis vector is orthogonal to the first SD basis vector in one dimension (either horizontal or vertical), a beam-offset index and to a beam-cluster index for the SD basis vector are needed to indicate, where the beam-offset index is selected from 3 beam-offset candidates (i.e., there are only 3 beam-offset

candidates in the dimension since the vector that is orthogonal in both dimensions is excluded). Also, each of the other remaining SD basis vectors is orthogonal to the first SD basis vector in the same dimension or in both of the two dimensions.

[0657] Utilizing those two properties, the following additional indicator with 2-bit (4 cases) in CSI part 1 are provided as in Table 3 below:

Bit-field	00 (case 1)	01 (case 2)	10 (case 3)	11 (case 4)
The number of SD basis vectors (out of the L SD basis vectors) that are orthogonal to the first SD basis vector in both dimensions	0	0	1	≥ 2
Dimension direction where the L SD basis vectors are orthogonal to the first SD basis vector	Horizontal (N1-dim direction)	Vertical (N2-dim direction)	—	—

[0658] For each case, the total payload of the beam-cluster/beam-offset indicators in addition to the 2-bit for the additional indicator above is given by, for L=2 or 3,

[0659] case 1/2:

$$\left\lceil \log_2 \binom{N_1 N_2 - 1}{L} \right\rceil + \lceil \log_2 3^L \rceil + 2 \text{ bits}$$

[0660] case 3:

$$\left\lceil \log_2 \binom{N_1 N_2 - 1}{L} \right\rceil + \lceil \log_2 6 \times 3^{L-2} \rceil + 2 \text{ bits}$$

[0661] case 4:

$$\left\lceil \log_2 \binom{N_1 N_2 - 1}{L} \right\rceil + 1_{\{L=3\}} \lceil \log_2 6 \rceil + 2 \text{ bits},$$

where 1_A is 1 only when event A satisfies.

[0662] This is because, once the beam-cluster indexes for the L SD basis vectors are indicated via

$$\left\lceil \log_2 \binom{N_1 N_2 - 1}{3} \right\rceil ;$$

[0663] for case 1 or 2, the beam-offset index for each of the L SD basis vectors can be selected from the 3 beam-offset candidates and all of the beam-offset indexes are indicated via a joint indicator with size of $\lceil \log_2 3^L \rceil$ bits;

[0664] for case 3, the beam-offset index for the SD basis vector orthogonal in the both dimensions is not needed to indicate, the beam-offset index for one SD basis vector out of the L=1 SD basis vectors can be selected from the 6 beam-offset candidates, and the beam-offset index for each of the remaining L=2 SD basis vectors can be selected from 3 beam-offset candidates since the

dimension direction where the L=1 SD basis vectors are orthogonal to the first SD basis vector is the same, hence all of the beam-offset indexes are indicated via joint indicator with size of $\lceil \log_2 6 \times 3^{L-2} \rceil$ bits.

[0665] for case 4, when L=2, there is no need to indicate the beam-offset indexes since all beam-offset indexes for the L SD basis vectors are the same as the one of the first SD basis vector, but when L=3, the beam-offset index for the last SD basis vector is needed to indicate and can be selected from the 6 beam-offset candidates, hence indicated via size of $\lceil \log_2 6 \rceil$ bits.

TABLE 4

Overhead comparison for RI = 7 or 8						
(N1, N2)	(8, 3)	(6, 4)	(16, 2)	(8, 4)	(16, 4)	(8, 8)
Method 1	24	24	24	24	27	27
Method 2	19	19	20	21	24	24
Method 3	20	20	22	22	25	25
Method 4 (case 1/2)	18	18	20	20	23	23
Method 4 (case 3)	18	18	20	20	23	23
Method 4 (case 4)	16	16	18	18	21	21

TABLE 5

Overhead comparison for RI = 5 or 6						
(N1, N2)	(8, 3)	(6, 4)	(16, 2)	(8, 4)	(16, 4)	(8, 8)
Method 1	16	16	16	16	18	18
Method 2	14	14	14	15	17	17
Method 3	14	14	15	15	17	17
Method 4 (case 1/2)	14	14	15	15	17	17
Method 4 (case 3)	13	13	14	14	16	16
Method 4 (case 4)	10	10	11	11	13	13

[0666] Table 4 and Table 5 show overhead comparison for RI=7/8 and RI=5/6, respectively. It is observed that the overhead of Method 4 for all cases for any (N1,N2) configuration is smaller than or equal to that of the other methods (Method 1/2/3), except only the following scenario:

[0667] Case 1/2 of Method 4 for the configuration of (N1,N2)=(16,2) with RI=5/6.

[0668] In one sub-embodiment (or another embodiment), an enhanced Type-I codebook for Rel-19 Type-I CSI reporting is designed based on a mixture of Scheme 1) a Rel-15 single-panel Type-I codebook and Scheme 2) one of the examples/embodiments in this disclosure.

[0669] In one example, when RI=1 (i.e., rank value=1), a Rel-15 single-panel Type-I codebook with codebookMode=1 is used for Rel-19 Type-I CSI reporting, and when RI=2-4, (i.e., rank value=2,3, or 4), an enhanced Type-I codebook described in this disclosure (e.g., in embodiments herein), is used for Rel-19 Type-I CSI reporting.

[0670] In one example, when RI=1 (i.e., rank value=1), a Rel-15 single-panel Type-I codebook with codebookMode=2 is used for Rel-19 Type-I CSI reporting, and when RI=2-4, (i.e., rank value=2,3, or 4), an enhanced Type-I codebook described in this disclosure (e.g., in embodiments herein), is used for Rel-19 Type-I CSI reporting.

[0671] In one example, when RI=1-2 (i.e., rank value=1 or 2), a Rel-15 single-panel Type-I codebook with codebookMode=1 is used for Rel-19 Type-I CSI reporting, and

when RI=3-4, (i.e., rank value=3 or 4), an enhanced Type-I codebook described in this disclosure (e.g., in embodiments herein), is used for Rel-19 Type-I CSI reporting.

[0672] In one example, when RI=1-2 (i.e., rank value=1 or 2), a Rel-15 single-panel Type-I codebook with codebookMode=2 is used for Rel-19 Type-I CSI reporting, and when RI=3-4, (i.e., rank value=3 or 4), an enhanced Type-I codebook described in this disclosure (e.g., in embodiments herein), is used for Rel-19 Type-I CSI reporting.

[0673] In one example, when RI is in a set A, a Rel-15 single-panel Type-I codebook with codebookMode=1 is used for Rel-19 Type-I CSI reporting, and when RI is in a set B, an enhanced Type-I codebook described in this disclosure (e.g., in embodiments herein), is used for Rel-19 Type-I CSI reporting, where A, B are a subset of {1, 2, 3, 4, 5, 6, 7, 8}.

[0674] In one example, A includes 1, 2, 3, 4 and B includes 5, 6, 7, 8.

[0675] In one example, A includes 1, 2 and B includes 3, 4, 5, 6, 7, 8.

[0676] In one example, A includes 1, 2 and B includes 3, 4.

[0677] In one example, A includes 1. In another example, A does not include 1.

[0678] In one example, A includes 2. In another example, A does not include 2.

[0679] In one example, A includes 3. In another example, A does not include 3.

[0680] In one example, A includes 4. In another example, A does not include 4.

[0681] In one example, A includes 5. In another example, A does not include 5.

[0682] In one example, A includes 6. In another example, A does not include 6.

[0683] In one example, A includes 7. In another example, A does not include 7.

[0684] In one example, A includes 8. In another example, A does not include 8.

[0685] In one example, B includes 1. In another example, B does not include 1.

[0686] In one example, B includes 2. In another example, B does not include 2.

[0687] In one example, B includes 3. In another example, B does not include 3.

[0688] In one example, B includes 4. In another example, B does not include 4.

[0689] In one example, B includes 5. In another example, B does not include 5.

[0690] In one example, B includes 6. In another example, B does not include 6.

[0691] In one example, B includes 7. In another example, B does not include 7.

[0692] In one example, B includes 8. In another example, B does not include 8.

[0693] In one example, when RI is in a set A, a Rel-15 single-panel Type-I codebook with codebookMode=2 is used for Rel-19 Type-I CSI reporting, and when RI is in a set B, an enhanced Type-I codebook described in this disclosure (e.g., in embodiments herein), is used for Rel-19 Type-I CSI reporting, where A, B are a subset of {1, 2, 3, 4, 5, 6, 7, 8}.

[0694] In one example, A includes 1, 2, 3, 4 and B includes 5, 6, 7, 8.

[0695] In one example, A includes 1, 2 and B includes 3, 4, 5, 6, 7, 8.

[0696] In one example, A includes 1, 2 and B includes 3, 4.

[0697] In one example, A includes 1. In another example, A does not include 1.

[0698] In one example, A includes 2. In another example, A does not include 2.

[0699] In one example, A includes 3. In another example, A does not include 3.

[0700] In one example, A includes 4. In another example, A does not include 4.

[0701] In one example, A includes 5. In another example, A does not include 5.

[0702] In one example, A includes 6. In another example, A does not include 6.

[0703] In one example, A includes 7. In another example, A does not include 7.

[0704] In one example, A includes 8. In another example, A does not include 8.

[0705] In one example, B includes 1. In another example, B does not include 1.

[0706] In one example, B includes 2. In another example, B does not include 2.

[0707] In one example, B includes 3. In another example, B does not include 3.

[0708] In one example, B includes 4. In another example, B does not include 4.

[0709] In one example, B includes 5. In another example, B does not include 5.

[0710] In one example, B includes 6. In another example, B does not include 6.

[0711] In one example, B includes 7. In another example, B does not include 7.

[0712] In one example, B includes 8. In another example, B does not include 8.

[0713] In another embodiment, an enhanced Type-I codebook for e.g., Rel-19 type-I CSI reporting can be designed (or extended) based on multiple frameworks, where each of the multiple frameworks is described in embodiments herein.

[0714] In one example, a first framework/scheme is based on one of the examples described in embodiments herein, and a second framework/scheme is based on one of the examples described in embodiments herein.

[0715] In one example, a first framework/scheme is based on one of the examples described in embodiments herein, and a second framework/scheme is based on another one of the examples described in embodiments herein.

[0716] In one example, a first framework/scheme is based on one of the examples described in embodiments herein, and a second framework/scheme is based on another one of the examples described in embodiments herein.

[0717] In another example, when rank is in a set A, an enhanced Type-I codebook is based on a first framework (or a first scheme), and when rank is in a set B, the enhanced Type-I codebook is based on a second framework (or a second scheme), where A, B are a subset of {1, 2, 3, 4, 5, 6, 7, 8}. The enhanced Type-I codebook can be according to at least one of the following examples.

[0718] In one example, A includes 1, 2, 3, 4 and B includes 5, 6, 7, 8.

[0719] In one example, A includes 1, 2 and B includes 3, 4, 5, 6, 7, 8.

[0720] In one example, A includes 1, 2 and B includes 3, 4.

[0721] In one example, A includes 1. In another example, A does not include 1.

[0722] In one example, A includes 2. In another example, A does not include 2.

[0723] In one example, A includes 3. In another example, A does not include 3.

[0724] In one example, A includes 4. In another example, A does not include 4.

[0725] In one example, A includes 5. In another example, A does not include 5.

[0726] In one example, A includes 6. In another example, A does not include 6.

[0727] In one example, A includes 7. In another example, A does not include 7.

[0728] In one example, A includes 8. In another example, A does not include 8.

[0729] In one example, B includes 1. In another example, B does not include 1.

[0730] In one example, B includes 2. In another example, B does not include 2.

[0731] In one example, B includes 3. In another example, B does not include 3.

[0732] In one example, B includes 4. In another example, B does not include 4.

[0733] In one example, B includes 5. In another example, B does not include 5.

[0734] In one example, B includes 6. In another example, B does not include 6.

[0735] In one example, B includes 7. In another example, B does not include 7.

[0736] In one example, B includes 8. In another example, B does not include 8.

[0737] In one example, a first framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0738] In one example, a first framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0739] In one example, a second framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0740] In one example, a second framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0741] In another example, when (N_1, N_2) is in a set A, an enhanced Type-I codebook is based on a first framework (or a first scheme), and when (N_1, N_2) is in a set B, the enhanced Type-I codebook is based on a second framework (or a second scheme), where A, B are a subset of the (N_1, N_2) pair values described in Table 1. The enhanced Type-I codebook can be according to at least one of the following examples.

[0742] In one example, A is a set including (N_1, N_2) values such that $N_2=1$, e.g., (64,1) (32,1), (16,1), (8,1), (4,1), (2,1).

[0743] In one example, B is a set including (N_1, N_2) values such that $N_2>1$, e.g., (16,4) (8,8), (16,2), (8,4), etc.

[0744] In one example, a first framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0745] In one example, a first framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0746] In one example, a second framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0747] In one example, a second framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0748] In another example, when (O_1, O_2) is in a set A, an enhanced Type-I codebook is based on a first framework (or a first scheme), and when (O_1, O_2) is in a set B, the enhanced Type-I codebook is based on a second framework (or a second scheme), where A, B are a subset of (4,4), (2,2), (4,1), (2,1). The enhanced Type-I codebook can be according to at least one of the following examples.

[0749] In one example, A includes $\{(4,4)\}$, and B includes $\{(4,1)\}$.

[0750] In one example, a first framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0751] In one example, a first framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0752] In one example, a second framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0753] In one example, a second framework (scheme) is based on one of the examples/embodiments described in embodiments herein.

[0754] In another example, an enhanced Type-I codebook based on either a first framework (or a first scheme) or a second framework (or a second scheme) can be configured via higher-layer signaling (e.g., codebookMode or codebook type, e.g., typeI-SinglePanel-r19, eTypeI-SinglePanel-r19). In one example, an enhanced Type-I codebook based on a first scheme/framework can be configured by higher-layer signaling e.g., 'codebookMode=1' (or 'CodebookMode=ModeA' or another name/terminology of information element (IE)) and an enhanced Type-I codebook based on a second scheme/framework can be configured by higher-layer signaling e.g., 'codebookMode=2' (or 'CodebookMode=Mode2' or another name/terminology of IE).

[0755] FIG. 20 illustrates an example method 2000 performed by a UE in a wireless communication system according to embodiments of the present disclosure. The method 2000 of FIG. 20 can be performed by any of the UEs 111-116 of FIG. 1, such as the UE 116 of FIG. 3, and a corresponding method can be performed by any of the BSs 101-103 of FIG. 1, such as BS 102 of FIG. 2. The method 2000 is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0756] The method begins with the UE receiving information about a CSI report (2010). For example, in 2010, the information indicates codebookType set to 'typeI-SinglePanel-r19', codebookMode set to either 'ModeA' or 'ModeB', and P_{CSI-RS} CSI-RS antenna ports, where $P_{CSI-RS} > 32$. The UE then determines the CSI report associated with the CSI-RS antenna ports based on the information (2020). For example, in 2020, the CSI report includes a PMI. The UE then transmits the CSI report (2030).

[0757] In various embodiments, when codebookMode is set to 'ModeA', for a rank value $v=1$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, and i_2 that indicate a precoding matrix. In various embodiments, when codebookMode is set to 'ModeA', for a rank value $v=2$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix. In various embodiments, when codebookMode is set to

'ModeA', the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix, for a rank value $v=3$ or the rank value $v=4$.

[0758] In various embodiments, when codebookMode is set to 'ModeB', for a rank value $v=1, 2, 3$, or 4 , the PMI includes a first indicator for SD basis vector b_l for each layer $l=1, 2, \dots, v$ and a second indicator for inter-polarization coefficient value c_l for each layer $l=1, 2, \dots, v$. The first and second indicators indicate a precoding matrix as follows

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_2 & \dots & b_v \\ c_1 b_1 & c_2 b_2 & \dots & c_v b_v \end{bmatrix},$$

b_l is a 2D DFT vector, and $c_l \in \{1, -1, j, -j\}$ for $l=1, 2, \dots, v$.

[0759] In various embodiments, when codebookMode is set to 'ModeB', for a rank value $v=6$ or $v=8$, the PMI includes a first indicator for SD basis vector b_l for each layer pair $l=1, 2, \dots, v'$ and a second indicator for inter-polarization coefficient value c_l associated with each layer pair $l=1, 2, \dots, v'$. The first and second indicators indicate a precoding matrix as follows

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v'} & b_{v'} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v'} b_{v'} & -c_{v'} b_{v'} \end{bmatrix},$$

b_l is a 2D DFT vector and $c_l \in \{1, j\}$ for $l=1, 2, \dots, v'$.

[0760] In various embodiments, when codebookMode is set to 'ModeB', for a rank value $v=5$ or $v=7$, the PMI includes a first indicator for SD basis vector b_l for each layer group $l=1, 2, \dots, v^*$, and a second indicator for inter-polarization coefficient value c_l associated with each layer group $l=1, 2, \dots, v^*$. The indicators indicate a precoding matrix as follows

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v^*-1} & b_{v^*-1} & b_{v^*} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v^*-1} b_{v^*-1} & -c_{v^*-1} b_{v^*-1} & c_{v^*} b_{v^*} \end{bmatrix},$$

b_l is a 2D DFT vector for $l=1, 2, \dots, v^*$, $c_l \in \{1, j\}$ for $l=1, 2, \dots, v^*-1$, and $c_{v^*} \in \{1, -1, j, -j\}$.

[0761] Any of the above variation embodiments can be utilized independently or in combination with at least one other variation embodiment. The above flowcharts illustrate example methods that can be implemented in accordance with the principles of the present disclosure and various changes could be made to the methods illustrated in the flowcharts herein. For example, while shown as a series of steps, various steps in each figure could overlap, occur in parallel, occur in a different order, or occur multiple times. In another example, steps may be omitted or replaced by other steps.

[0762] Although the present disclosure has been described with exemplary embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims. None of the descriptions in this application should be read as implying that any particular element, step, or func-

tion is an essential element that must be included in the claims scope. The scope of patented subject matter is defined by the claims.

What is claimed is:

1. A user equipment (UE) comprising:

a transceiver configured to receive information about a channel state information (CSI) report, the information indicating (i) codebookType set to 'typeI-SinglePanel-r19', (ii) codebookMode set to either 'ModeA' or 'ModeB', and (iii) P_{CSI-RS} CSI reference signal (CSI-RS) antenna ports, where $P_{CSI-RS} > 32$; and

a processor operably coupled to the transceiver, the processor, based on the information, configured to determine the CSI report associated with the CSI-RS antenna ports, the CSI report including a precoding matrix indicator (PMI),

wherein the transceiver is further configured to transmit the CSI report.

2. The UE of claim 1, when codebookMode is set to 'ModeA', for a rank value $v=1$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, and i_2 that indicate a precoding matrix shown in Table 1:

TABLE 1

when the rank value = 1			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1 O_1 - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1, 2, 3$	$W_{i_{1,1}, i_{1,2}, i_2}^{(1)}$
where $W_{i_{1,1}, i_{1,2}}^{(1)} = \frac{1}{\sqrt{P_{CSI-RS}}} \begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \end{bmatrix}$			

where

$$v_{l,m} = \begin{bmatrix} u_m e^{j \frac{2\pi l}{O_1 N_1}} & u_m & \dots & e^{j \frac{2\pi l (N_1 - 1)}{O_1 N_1}} u_m \end{bmatrix}^T,$$

$$u_m = \begin{bmatrix} 1 & e^{j \frac{2\pi m}{O_2 N_2}} & \dots & e^{j \frac{2\pi m (N_2 - 1)}{O_2 N_2}} \end{bmatrix},$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively, and

$$\varphi_n = e^{j \frac{2\pi n}{2}}.$$

3. The UE of claim 1, when codebookMode is set to 'ModeA', for a rank value $v=2$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix shown in Table 2:

TABLE 2

when the rank value = 2			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(2)}$

where $W_{i,l',m,m',n}^{(2)} = \frac{1}{\sqrt{2P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 3

where

$$v_{l,m} = \begin{bmatrix} u_m e^{j \frac{2\pi l}{O_1 N_1}} u_m \dots e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T,$$

$$u_m = \begin{bmatrix} 1 e^{j \frac{2\pi m}{O_2 N_2}} \dots e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix},$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively,

$$\varphi_n = e^{j \frac{\pi n}{2}},$$

and

the Table 3 is given as follows:

$i_{1,3}$	k_1	k_2
0	0	0
1	O_1	0
2	0	O_2
3	$2O_1$	0

4. The UE of claim 1, when codebookMode is set to 'ModeA', the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix shown in Table 4 or Table 5, for a rank value $v=3$ or the rank value $v=4$, respectively:

TABLE 4

when the rank value = 3			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(3)}$

where $W_{i,l',m,m',n}^{(3)} = \frac{1}{\sqrt{3P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} & v_{l,m} \\ \varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l,m} \end{bmatrix}$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 6

TABLE 5

when the rank value = 4			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(4)}$

where $W_{i,l',m,m',n}^{(4)} = \frac{1}{\sqrt{4P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} & v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 6

where

$$v_{l,m} = \begin{bmatrix} u_m e^{j \frac{2\pi l}{O_1 N_1}} u_m \dots e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T,$$

$$u_m = \begin{bmatrix} 1 e^{j \frac{2\pi m}{O_2 N_2}} \dots e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix},$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively,

$$\varphi_n = e^{j \frac{\pi n}{2}},$$

and

the Table 6 is given as follows:

$i_{1,3}$	k_1	k_2
0	O_1	0
1	0	O_2
2	O_1	O_2
3	$2O_1$	0

5. The UE of claim 1, when codebookMode is set to 'ModeB', for a rank value $v=1, 2, 3$, or 4, the PMI includes: a first indicator for spatial domain (SD) basis vector b_l for each layer $l=1, 2, \dots, v$, and a second indicator for inter-polarization coefficient value c_l for each layer $l=1, 2, \dots, v$, where:

the first and second indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_2 & \dots & b_v \\ c_1 b_1 & c_2 b_2 & \dots & c_v b_v \end{bmatrix},$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector, and

$c_l \in \{1, -1, j, -j\}$ for $l=1, 2, \dots, v$.

6. The UE of claim 1, when codebookMode is set to 'ModeB', for a rank value $v=6$ or $v=8$, the PMI includes: a first indicator for spatial domain (SD) basis vector b_l for each layer pair $l=1, 2, \dots, v'$, and

a second indicator for inter-polarization coefficient value c_l associated with each layer pair $l=1, 2 \dots, v'$, where:

$$v' = \frac{v}{2}$$

the first and second indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v'} & b_{v'} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v'} b_{v'} & -c_{v'} b_{v'} \end{bmatrix}$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector, and

$c_l \in \{1, j\}$ for $l=1, 2 \dots, v'$.

7. The UE of claim 1, when codebookMode is set to 'ModeB', for a rank value $v=5$ or $v=7$, the PMI includes:

a first indicator for spatial domain (SD) basis vector b_l for each layer group $l=1, 2 \dots, v^*$, and

a second indicator for inter-polarization coefficient value c_l associated with each layer group $l=1, 2 \dots, v^*$, where:

$$v^* = \left\lceil \frac{v}{2} \right\rceil$$

the indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v^*-1} & b_{v^*-1} & b_{v^*} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v^*-1} b_{v^*-1} & -c_{v^*-1} b_{v^*-1} & c_{v^*} b_{v^*} \end{bmatrix}$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector for $l=1, 2 \dots, v^*$,

$c_l \in \{1, j\}$ for $l=1, 2 \dots, v^*-1$, and

$c_{v^*} \in \{1, -1, j, -j\}$.

8. A base station comprising:

a processor; and

a transceiver operably coupled to the processor, the transceiver configured to:

transmit information about a channel state information (CSI) report, the information indicating (i) codebookType set to 'typeI-SinglePanel-r19', (ii) codebookMode set to either 'ModeA' or 'ModeB', and (iii) P_{CSI-RS} CSI reference signal (CSI-RS) antenna ports, where $P_{CSI-RS} > 32$; and

receive the CSI report associated with the CSI-RS antenna ports, the CSI report including a precoding matrix indicator (PMI).

9. The BS of claim 8, when codebookMode is set to 'ModeA', for a rank value $v=1$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, and i_2 that indicate a precoding matrix shown in Table 1:

TABLE 1

when the rank value = 1			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1 O_1 - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1, 2, 3$	$W_{i_{1,1}, i_{1,2}, i_2}^{(1)}$

where $W_{i_{1,1}, i_{1,2}, i_2}^{(1)} = \frac{1}{\sqrt{P_{CSI-RS}}} \begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \end{bmatrix}$

where

$$v_{l,m} = \begin{bmatrix} u_m e^{j \frac{2\pi l}{O_1 N_1}} u_m \dots e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T$$

$$u_m = \begin{bmatrix} 1 e^{j \frac{2\pi m}{O_2 N_2}} \dots e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix}$$

where:

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively, and

$$\varphi_n = e^{j \frac{\pi n}{2}}$$

10. The BS of claim 8, when codebookMode is set to 'ModeA', for a rank value $v=2$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix shown in Table 2:

TABLE 2

when the rank value = 2			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1 O_1 - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(2)}$

where $W_{i_{1,1}, i_{1,1}+k_1, i_{1,2}, i_{1,2}+k_2, i_2}^{(2)} = \frac{1}{\sqrt{2P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 3

where

$$v_{l,m} = \begin{bmatrix} u_m e^{j \frac{2\pi l}{O_1 N_1}} u_m \dots e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T$$

$$u_m = \begin{bmatrix} 1 e^{j \frac{2\pi m}{O_2 N_2}} \dots e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix}$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively,

$$\varphi_n = e^{\frac{j\pi n}{2}},$$

and
the Table 3 is given as follows:

$i_{1,3}$	k_1	k_2
0	0	0
1	O_1	0
2	0	O_2
3	$2O_1$	0

11. The BS of claim 8, when codebookMode is set to 'ModeA', the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix shown in Table 4 or Table 5, for a rank value $v=3$ or the rank value $v=4$, respectively:

TABLE 4

when the rank value = 3			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1} i_{1,1} + k_1 i_{1,2} + k_2 i_2}^{(3)}$

$$\text{where } W_{i_{1,1} i_{1,2} i_2}^{(3)} = \frac{1}{\sqrt{3P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} & v'_{l,m'} & v_{l,m} \\ \varphi_n v_{l,m} & \varphi_n v'_{l,m'} & -\varphi_n v_{l,m} \end{bmatrix}$$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 6

TABLE 5

when the rank value = 4			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1} i_{1,1} + k_1 i_{1,2} + k_2 i_2}^{(4)}$

$$\text{where } W_{i_{1,1} i_{1,2} i_2}^{(4)} = \frac{1}{\sqrt{4P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} & v'_{l,m'} & v_{l,m} & v'_{l,m'} \\ \varphi_n v_{l,m} & \varphi_n v'_{l,m'} & -\varphi_n v_{l,m} & -\varphi_n v'_{l,m'} \end{bmatrix}$$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 6

where

$$v_{l,m} = \begin{bmatrix} u_m e^{j\frac{2\pi l}{O_1 N_1}} u_m \dots e^{j\frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T,$$

$$u_m = \begin{bmatrix} 1 e^{j\frac{2\pi m}{O_2 N_2}} \dots e^{j\frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix},$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively,

$$\varphi_n = e^{\frac{j\pi n}{2}},$$

and
the Table 6 is given as follows:

$i_{1,3}$	k_1	k_2
0	O_1	0
1	0	O_2
2	O_1	O_2
3	$2O_1$	0

12. The BS of claim 8, when codebookMode is set to 'ModeB', for a rank value $v=1, 2, 3$, or 4, the PMI includes: a first indicator for spatial domain (SD) basis vector b_l for each layer $l=1, 2, \dots, v$, and a second indicator for inter-polarization coefficient value c_l for each layer $l=1, 2, \dots, v$, where:

the first and second indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{\text{CSI-RS}}}} \begin{bmatrix} b_1 & b_2 & \dots & b_v \\ c_1 b_1 & c_2 b_2 & \dots & c_v b_v \end{bmatrix},$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector, and

$c_l \in \{1, -1, j, -j\}$ for $l=1, 2, \dots, v$.

13. The BS of claim 8, when codebookMode is set to 'ModeB', for a rank value $v=6$ or $v=8$, the PMI includes: a first indicator for spatial domain (SD) basis vector b_l for each layer pair $l=1, 2, \dots, v'$, and a second indicator for inter-polarization coefficient value c_l associated with each layer pair $l=1, 2, \dots, v'$, where:

$$v' = \frac{v}{2}$$

the first and second indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{\text{CSI-RS}}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v'} & b_{v'} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v'} b_{v'} & -c_{v'} b_{v'} \end{bmatrix},$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector, and

$c_l \in \{1, j\}$ for $l=1, 2, \dots, v'$.

14. The BS of claim 8, when codebookMode is set to 'ModeB', for a rank value $v=5$ or $v=7$, the PMI includes: a first indicator for spatial domain (SD) basis vector b_l for each layer group $l=1, 2, \dots, v^*$, and a second indicator for inter-polarization coefficient value c_l associated with each layer group $l=1, 2, \dots, v^*$, where:

$$v^* = \left\lceil \frac{v}{2} \right\rceil,$$

the indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v^*-1} & b_{v^*-1} & b_{v^*} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v^*-1} b_{v^*-1} & -c_{v^*-1} b_{v^*-1} & c_{v^*} b_{v^*} \end{bmatrix},$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector for $l=1, 2, \dots, v^*$,

$c_l \in \{1, j\}$ for $l=1, 2, \dots, v^*-1$, and

$c_{v^*} \in \{1, -1, j, -j\}$.

15. A method performed by a user equipment (UE), the method comprising:

receiving information about a channel state information (CSI) report, the information indicating (i) codebook-Type set to 'typeI-SinglePanel-r19', (ii) codebookMode set to either 'ModeA' or 'ModeB', and (iii) P_{CSI-RS} CSI reference signal (CSI-RS) antenna ports, where $P_{CSI-RS} > 32$;

based on the information, determining the CSI report associated with the CSI-RS antenna ports, the CSI report including a precoding matrix indicator (PMI); and

transmitting the CSI report.

16. The method of claim **15**, when codebookMode is set to 'ModeA', for a rank value $v=1$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, and i_2 that indicate a precoding matrix shown in Table 1:

TABLE 1

when the rank value = 1			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1 O_1 - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1, 2, 3$	$W_{i_{1,1}, i_{1,2}, i_2}^{(1)}$
where $W_{i_{1,1}, i_{1,2}, i_2}^{(1)} = \frac{1}{\sqrt{P_{CSI-RS}}} \begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \end{bmatrix}$			

where:

$$v_{l,m} = \left[u_m \ e^{j \frac{2\pi l}{O_1 N_1}} u_m \ \dots \ e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \right]^T,$$

$$u_m = \left[1 \ e^{j \frac{2\pi m}{O_2 N_2}} \ \dots \ e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \right],$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively, and

$$\varphi_n = e^{j \frac{\pi n}{2}}.$$

17. The method of claim **15**, when codebookMode is set to 'ModeA', for a rank value $v=2$, the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix shown in Table

TABLE 2

when the rank value = 2			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, 1, \dots, N_1 O_1 - 1$	$0, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,2}, i_2}^{(2)}$
where $W_{i_{1,1}, i_{1,2}, i_2}^{(2)} = \frac{1}{\sqrt{2P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$			
and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 3			

where

$$v_{l,m} = \left[u_m \ e^{j \frac{2\pi l}{O_1 N_1}} u_m \ \dots \ e^{j \frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \right]^T,$$

$$u_m = \left[1 \ e^{j \frac{2\pi m}{O_2 N_2}} \ \dots \ e^{j \frac{2\pi m(N_2-1)}{O_2 N_2}} \right],$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively,

$$\varphi_n = e^{j \frac{\pi n}{2}},$$

and
the Table 3 is given as follows:

$i_{1,3}$	k_1	k_2
0	0	0
1	O_1	0
2	0	O_2
3	$2O_1$	0

18. The method of claim **15**, when codebookMode is set to 'ModeA', the PMI includes indicators $i_{1,1}$, $i_{1,2}$, $i_{1,3}$, and i_2 that indicate a precoding matrix shown in Table 4 or Table 5, for a rank value $v=3$ or the rank value $v=4$, respectively:

TABLE 4

when the rank value = 3			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,2}, i_2}^{(3)}$
where $W_{i_{1,1}, i_{1,2}, i_2}^{(3)} = \frac{1}{\sqrt{3P_{CSI-RS}}} \begin{bmatrix} v_{l,m} & v_{l',m'} & v_{l,m} \\ \varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l,m} \end{bmatrix}$			
and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 6			

TABLE 5

when the rank value = 4			
$i_{1,1}$	$i_{1,2}$	i_2	
$0, \dots, N_1 O_1 - 1$	$0, 1, \dots, N_2 O_2 - 1$	$0, 1$	$W_{i_{1,1}, i_{1,1} + k_1, i_{1,2}, i_{1,2} + k_2, i_2}^{(4)}$

where $W_{i, i', m, m', n}^{(4)} = \frac{1}{\sqrt{4P_{CSI-RS}}} \begin{bmatrix} v_{i,m} & v_{i',m'} & v_{i,m} & v_{i',m'} \\ \varphi_n v_{i,m} & \varphi_n v_{i',m'} & -\varphi_n v_{i,m} & -\varphi_n v_{i',m'} \end{bmatrix}$

and a mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 6

where:

$$v_{i,m} = \begin{bmatrix} u_m & e^{j\frac{2\pi i}{O_1 N_1}} u_m & \dots & e^{j\frac{2\pi i(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T,$$

$$u_m = \begin{bmatrix} 1 & e^{j\frac{2\pi m}{O_2 N_2}} & \dots & e^{j\frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix},$$

N_1 and N_2 are numbers of antenna ports in a first dimension and a second dimension of the antenna ports, respectively,

O_1 and O_2 are values of oversampling factors for the first and second dimensions, respectively,

$$\varphi_n = e^{j\frac{in}{2}},$$

and
the Table 6 is given as follows:

$i_{1,3}$	k_1	k_2
0	O_1	0
1	0	O_2
2	O_1	O_2
3	$2O_1$	0

19. The method of claim **15**, when codebookMode is set to ‘ModeB’, for a rank value $v=1, 2, 3$, or 4 , the PMI includes:

a first indicator for spatial domain (SD) basis vector b_l for each layer $l=1, 2, \dots, v$, and
a second indicator for inter-polarization coefficient value c_l for each layer $l=1, 2, \dots, v$,

where:

the first and second indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_2 & \dots & b_v \\ c_1 b_1 & c_2 b_2 & \dots & c_v b_v \end{bmatrix},$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector, and

$c_l \in \{1, -1, j, -j\}$ for $l=1, 2, \dots, v$.

20. The method of claim **15**, when codebookMode is set to ‘ModeB’, for a rank value $v=6$ or $v=8$, the PMI includes:

a first indicator for spatial domain (SD) basis vector b_l for each layer pair $l=1, 2, \dots, v'$, and
a second indicator for inter-polarization coefficient value c_l associated with each layer pair $l=1, 2, \dots, v'$,

where:

$$v' = \frac{v}{2}$$

the first and second indicators indicate a precoding matrix as follows:

$$W = \frac{1}{\sqrt{vP_{CSI-RS}}} \begin{bmatrix} b_1 & b_1 & b_2 & b_2 & \dots & b_{v'} & b_{v'} \\ c_1 b_1 & -c_1 b_1 & c_2 b_2 & -c_2 b_2 & \dots & c_{v'} b_{v'} & -c_{v'} b_{v'} \end{bmatrix},$$

b_l is a two-dimensional (2D) discrete Fourier transform (DFT) vector, and

$c_l \in \{1, j\}$ for $l=1, 2, \dots, v'$.

* * * * *